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USE AND MAINTENANCE INSTRUCTION MANUAL





2K23-048

FOTOMETRIA

AIMING S220

MM INDIA

TRANSLATION OF THE ORIGINAL INSTRUCTIONS

CE

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DECLARATION OF CONFORMITY

IN ACCORDANCE WITH EUROPEAN DIRECTIVES

MANUFACTURER'S NAME:	INFIND SRL – IN QUALITÀ DI COSTRUTTORE
ADDRESS:	VIA E. FERMI, 21 – Z.I. SETTIMO 37026 PESCANTINA – VERONA – ITALY
PRODUCT NAME:	FOTOMETRIA AIMING S220
MODEL:	2K23048_01
MANUFACTURING YEAR:	08/2023

IT IS HEREBY CERTIFIED THAT THE PRODUCT COMPLIES WITH THE FOLLOWING EUROPEAN DIRECTIVES:

ELECTRICAL SAFETY: 2014/35/UE

MACHINERY DIRECTIVE: 2006/42/EC

EMC DIRECTIVE: 2014/30/EU

FOR THE SAFETY REGULATIONS OF THE INSTALLED DEVICES, REFER TO THE RELEVANT DECLARATION MADE BY THE MANUFACTURER

IT IS HEREBY CERTIFIED THAT THE TECHNICAL FILE RESPONSIBLE IS AT THE COMPANY:
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1 INTRODUCTION

1.1 SCOPE OF THIS MANUAL OF INSTRUCTIONS FOR USE AND MAINTENANCE

This instruction manual is an integral part of the machine and serves to provide all the information necessary for:

- Correctly informing the operators about the problems of safety.
- Handling the machine, packed and unpacked, under safe conditions.
- Correct installation of the machine.
- Thorough knowledge of its operation and limits.
- Correct use under safe conditions.
- Instructions for performing maintenance safely and correctly.



The managers of the company departments where this machine will be installed are required, according to the legislation in force, to read the contents of this instruction manual carefully and have the operators and maintenance personnel read the parts of it that are pertinent to their job.
This document assumes that the standards of safety and hygiene in the workplace are regularly complied with in the plant where the machine will be installed.
The instructions, drawings and documents contained in this manual are strictly confidential technical information and the property of the manufacturer. They may not be copied in any way, either wholly or in part.
The client is also responsible for ensuring that, if this document should be amended by the manufacturer, only the updated versions of the manual will be effectively present in the place of use.

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2 GENERAL INFORMATION

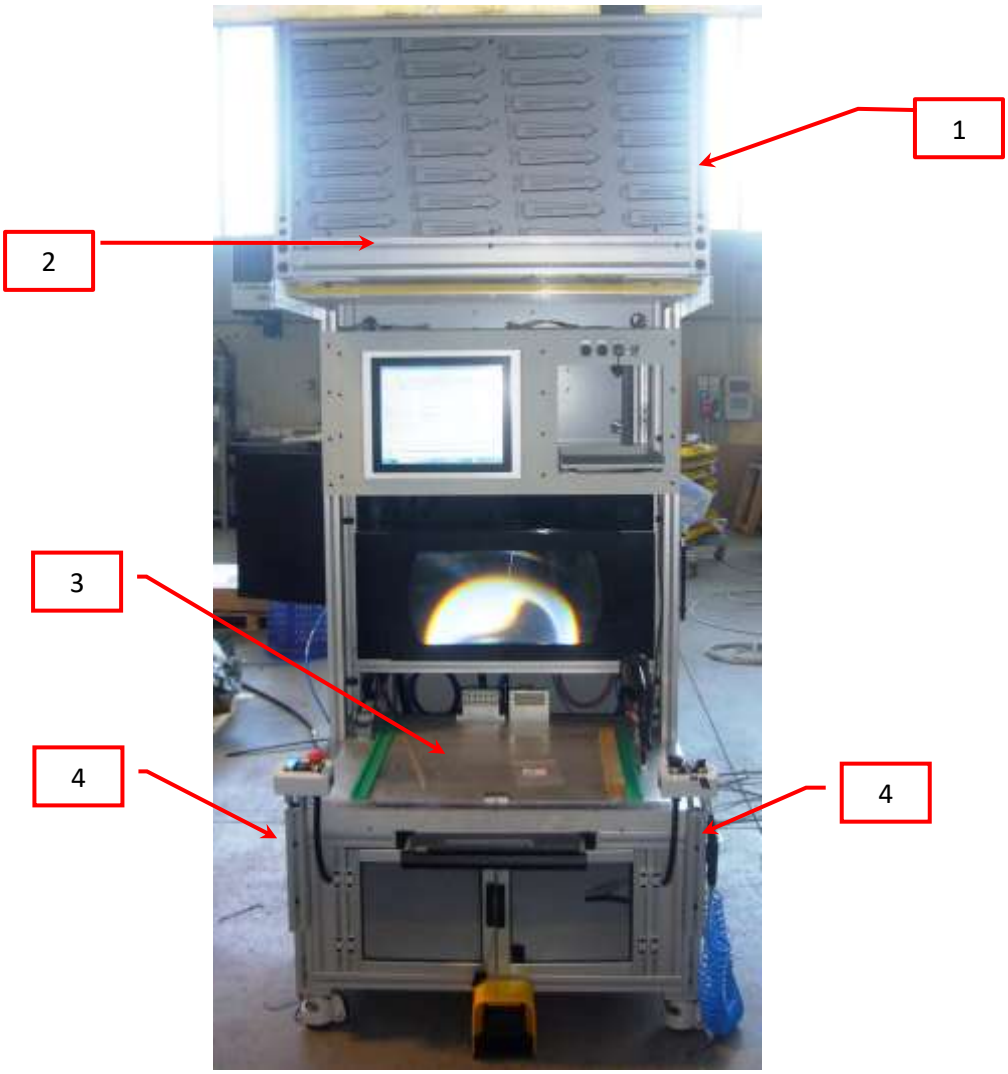
2.1 OPERATING PRINCIPLE

The Photometric Tester series "WXE2012" has the ability to perform all controls of the finished headlamp in a rapid, accurate manner at its production island. The unit can be supplied in a range of versions so as to be perfectly integrated in the production island.
All models follow the same algorithm of control and adjustment of the headlamp.
The main features of the control are:

- GEOMETRICAL analysis of the light beam (Quality analysis)
- PHOTOMETRIC analysis of the light beam with measurement of photometric points
- Control of ABSORPTION of 9 Functions (Low beam, High beam, Spotlight, Direction, Parking light, Fog light, DRL, SBL, Side Mark)
- Check of CORRECT ATTACHMENT OF THE PARABOLA
- Functional test of PARABOLIC DIP (Corrector) with control of absorption
- Check of high-low beam alignment
- EMAX search
- Recognition and check of horizontality of the line of demarcation
- Control of CIRCULATION (ECE, UK, SAE)
- Ignition and control of discharge lamps and ECU control
- Check of horizontality and linearity of fog light beam
- Adjustment of headlamp within programmable limits of acceptability
- Check of movement of the low beam in AFS systems.

The system is based on comparison with respect to a sample piece. Each recipe is “Constructed” around a piece previously centered and measured in the photometric laboratory. After determining the thresholds of acceptable deviation with respect to that sample, the system “Passes” or “Fails” the pieces under production, comparing them with the settings of the recipe constructed in that way.

2.2 STRUCTURE OF THE MACHINE



Description:

1. Structure group consisting of an aluminum profile structure and aluminum components
2. Control panel
3. Test piece area
4. Button panel

2.3 DECLARATIONS

The machine is produced in accordance with the Community directives applicable and pertinent to it at the time of its sale.

2.4 ELECTROMAGNETIC COMPATIBILITY

All commercial components are provided with the electromagnetic compatibility declaration, provided by the manufacturer, and have been installed according to the instructions given by the component manufacturer.

2.5 CERTIFICATES OF THIRD PARTY PRODUCTS INSTALLED

You can request or consult the certifications of the main third-party products installed at the following links:

SIEMENS:
https://sieportal.siemens.com/it-it/search?scope=knowledgebase&SearchTerm=&SortingOption=CreationDateDesc&EntryTypes=all&Page=0&PageSize=20

Schneider electric:
https://www.se.com/it/it/work/support/customer-service/normative/conformity-ec-marcatura/conformity-ec-marcatura.jsp

2.6 INFORMATION ABOUT TECHNICAL ASSISTANCE

The machines are covered by a warranty, as envisaged in the general conditions of sale.
If during the period of validity, the machine should experience defective function or breakdowns of parts, in the cases covered by the warranty the manufacturer, after performing the necessary checks of the machine, will repair it or replace the defective parts.
Any alterations made by the user without the explicit written authorization of the manufacturer will invalidate the warranty and relieve the manufacturer of any liability for damages caused by a defective product.
This applies in particular when the aforementioned alterations are made on safety devices, interfering with their effectiveness.
The same considerations apply to the use of parts other than original spare parts, or parts different from those explicitly recommended by the manufacturer.
For all these reasons we advise our clients always to contact our assistance service.

2.7 PREPARATIONS TO BE MADE BY THE CLIENT

Except as may be contractually agreed otherwise, the client is responsible for the following:

- Preparation of the site, including any works of masonry and/or raceways that may be required.
- Supplying electrical power to the machine in accordance with the standards in force in the country of use.
- Supplying the pneumatic system with compressed air.

3 SAFETY

3.1 GENERAL SAFETY WARNINGS



Before commissioning the machine read the instructions in this manual carefully, and make sure you understand all the information it contains.

The manufacturer has put maximum effort into designing the machine, considering both functional and safety aspects.
It has been equipped with all the protections and safety devices deemed necessary and is also equipped with sufficient information for its safe and correct use.
The user can appropriately integrate the information provided by the manufacturer with any additional work instructions, obviously not in conflict with those contained in this instruction manual, to contribute to its safety of use.
For example, all people working on the machine should pay special attention to their clothing:

- Do not wear clothing with loose ends that could catch on parts of the machine.
- Do not wear ties, scarves or other loose clothing.
- Do not wear large rings or bracelets that could get caught in moving parts of the machine.

In any case it is essential to diligently follow these instructions:

- It is strictly forbidden to operate the machine in automatic mode with the fixed and/or mobile guards removed.
- It is strictly forbidden to circumvent the safety devices installed on the machine.
- Operations carried out with reduced safety must be carried out scrupulously respecting the instructions given in the relevant descriptions.
- Do not alter parts of the machine for any reason; in case of malfunction due to failure to comply with these regulations, the manufacturer is not responsible for the consequences. We recommend requesting any changes directly from the manufacturer.
- Clean the machine casings, panels and electrical panels with a soft, dry cloth or rag moistened with a mild detergent solution; do not use any type of solvent such as alcohol or petrol as the surfaces could be damaged.
- Position the machine as agreed upon when ordering, according to the diagrams provided by the manufacturer. Otherwise, the manufacturer will not be responsible for any problems.

IMPORTANT! The manufacturer does not accept any liability for damages caused by the machine to persons, animals or property in the following cases:

- **Use of the machine by inadequately trained personnel.**
- **Improper use of the machine.**
- **Defective electrical, hydraulic or pneumatic power supplies.**
- **Erroneous installation.**
- **Lack of regular maintenance.**
- **Unauthorized alterations or repairs.**
- **Use of non-original spare parts or parts not specific for the model.**
- **Total or partial disregard of the instructions.**
- **Use contrary to specific national regulations.**
- **Calamities and exceptional events.**

General Requisites

The movable elements must always be used as indicated by the manufacturer, following the instructions in this manual, which must always be available in the workplace.
All safety devices on the movable elements to prevent accidents and ensure safety must be left unaltered and not removed, and must be adequately protected.
The user is required to inform the employer or a direct supervisor promptly about any defects or malfunctions observed regarding the movable elements.

Controls and inspections

These should be performed as indicated in the relative section of the manual.
If worn or defective parts are not promptly replaced, the manufacturer is not liable for any possible resulting accidents or damage.
Inspections must be made by an expert; they must be both visual and functional, in order to guarantee the safety of the machine.
They will include:

- Inspection of all load-bearing structures, which should not show any signs of breakage, damage, distortion, corrosion, wear or alteration with respect to the original characteristics.
- Inspection of all mechanical parts.
- Inspection of all the safety devices installed on the machine.
- Inspection of all connections with screws and bolts.
- Functional inspection of the machine.
- Inspection of the machine's status.
- Control of the seal and efficiency of the pneumatic and/or hydraulic system.
- The results of this inspection are reported on a specific inspection report form.

IMPORTANT! If any malfunctions are found, they must be eliminated before the machine is used again, and the expert who reported them shall note on the form that they have been satisfactorily repaired, giving his approval for use of the machine.

The inspector shall report promptly to the manufacturer if any breakage or hazardous malfunctions are found.

Take the machine out of use when any malfunctions are found in operation, providing for the necessary inspection and/or repairs.

Make sure there are no extraneous objects in the working parts of the machine.

Always check after any maintenance work that no object is left among the moving parts.

To guarantee the maximum safety in handling the machine it is, in any case, **strictly forbidden:**

- To tamper with any part of the machine.
- To leave movable elements unwatched.
- To use the machine working but not fully efficient.
- To alter the machine so as to change the type of use originally established, without the manufacturer's explicit authorization or without taking complete responsibility as required by Machine Directive 2006/42/CE.
- To handle the movable parts manually in case of power outage.

3.2 PROPER USE

Machine operator 1st level: operator lacking specific skills, only able to perform simple tasks such as operate the machine using the keys on the keyboard, loading and unloading the materials used during production with the safety devices in place and in operation, simple operations of assembly and fastening screws.

Before being able to work on the machine, the operator must be informed, by the company that commissioned the machine, on the characteristics of the product he will have to handle, on the safety regulations in force in the workplace and must be provided with a copy of this manual.



The use of different products / materials from those specified by the manufacturer, which can cause damage to the machine and situations of hazard for the operator and / or persons near the machine, is considered erroneous and improper.

During use, the operator stands in front of the bench.

The procedure envisaged for the compliant use of the machinery is the following:

1. Make sure all connections are properly connected
2. Turn on the bench by turning the switch on the electrical panel (see example fig below)



3. Rotate the automatic/manual machine selection lever to the “AUT” position (see chapter 5)
4. Press the start button to arm the machine (see chapter 5)
5. Load the piece into the machine as indicated on the operator panel
6. Move away from the handling area by exiting the photoelectric barriers.
7. Press the cycle start button (see chapter 5).
8. Wait for the end of the work cycle. When all movements are stopped and the result of the processing appears on the display, remove the worked piece.
9. Take a new piece and repeat the procedure from step 4.

Any machine failures during production are shown on the display.

3.3 SAFETY DEVICES

The following safety devices are installed in the machine:

1. Structure in extruded aluminum profile with panels in rigid transparent Lexan material and/or rigid gray Ecobond material.
2. Logic blocks to ensure safety functions, for example Pilz or Sick.
3. Safety barrier
4. Emergency stops, rotary mushroom buttons located on the control panels. They are mushroom buttons with the purpose of interrupting the power and control circuits.
5. Pressure switches to control the presence of air in the pneumatic circuits

3.4 SIGNS

Signs must be installed near the machine and in the work zone, as follows:

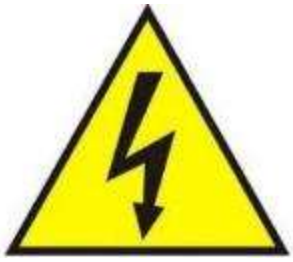
Prohibition signs

1. Sign prohibiting removal of the safety devices or guards on the machine.
2. Sign prohibiting repairs and/or adjustments while the machine is running.

Sign 1	Sign 2
	

Danger Signs

1. Sign indicating electrical danger
2. Moving organs

Cartello 1	Cartello 2
	

Requirement signs

1. Requirement to wear protective footwear.
2. Requirement to wear gloves.
3. La macchina emette un rumore sempre <75dB

Cartello 1	Cartello 2
	

3.5 RESIDUAL RISKS

Although the manufacturer has made the utmost effort in designing this machine and has equipped it with all the protections and safety devices deemed necessary, the operator must always pay the utmost attention when using the machine.
The movement speed of the electric axes is adequately regulated to avoid creating danger if the photoelectric barriers are interrupted, causing the operator to invade the work area with the moving axes.

It is necessary to pay attention to the following residual risks which are present when using the machine and which cannot be eliminated.

	**IMPORTANT: HAZARDOUS MOVEMENTS** No one except the operator may stand near the machine. If any third parties are present they should be asked to move away to prevent hindrance during production and possible damage.

	**IMPORTANT: ELECTRICAL RISKS DUE TO POWERED PARTS** Any works on the general power panel of the machine must be carried out by qualified personnel only, and in any case with the power disconnected.

	**WARNING: IMPROPER USE** The machine is made in compliance with the relevant community directives. Use that does not comply with what is indicated in this manual is to be considered dangerous for the safety of the operator. Use with malfunctioning or tampered with safety devices generates serious danger for the operator.

4 INSTALLATION

4.1 MOVEMENT AND HANDLING

Driver of hoisting and moving equipment: an operator qualified to use equipment for hoisting and moving materials and machinery (scrupulously following the manufacturer's instructions), in accordance with the laws in force in the country in which the machine is used.

The machine consists of many parts assembled in a single structure; it can be moved using equipment capable of withstanding its size and weight.
It is preferable during handling to have several skilled workers present so as to prevent any damage to the machine or persons and/or surrounding property.
For movement in the plant, the parts can be carried on a forklift or bridge crane, providing appropriate harnessing with the use of cables having adequate characteristics of tensile strength in relation to the weight to be moved.



For movement within the factory it is possible to use the wheels supplied with the counter

The manufacturer is not responsible for any damage to persons and/or property in case of the use of different hoisting systems from those described above.

Peso (kg):	400
Dimensioni (lxpxh m):	1x2x2,3

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4.2 PREPARATIONS

Preparations for installation

Prepare a handling zone adequate for the dimensions of the machine and hoisting equipment used to move it.

Preparations shall be made for the machine in such a way as to optimize ergonomics and safety in the workplace: leave sufficient room around the machine to facilitate use and handling of materials for processing and for operations of maintenance and adjustment.

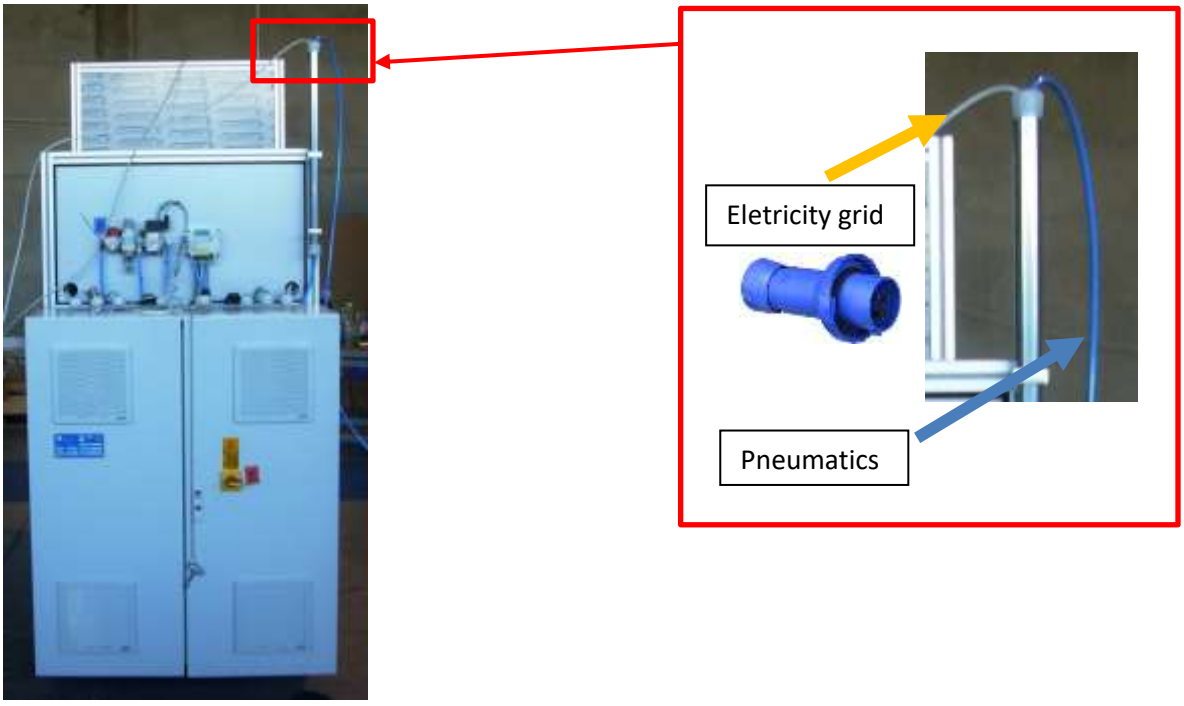
Preparation of the electrical system

Electrical maintenance expert: a qualified electrician able to operate the machine under normal conditions, to run it and carry out activities of adjustment, maintenance and repairs to the electrical parts. **He is qualified to work with the power on inside cabinets and shunt boxes.**

Connection of the electrical system is made by specialized and qualified personnel according to the wiring diagram and provisions required by law and/or technical standards in force on the subject of safety in the workplace and electrical systems.

Location of connections

The electrical and pneumatic mains connections are located in the top left corner of the machine. They come out of a special channel with a cable or tube long enough to be reached by the operator.



Our company has no liability for damage to property, persons and/or animals caused by failure to comply with these provisions.

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To ensure an adequate level of safety, the electrical system supplying the machine should provide, entirely for the user's account, a grounding system in accordance with the regulations in the user's country, and anything else required for correct compliance in a workmanlike manner with the laws and/or technical standards on the subject of safety in the workplace and electrical systems.
Prepare connections for grounding the machine chassis.

IMPORTANT!
These preparations are always for the account and under the complete responsibility of the user and nothing can be charged to the manufacturer for damage to persons, property and/or animals caused by improper electrical connection.

4.3 POSITIONING

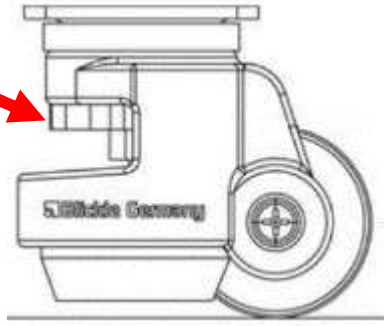
Mechanical maintenance expert: a qualified expert, able to operate the machine under normal conditions, to run it and carry out any necessary activities of adjustment, maintenance and repairs on mechanical parts. **Typically, he is not qualified to work on the electrical system with the power on.**

Considering the structure in a single unit of the machine, and its moderate weight, it is equipped with adjustable feet for leveling. We recommend using suitable equipment to perform the leveling operation so as not to endanger machine operation.





Leveling ring nut



ATTENTION!
The bench must be perfectly level to work correctly.

4.4 CONNECTIONS

Manufacturer's expert: a qualified expert made available by the manufacturer to perform complex operations under particular conditions or, in any case, as agreed with the user. His skills are, depending on the case, mechanical and/or electrical and/or electronic and/or software.

Electric connection

The connections inside the machine are made by the manufacturer's qualified expert.

The electric connection between the machine power panel and the client's mains connection of electrical distribution must be made by the client's qualified personnel.

Make sure of the following:

- The characteristics of the electrical system comply with the information on the rating plate showing the main technical data.
- The system is equipped with an efficient grounding system that complies with the regulations and laws in force (grounding is mandatory by law).

In case of prolonged interruption of use of the station, disconnect it from the power supply using the main circuit breaker on the power panel.

In case of outage of malfunction of the electrical system check the connection with the main power panel and, if necessary, contact our assistance service.



IMPORTANT!
The power line for the station must be protected by a magnetothermal switch with a differential disconnection mechanism of suitable size to perform this task.

Technical data for electric connection:

Supply voltage	220 Vac P+N+T
Frequency	50/60 Hz
Nominal power	16 A
Degree of protection	IP67

Technical data for pneumatic connection:

The air line must be inert, filtered, dehumidified and unlubricated.

Minimum pressure	4,0 bar
Maximum pressure	6,5 bar
Filtration	5,0 µm

4.5 PRELIMINARY CHECKLIST

Every time the machine is started, a few preliminary checks have to be made:

- Check of all the safety system.
- Check of the guards.
- Check of the signs.

It is essential to make a series of inspections and controls before starting the machine, in order to prevent errors or accidents during startup:

- Make sure the machine did not suffer any damage during assembly.
- Carefully inspect the power panels, control panels, electric wires and pipes.
- Check the correct connection to all the external energy sources.
- Make sure all parts of the machine can move and rotate freely.
- Make sure the pneumatic connections are tight so as not to cause hazardous leaks.

4.6 ENVIRONMENTAL CONDITIONS

The machine does not require particular environmental conditions.
It must be installed inside a lighted, ventilated industrial building with a solid and level floor.
For its correct functioning, the machine requires an environment characterized by the following parameters:

Temperature	+5°C ÷ +40°C
Relative humidity	30% ÷ 80%

4.7 LIGHTING

The lighting of the installation room must comply with the laws in force in the country in which the machine is installed and must in any case guarantee good visibility at every point, not create dangerous reflections and allow clear reading of the control panels, as well as identification emergency buttons, if provided.

5 ELECTRICAL PANEL AND BUTTONS

The composition of the machine for its operational functions includes:

- N°1 General electrical panel
- N°1 operator panel for managing settings and the interface towards the operator, buttons and control selectors
- N° 1 PC for managing the leak test and functional vision
- N°2 pushbutton panels
- N°1 Manual barcode reader

5.1 GENERAL ELECTRICAL PANEL

The general electrical panel contains all the power and control devices necessary for the operation of the machine.

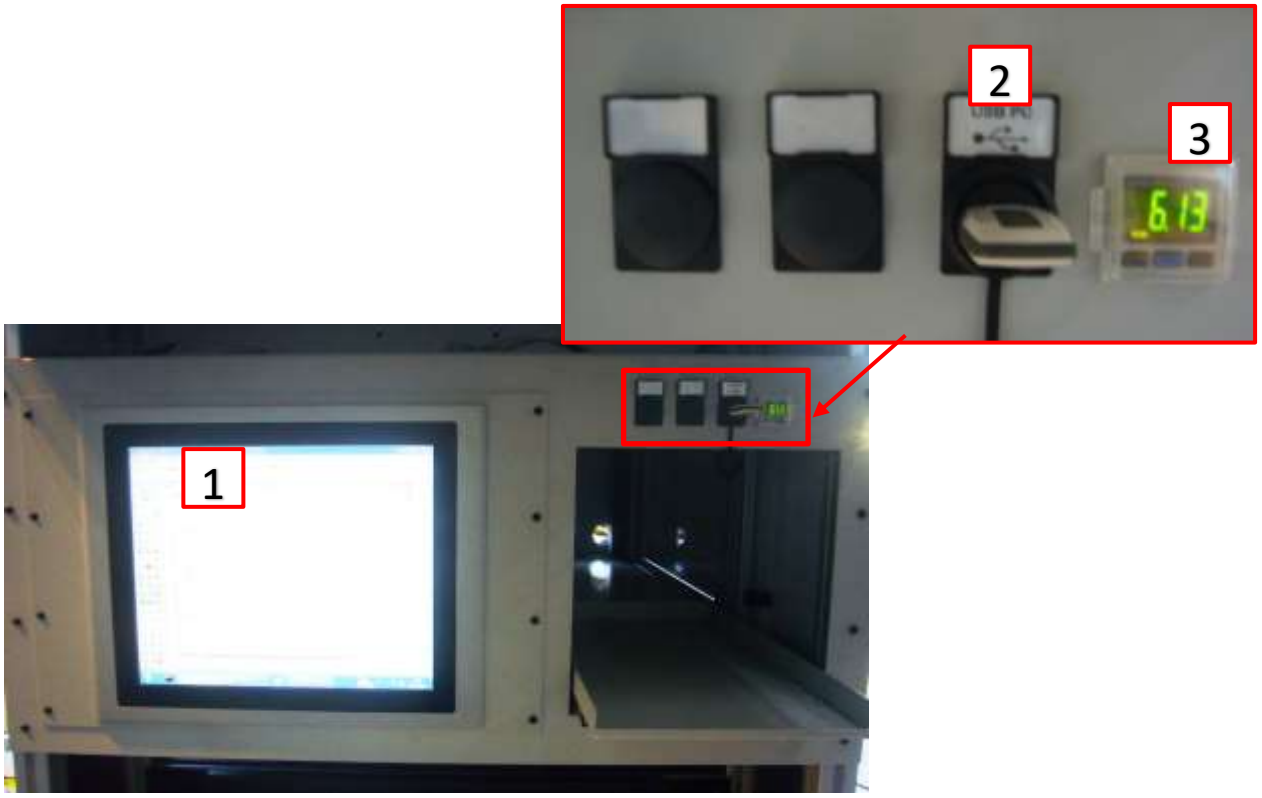
The main devices are:

- Main disconnect switch with return control on the door.
- Automatic, circuit breaker and differential circuit breakers, for the power supplies of the various users.
- Sensor and valve power supplies.
- Controller modules for logical safety during operation.
- Terminal blocks for connecting signals.

The wiring of the electrical system was carried out in compliance with current regulations.

5.2 OPERATOR PANEL

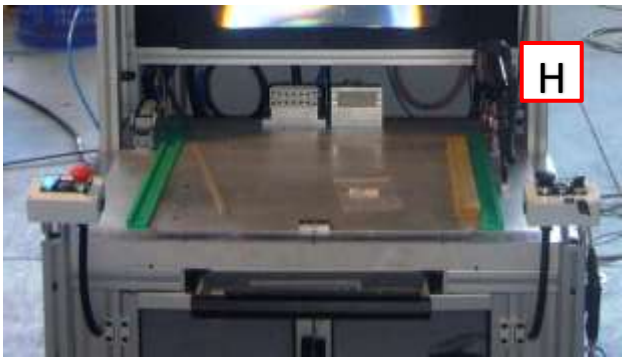
A panel consisting of a panel and control and signaling commands is installed on the front of the machine. The function of this panel is to allow the control and piloting of the machine.

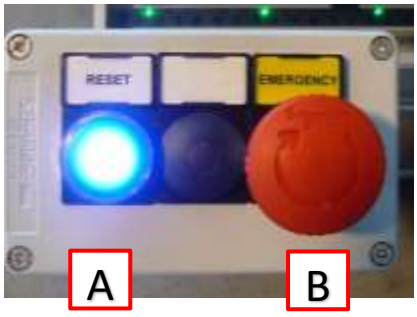


N.	REPORT	COLOR	TYPE	FUNCTION
1	Operator panel		Monitor touch screen	Shows the screens for commanding operations.
2	USB	Black	Socket	Connection for external USB devices
3	Pressure switch			Indicates the pressure in the pneumatic circuit

5.3 PUSHBUTTON

The machine is equipped with a double button panel located in front of the operator.
The composition of the pushbutton panels is as follows:



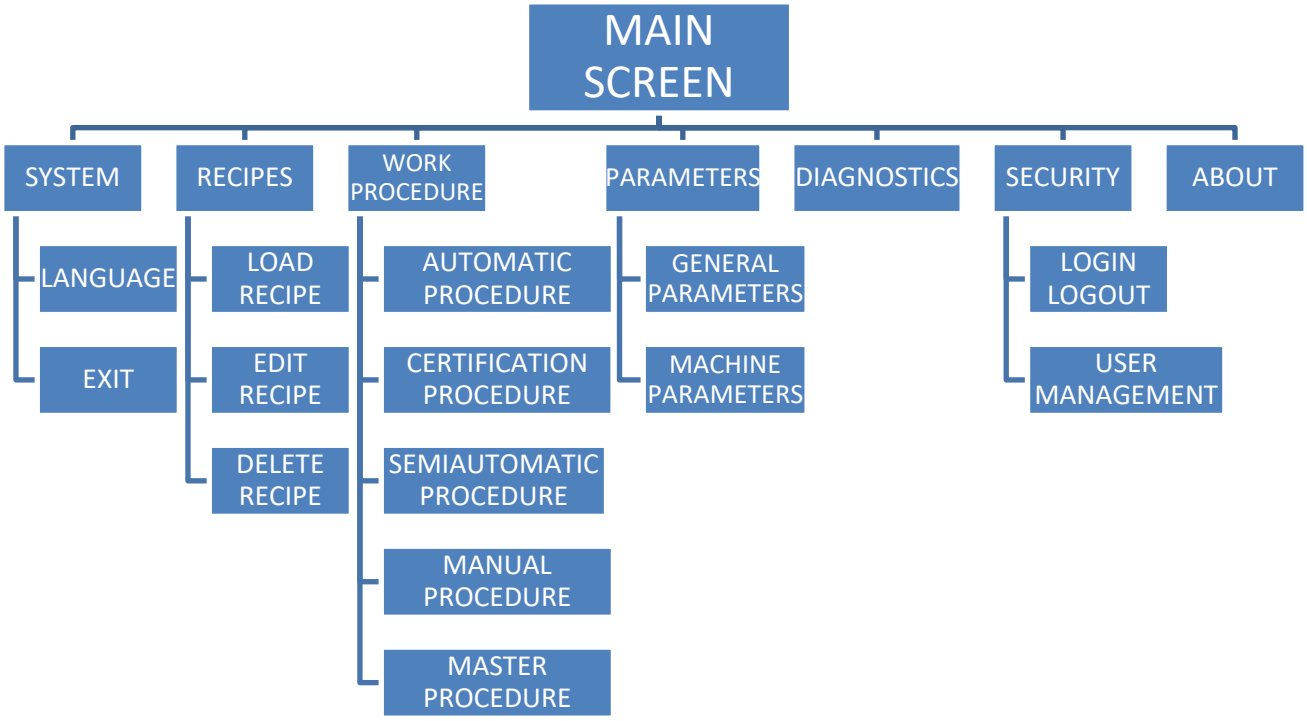




	COMMANDS	COLOR	TYPE	FUNCTION
A	Reset	Blue	Lighted button	Restoring the safety circuits. In Automatic, it repositions the machine itself to rest
B	Emergency	Red	Mushroom button with lock	Emergency button, blocks the machine, to reset rotate the button to unlock it from the emergency position
C	Start	Black	Mushroom button	Puts the car into gear
D	Step by step	Black	Button	Advance one operation at a time in manual mode
E	Aut/Man	Black	Selettore	Select the working method
H	Code reader	Black	Manual gun	Reads the datamatrix code of the piece to be processed.

6 USING THE MACHINE

6.1 PAGE LAYOUT



7 LOADING THE PROGRAM

When the program is started, the following window appears



At this time, the program checks the screens and peripherals required:

- I/O interface
- Acquisition by BurnIn screen (if required)
- Smart Motor addressing (if required)
- Printer

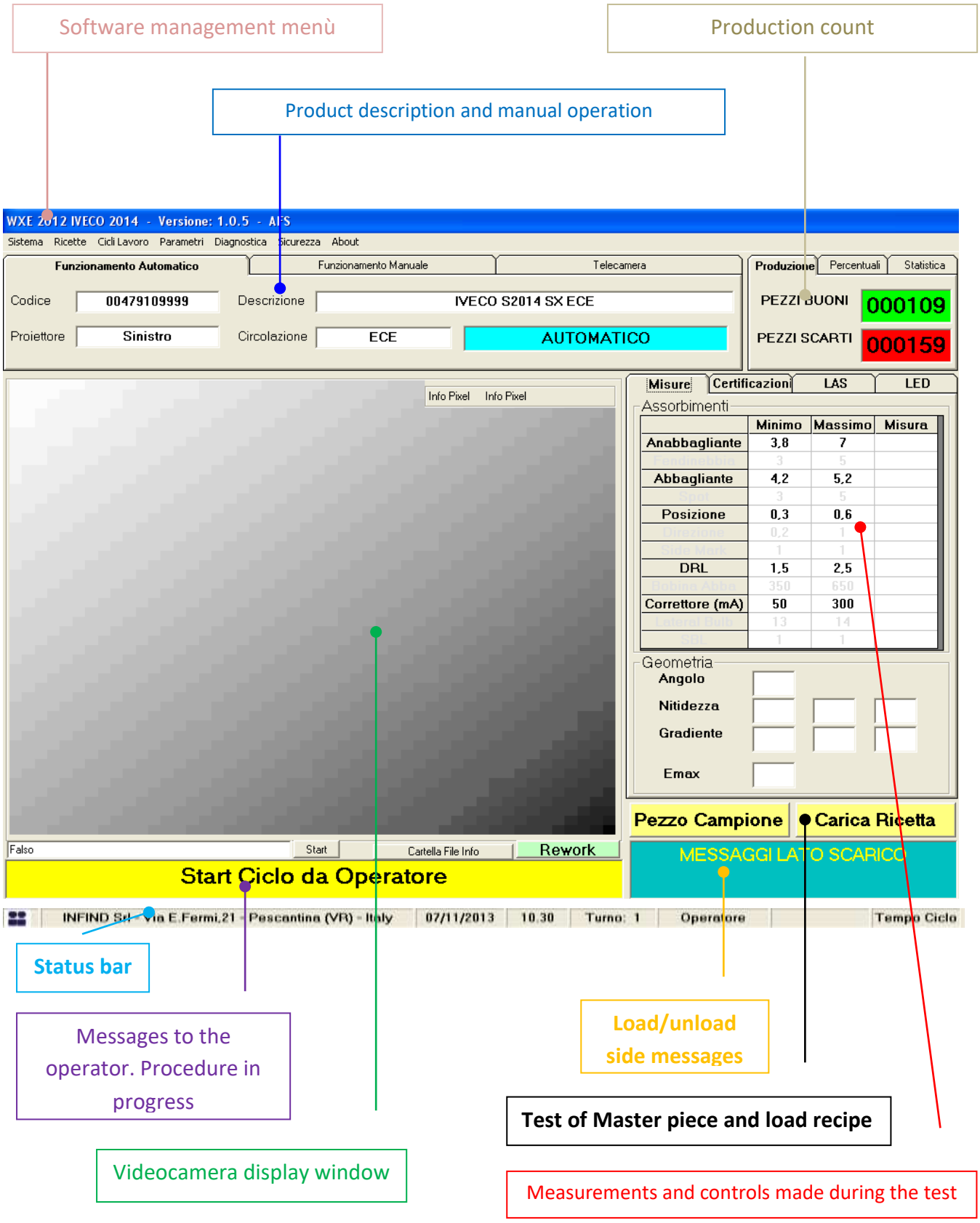
IMPORTANT: when the program is started up and during normal operation, the security access is set at the "operator" level (see safety section), so only certain screens and functions, as described hereafter, are available.

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8 MAIN SCREEN

After performing the entire startup procedure successfully, the main screen appears:



8.1 SOFTWARE MANAGEMENT MENU



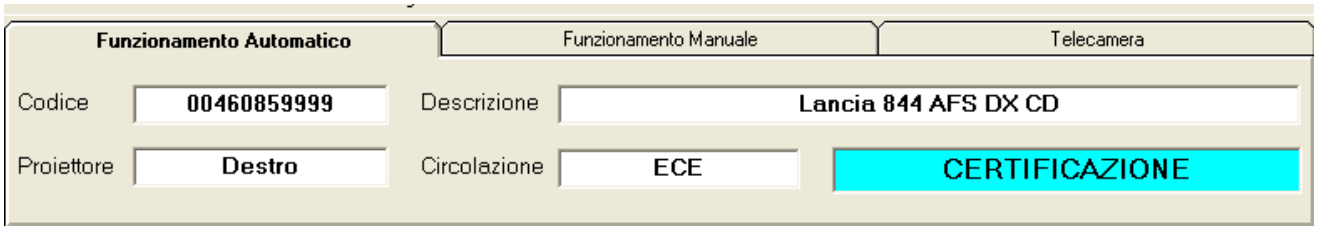
Use the menu to access the software management screens:

- **System:**
  - Selection of software language
  - Exit the program
- **Recipes**:
  - Load the recipe relative to each product
  - Amend the parameters relative to a recipe
  - Delete a recipe
- **Test procedure:**
  - Selection of one of the test procedures
- **Parameters:**
  - Configuration of general parameters
  - Configuration of machine parameters
- **Diagnostics:**
  - Access to diagnostic screens of the system
- **Safety:**
  - Access to safety levels
  - User management
- **About:**
  - Information about the software and system

8.2 DESCRIPTION OF THE PRODUCT AND MANUAL OPERATION

Automatic Function screen

When the program is started, the Automatic Function screen is automatically displayed.



The automatic function screen displays all the information relative to the headlamp being tested.

Code: indicates the product code; this field is the same for every headlamp, therefore there cannot be two products with the same code.

Description: the product description is displayed.

Headlamp: indicates whether the headlamp is Right or Left.

Circulation: indicates the type of circulation (ECE – UK – SAE).

Test procedure: The field with a blue background indicates the procedure you are using for the test. There are 5 different procedures that are fully settable: Automatic – Certification – Semiautomatic – Manual – Master.



IMPORTANT: The automatic test procedure starts only if the Automatic Function screen is enabled.

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Manual Function screen

The Manual Function screen is disabled in the Operator safety level.
By accessing different safety levels from the Operator level it is possible to display this screen.



Hold the Start Procedure button down until the headlamp is blocked and contacted.
At this point it is possible to enable the different headlamp functions by clicking on the relative keys on the screen:

Low beam-Foglight-High beam-Spotlight-Parking light-Direction light– Side Marker-DRL: switching on the different lights provided on the headlamp and viewing the absorption in the measurement window.

Cross Test: performs a test on the shape of the low beam, displaying in the window the measurements of the angle and sharpness of the line of demarcation, if required for in the test.

Emax Test: tests the position of the high beam.

Low Beam Certification: Measures the photometric points of the low beam setting.

High Beam Certification: Measures the photometric points of the high beam setting.

Motor Test: Tests the motor if the headlamp is equipped with an analogue motor.

Print Label: Prints the label relative to the product loaded.

Videocamera screen

The Videocamera screen is disabled in the Operator safety level.
By accessing different safety levels from the Operator level it is possible to display this screen.



The videocamera screen can be used to edit the videocamera settings used in order to perform tests, for example, on headlamps with less luminous beams, with less sharp lines of demarcation and whenever it is necessary to edit the videocamera settings before performing a test.

Live: serves to display the live image.

Stop: freezes the picture displayed by the videocamera.

Exp*:* edits videocamera exposure.

Gain: edits videocamera gain.

Offset: edits videocamera offset.

Reset: resets Exp – Gain and Offset to default values.

Videocamera 1..5: Can be used to select the videocamera desired, which then becomes the default videocamera enabled.

Grid: superimposes a grid on the videocamera image, also setting the block size.

Save picture: after freezing the frame, you can save the picture.

Load picture: after freezing the frame, you can load a picture saved previously.

IMPORTANT: The settings made on this screen are not saved.

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8.3 INFORMATION ABOUT PRODUCTION

This section of the main screen displays information about production performance.
At the “operator” security level only the “production” screen is enabled.



The Production screen displays the partial count of good pieces and rejects, i.e. produced since the last piece counter reset.
To reset these counters just double-click on the Production screen.
The following message appears:



The Percentage screen displays the following data:



Total number of pieces produced with the percentages of good pieces and rejects.
The partial count of pieces produced since the last reset with the percentages of good pieces and rejects.

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8.4 VIDEOCAMERA DISPLAY WINDOW

This section of the Main Screen displays the picture viewed by the videocamera.



If there is more than one camera, for example in case of AFS headlamps, the picture viewed on the active camera is displayed.

By moving the mouse over the image, a box at the top right will display the data relative to the pixel pointed by the mouse.



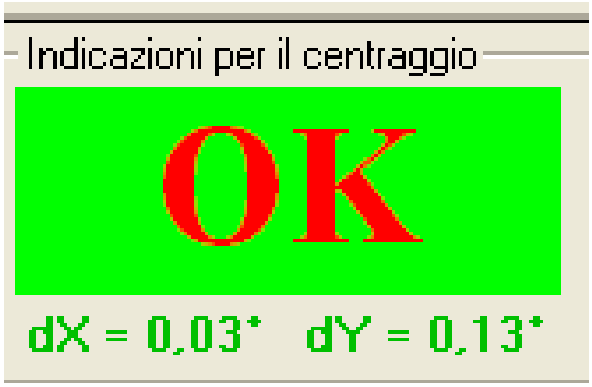
The first two numbers indicate the X – Y coordinates starting from the top left corner, the third number indicates the level of luminosity.
Depending on the type of videocamera installed, the luminosity range goes from 0 to 255 for an 8 bit camera and 0 to 65535 for a 12 bit camera.

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During the test procedure, when performing the "intersection search" step, a window will appear indicating the degree of centering.



When the low beam is correctly focused, OK appears on a green background, and any horizontal (dX) and vertical (dY) deviations with respect to the sample piece are also displayed.

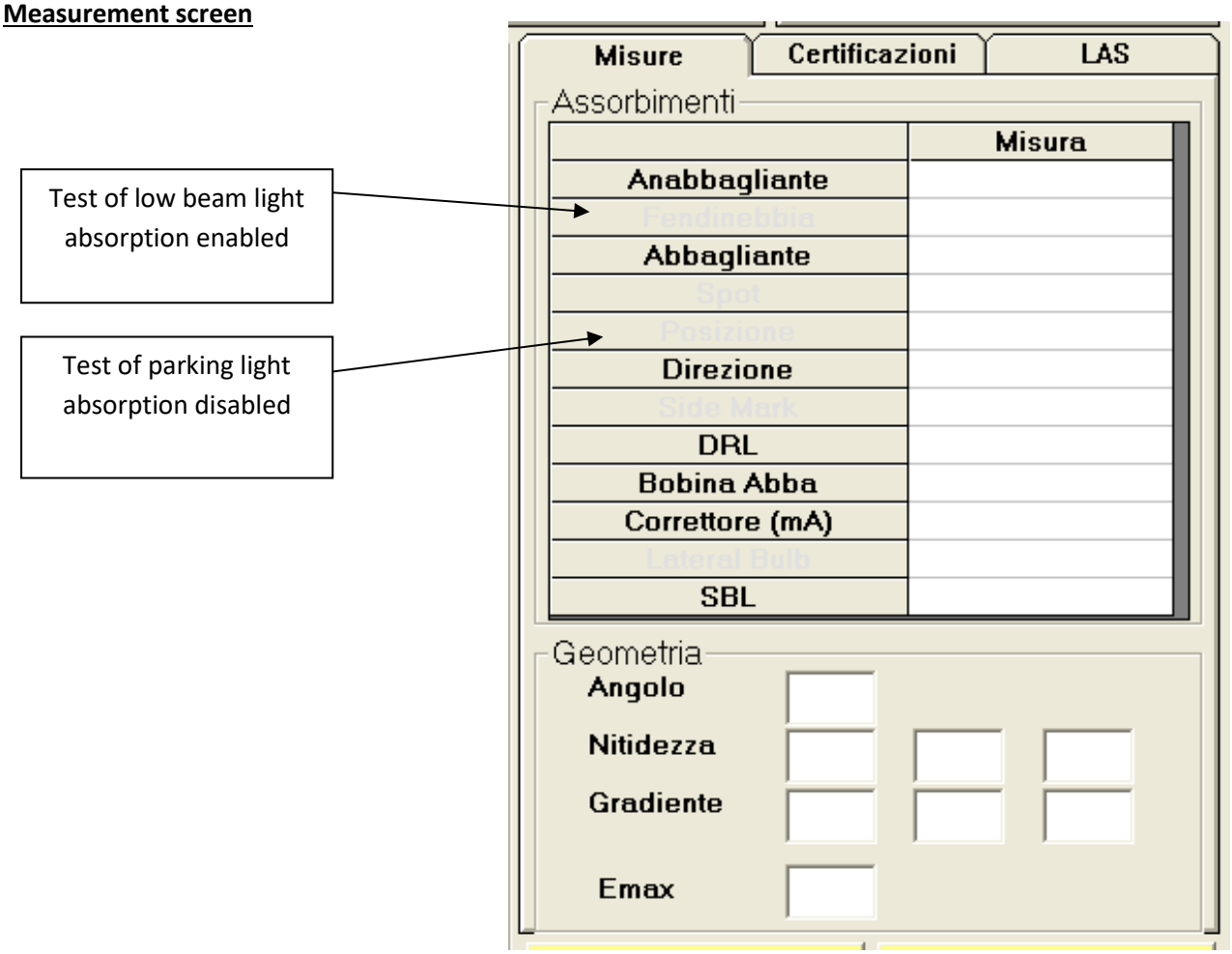


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8.5 MEASUREMENTS AND CONTROLS MADE DURING THE TEST

All the measurements and controls made during the test are displayed in this section of the main screen. Depending on whether the result of the test or measurement is positive or negative, the relative value is displayed in green or in red.

Measurement screen



This screen displays the following data:

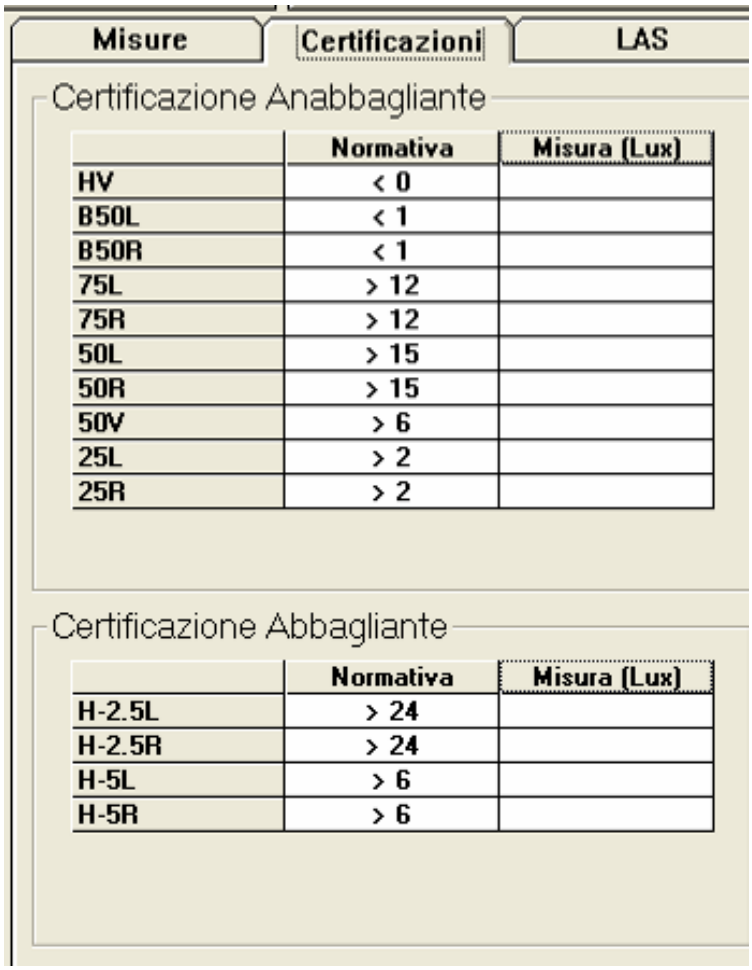
- Data relative to lamp absorption / functions enabled:

Low beam – Foglight – High beam – Spotlight - Parking light – Direction light - Side Mark – DRL – SBL - Analogue Corrector.

- Display of the voltage of the Lateral Bulb is enabled if required.
- The data relative to beam geometry of the low beam, if enabled:
  Angle of the line of demarcation – Sharpness of the line of demarcation – Position of the Gradient of the line of demarcation (Horizontality).
- The data relative to the position of the Emax of the High Beam.

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Certifications Screen



The Certifications screen displays the measurements of the photometric points of the high and low beams enabled in the test procedure set.

The first column of the table indicates the point of the beam, the second column (if enabled) indicates the value of the standard, the third column indicates the measurement obtained.

Example:

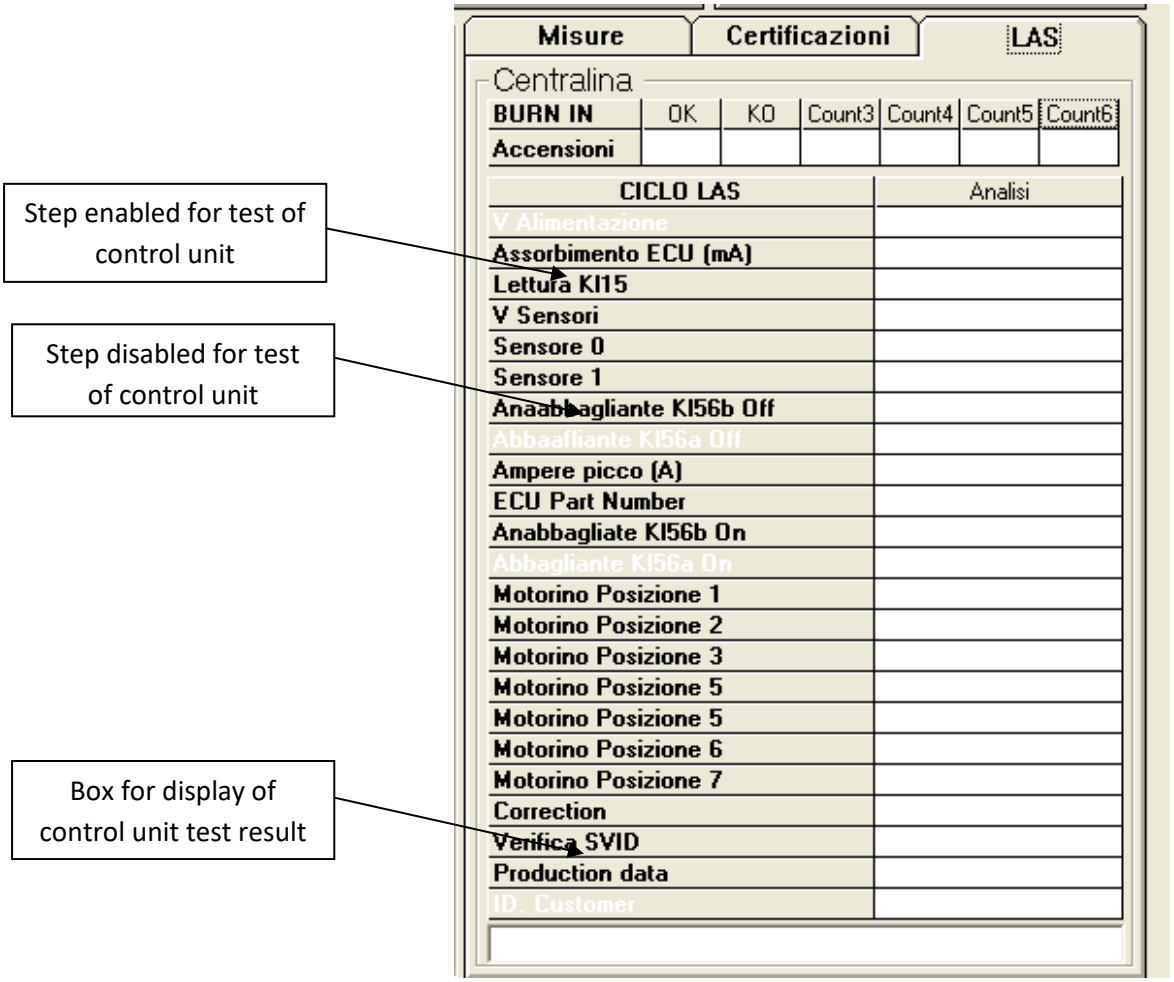


If the measurement in Lux obtained in point B50L of the low beam is lower than 1 the measurement is correct, otherwise it is considered erroneous.

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LAS screen

If the headlamp uses discharge lamps (Xenon) and control units that require testing, the following screen is displayed.



The first table indicates ignitions during BurnIn of the Discharge Lamp (if required)
Ignitions OK – Ignitions not OK – Any internal counters in the control unit.

The second table indicates the checks made on the control unit directly.

During the contact, the box beneath the tables displays the message in response to the various commands given by the control unit: if the response is the expected one, the box shows in green, otherwise it shows in red.

IMPORTANT: For the control breakdown, see the control unit specifications of the product loaded.

8.6 TEST OF MASTER PIECE

It is possible at any time to perform the test procedure of the sample (master) piece by pressing the "sample piece" key.



The master piece test procedure differs from a normal test procedure only because automatic adjustment of beam centering is not made with the screwdrivers. This serves to check machine operation preventing any movement of the headlamp beams from their position as centered in the workshop.
It is possible to force the sample piece test at every change of shift or startup of the program.
(See the section on editing recipes).

8.7 LOADING THE RECIPE

Use the Load Recipe key to load all the parameters for testing a specific headlamp.



The following window appears, where you can select the recipe to load either by entering the serial number of the product or a description of it.



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8.8 MESSAGES FOR THE OPERATOR

The section of the Main Screen relative to Messages for the Operator displays various types of messages, depending on the status of the program:

- Warning messages (yellow background).
- Error or Caution messages (red background).
- Good piece messages (green background).
- Step in progress messages (blue background).





IMPORTANT: In the rotary apron version, some of these messages relative to the piece on the loading and unloading station are displayed in the box shown below:



8.9 STATUS BAR

At the bottom of the main screen you will see the status bar with some general information:



- Client's logo and address.
- Date, time and work shift.
- Current security level and login timer (see security section).

9 HEADLAMP TEST PROCEDURE

After placing the headlamp on the fixture, the operator presses the start key to begin testing.

Rotary Apron version

On the ROTARY APRON version the apron is rotated (either manual or automatic mode) and the test begins. At this point it is possible to load a new piece on the station in front of the operator and set the next test.

When the piece on the inside station finishes the test, the apron is rotated, either manually or automatically, and if a piece has been set for testing on the inside station, the new test begins.

The headlamp test procedure consists of several steps that can be programmed at will, depending on the specific requirements.

Allineamento | **Ciclo Lavoro** | Varie

1	Blocco Pezzo	5	Disabilitato
2	Oggettivazioni Assemblaggio	30	Attesa Start da Operatore
3	Burn In Lampade	1	Disabilitato
4	Assorbimento Lampade	5	Disabilitato
5	Disponibile	5	Disabilitato
6	Disponibile	5	Disabilitato
7	Disponibile	5	Disabilitato
8	Disponibile	5	Disabilitato
9	Disponibile	5	Disabilitato
10	Accensione Anabbagliante	1	Disabilitato
11	Ricerca Incrocio	30	Segnalazione e Blocco del Ciclo
12	Sregolazione	30	Disabilitato
13	Abbattimento Motorino	5	Disabilitato
14	Assorbimento Motorino	5	Disabilitato
15	Certificazione Anabbagliante	5	Disabilitato
16	Accensione Abbagliante	1	Disabilitato
17	Ricerca Emax	5	Disabilitato
18	Certificazione Abbagliante	5	Disabilitato
19	Posizione Fendinebbia	5	Disabilitato
20	Verifica Luci Direzione	5	Disabilitato
21	Oggettivazioni Estetiche	10	Disabilitato
22	Assorbimenti Fine Ciclo	5	Disabilitato
23	Stampa Etichetta	1	Disabilitato
24	Timbro	1	Disabilitato

We shall now describe each step; to program them see the section entitled “editing recipes”.

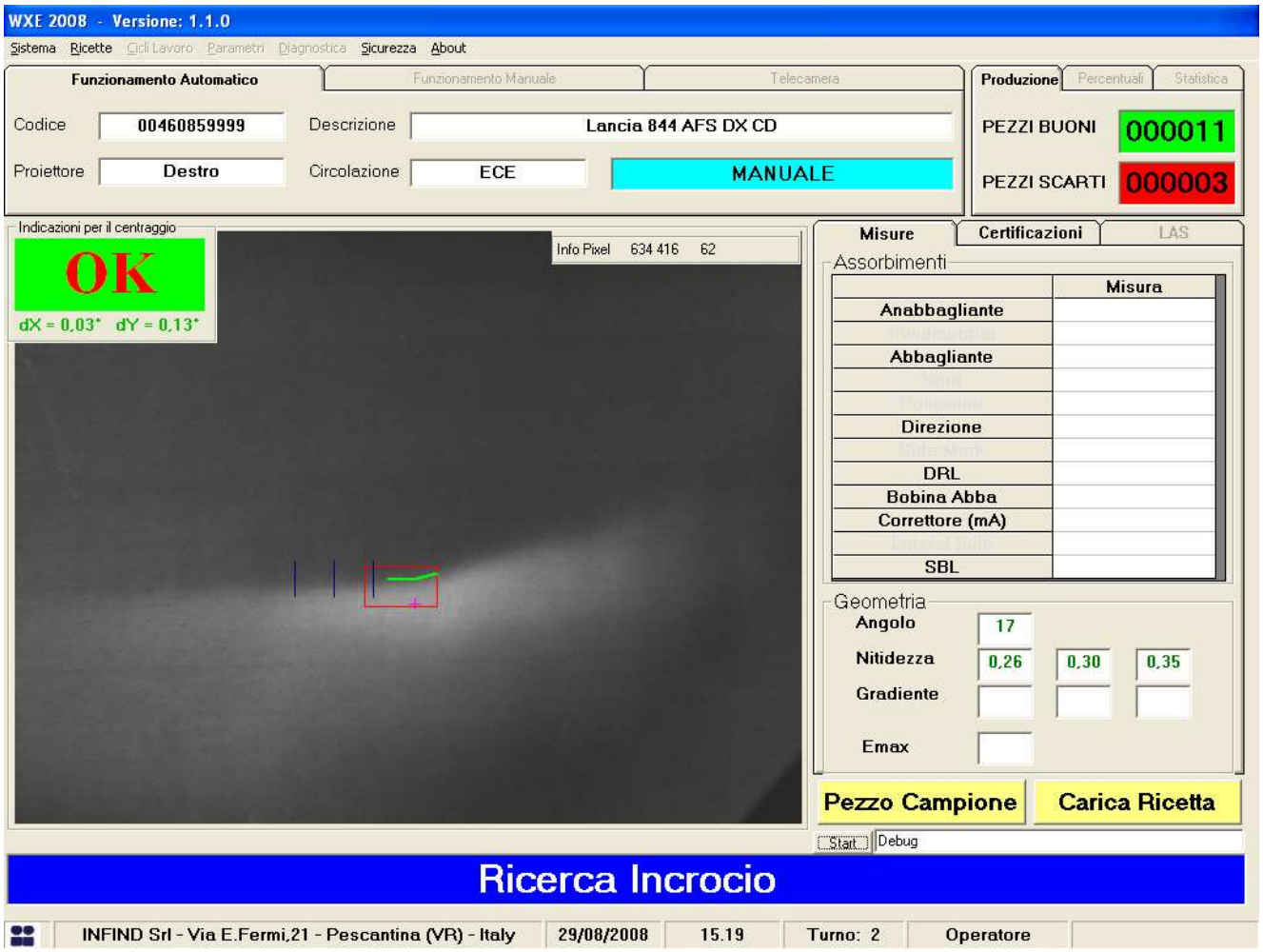
1. **Block piece:** The piece is blocked on the fixture and contacted electrically. If the system of automatic adjustment has been enabled, the screwdrivers are positioned and the tools are inserted in the screws on the headlamp.
   On the rotary apron version these operations take place after the apron has been rotated.
2. **Assembly recognition:** Any recognitions requested are made, such as the presence of vent caps or the absence of a specific control unit.
3. **Lamp BurnIn:** The lamps / functions are switched on and off. This step serves to stress the lamps / functions so as to reveal problems of early deterioration of the bulbs or LED.
4. **Lamp absorption:** The absorption of each lamp / function is measured.
5. **Preadjustment:** Makes a timed preadjustment of the beam.
6. **Not used:** This step is available for future use.
7. **Not used:** This step is available for future use.
8. **Not used:** This step is available for future use.
9. **Not used:** This step is available for future use.
10. **Low Beam Ignition:** The low beam is switched on. In case of a discharge lamp the control unit is initialized.
11. **Intersection Search:** Searches for the intersection point of the horizontal line of the 15° cut (V point). If the automatic adjustment system is enabled, the intersection is moved by the screwdrivers so as to fit within the tolerance window setting. If the system does not include screwdrivers this operation is performed manually by the operator.
    During this step the following checks can be set:
    a. Angle of the line of demarcation.
    b. Sharpness of the line of demarcation.
    c. Horizontality of the gradient of the line of demarcation.
    d. Position of the maximum low beam.

    In addition, the congruence of the geometry of the line of demarcation with the circulation setting (ECE, UK, SAE) is always checked.
12. **Maladjustment:** If required, the automatic adjustment system will make an automatic maladjustment of the position of the low beam. This serves to check correct movement of the parabola by means of the screws on the headlamp**.**
13. **Corrector Dip:** Tests the movement of the low beam position corrector.
    Depending on the type of corrector, different situations are possible:
    a. **Analogue Corrector:** the corrector lowers the beam by means of a voltage adjustment obtained with a potentiometer inside the control panel.
    b. **Bilitronic Corrector:** the corrector makes a downward adjustment and an upward adjustment of the low beam.
    c. **AFS Corrector:** the corrector moves the beam upward/downward and left/right according to a "hysteresis procedure".
14. **Motor absorption:** Measures the absorption of the motor during adjustment of the corrector.
15. **Low Beam Certification:** the Lux are measured for the low beam points enabled.
16. **Low Beam Ignition:** The low beam is switched on.
17. **Emax search:** checks the maximum high beam position. During this step you can enable a check of the spotlight function position.
18. **High Beam Certification:** the Lux are measured for the high beam points enabled.
19. **Foglight position:** checks the position of the foglight beam.
20. **Test of direction lights:** measures the intensity of the parking light.
21. **Esthetic recognition:** not implemented.
22. **End of procedure absorption:** other measurements of absorption are made.

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23. **Print Label:** if the test is positive, the good piece label is printed, otherwise the label with the reject motivation identified during the procedure is printed.
24. **Stamp:** if the piece is good the stamp that marks the piece is enabled.

During the test, the areas of the screen devoted to messages for the operator and display of the measurements are updated to keep pace with the test step.



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If the test is successfully completed the program displays: “Piece good – Remove piece”.



If not, the error detected is displayed.



In both cases, the value of the measurements made is displayed until the piece is removed.





If the test is correct, the headlamp is disconnected automatically and the operator can remove it from the fixture; if not, the piece remains locked on the fixture.
Depending on the requirements of the client, the operator has to use the appropriate command to disconnect it: the Start Procedure key, the Step-Step key or the Unlock key. It will be possible to remove the headlamp only after giving the command.

We shall now describe the different items on the menu in the order in which they are displayed on the main screen.
For each, the access level is specified.

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10 SYSTEM



Using the menu item “system” you can:

- Change program language.
- Exit the program.

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10.1 SELECT LANGUAGE

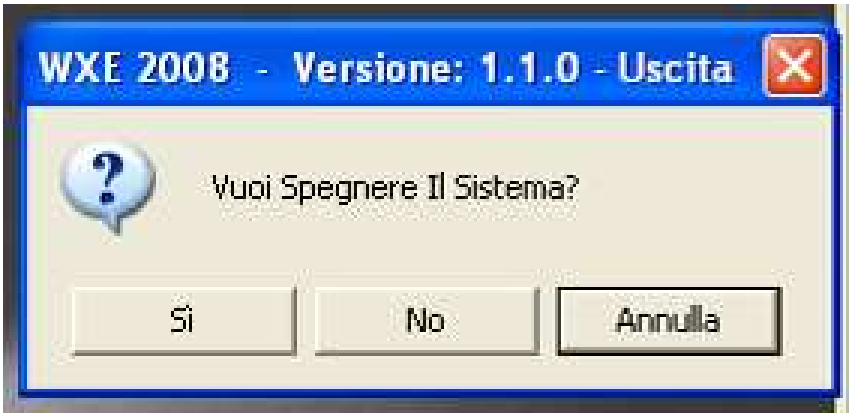
Access level: Operator.



Select the language desired in the window shown above and press “OK”.
Exit and start the program again to make the language setting effective.

10.2 EXIT THE PROGRAM

Access level: Operator.
At the request to exit the program you will be asked if you want to shut the system down.



If you press “Yes” : you exit the program and the Windows shutdown procedure begins.
If you press “No” : you exit the program and the system returns to the main screen of Windows.
If you press “Cancel” : you stay in the program WXE2012.

11 RECIPES



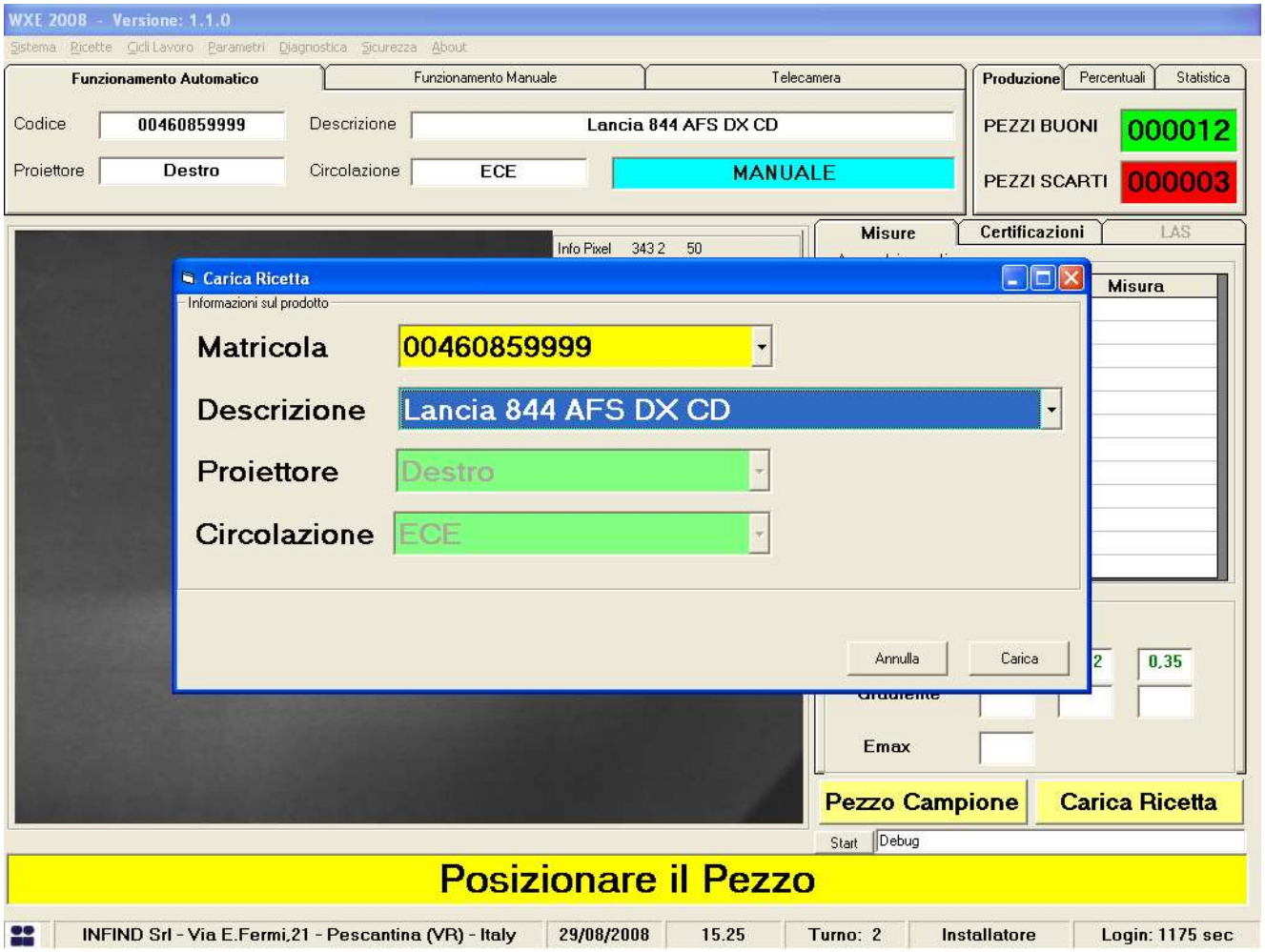
By "recipe" is meant the set of testing parameters for a headlamp.
You will therefore have a single recipe for every type of headlamp, identified by the same code.

Using the menu item “Recipes” you can:

- Load the desired recipe.
- Edit the parameters of the recipe loaded.
- Delete a recipe.

11.1 LOAD RECIPE

Access level: Operator.



Use the Load Recipe key to load all the parameters for testing a specific headlamp.

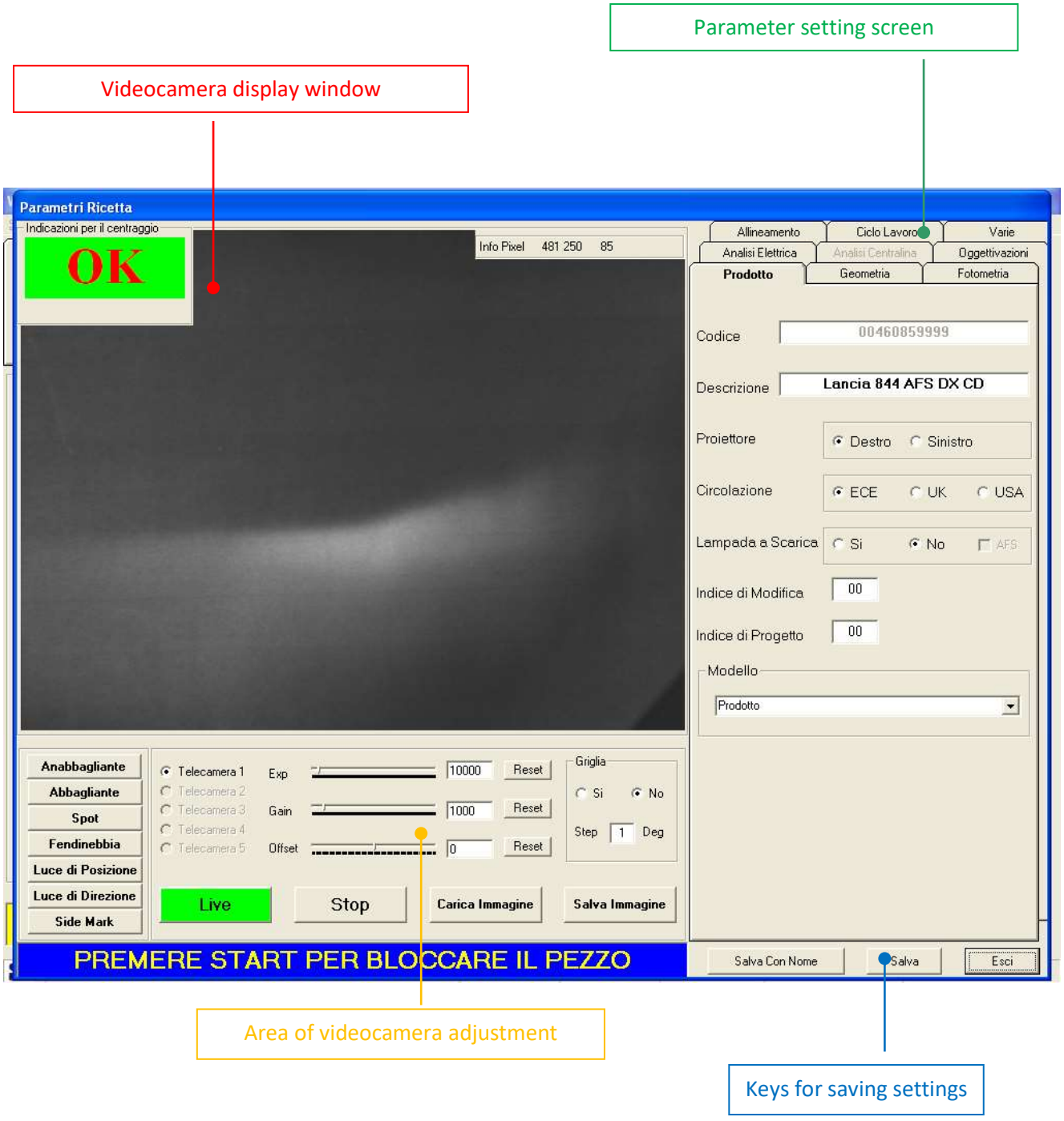
A window appears, where you can select the recipe to load either by entering the serial number of the product or a description of it.

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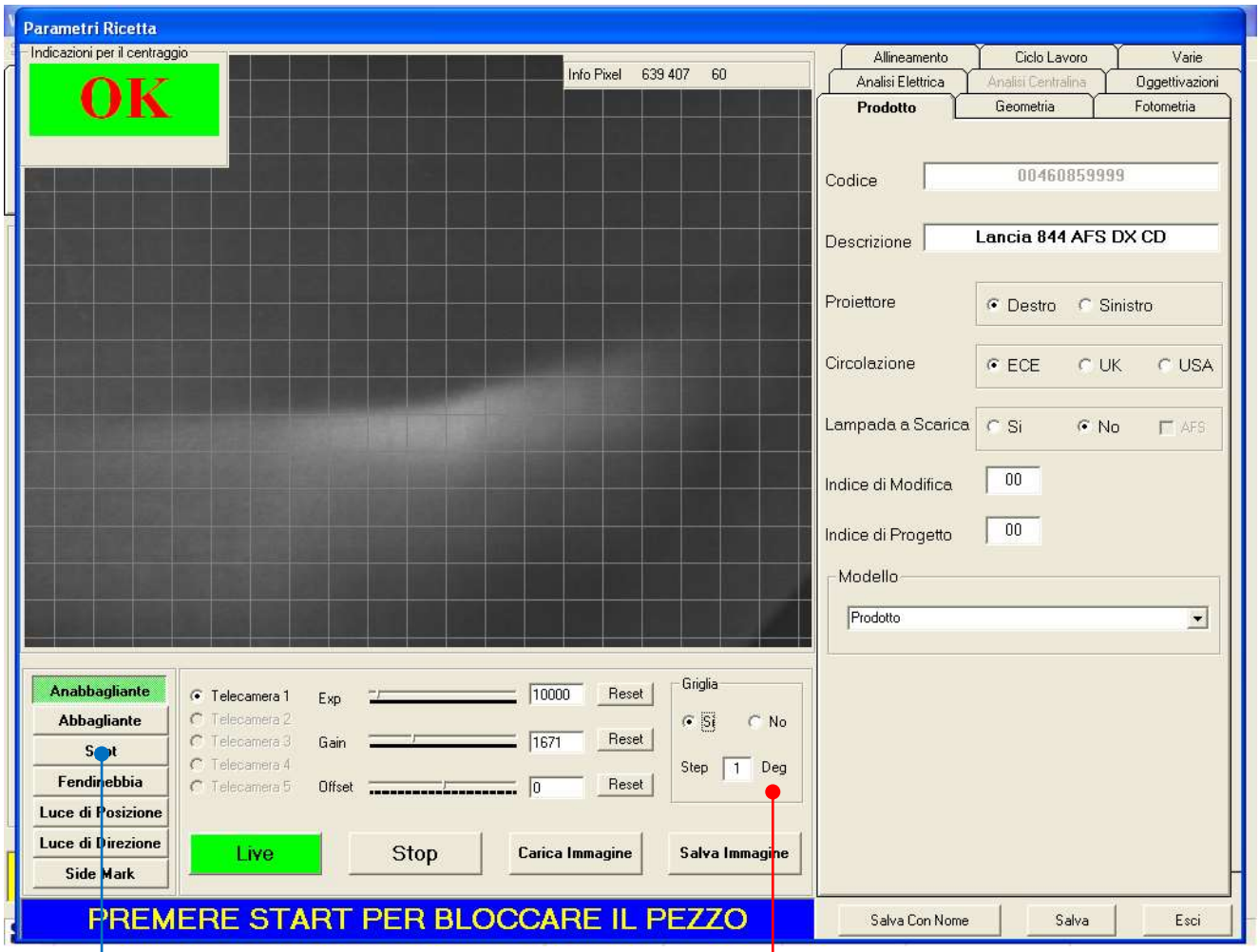
11.2 EDIT RECIPE

Access level: Supervisor.
Setting the proper parameters of the recipe is fundamental to ensure correct operation of the whole system.

We shall now provide some general indications about this screen.



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Grid definition area

Keys for switching on lamps

11.2.1 SAVING DATA

After editing a parameter you have to save it for it to take effect.
When you press “Save” the following message appears:



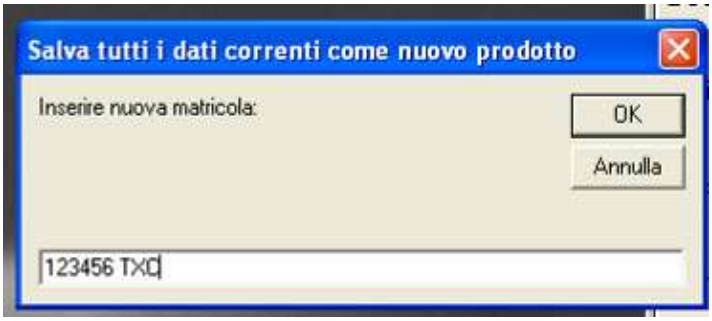
Press “Yes” to make the changes effective, otherwise press “No”.
If after editing a parameter you want to exit the screen without saving the data, the following message is displayed:



Press “yes” to exit the screen without saving the changes and “No” to return to the editing screen.

11.2.2 COPY A RECIPE

You can make a copy of a recipe and assign it another product code.
When you press “Save as...” the following message appears:



Enter the new code and press “OK”.
Now all the files are copied and you are asked if you want to copy the label file as well.



Enter the new description for the product you have just created and press “OK”.



IMPORTANT: the recipe you have just created will not be the active recipe so, in order to change the data of the new product, you have to exit the edit recipe screen, load the new recipe and edit the data if necessary.

11.2.3 GENERAL PROCEDURE FOR SETTING THE RECIPE



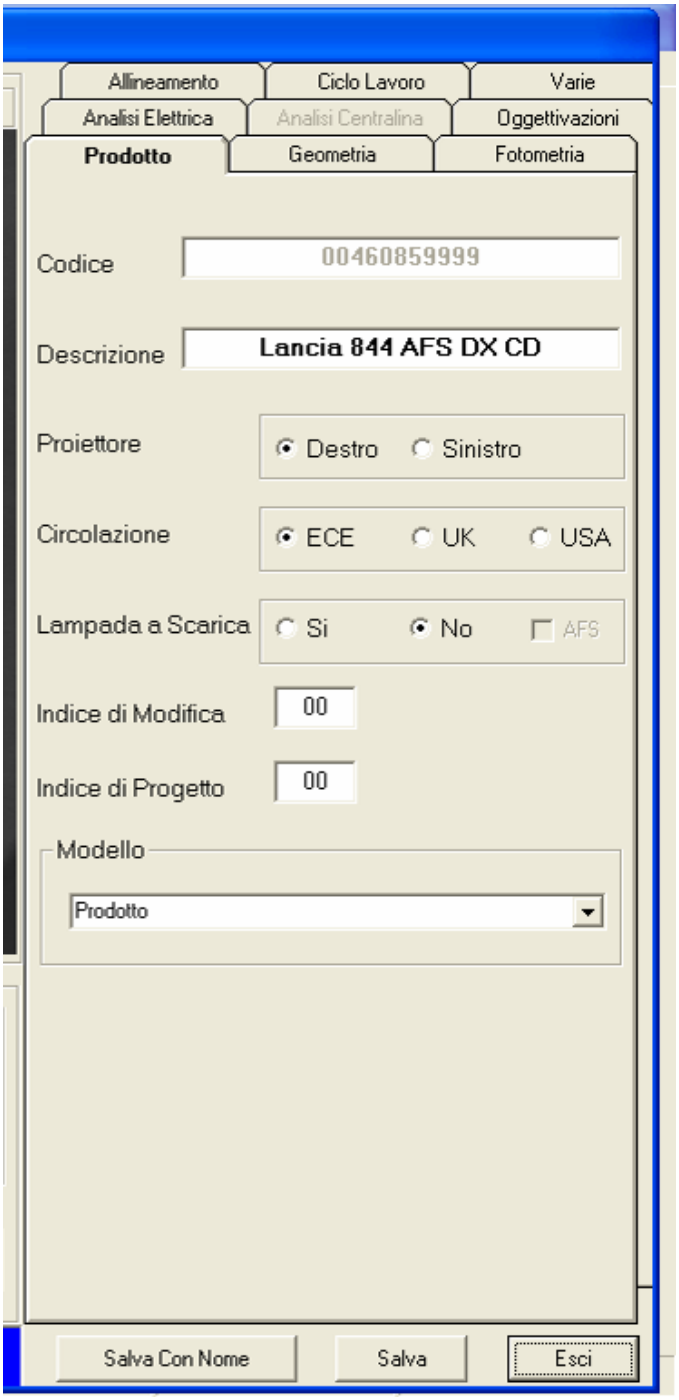
IMPORTANT: to set the recipe parameters correctly you must have one piece for each type of headlamp correctly oriented and, if necessary, with the measurement of the photometric points of interest.
This piece will be your sample piece, or master, and should be kept for checking the system.

Hereafter we shall describe the procedure for setting up the recipe.
For the details see the paragraphs that follow.

1. Load the sample piece on the fixture.
2. Press and hold the start button until the piece is fully locked in place and the electric connectors are inserted.
3. Enter the data for the product to be tested.
4. Adjust the picture on the videocamera.
5. Enter the data of the low beam geometry.
6. Calibrate the reading of the photometric points of the low beam.
7. Enter the data of the high beam geometry.
8. Calibrate the reading of the photometric points of the high beam.
9. Enter the data of the foglight geometry.
10. Enter the data relative to the BurnIn and absorption of the lamps.
11. Enter the parameters relative to analysis of the control unit.
12. Enable the required recognitions.
13. Enter the data of corrector movement.
14. Enter the beam alignment data.
15. Enter the steps of the work procedure.

11.2.4 PRODUCT DATA

The “Product” screen contains the data relative to the Headlamp description:



Code: this is the code (serial number) of the headlamp.
This code is not editable.

Description: contains the headlamp description.

Headlamp: indicates the position of the headlamp on the car.

Circulation: indicates whether the headlamp beam is oriented for driving on the right (ECE), driving on the left (UK) or driving in the U.S.(SAE).

Discharge lamp: indicates whether the headlamp has a discharge lamp (Xenon), and in this case whether it is an AFS or halogen version.

Version index: index relative to the product version.

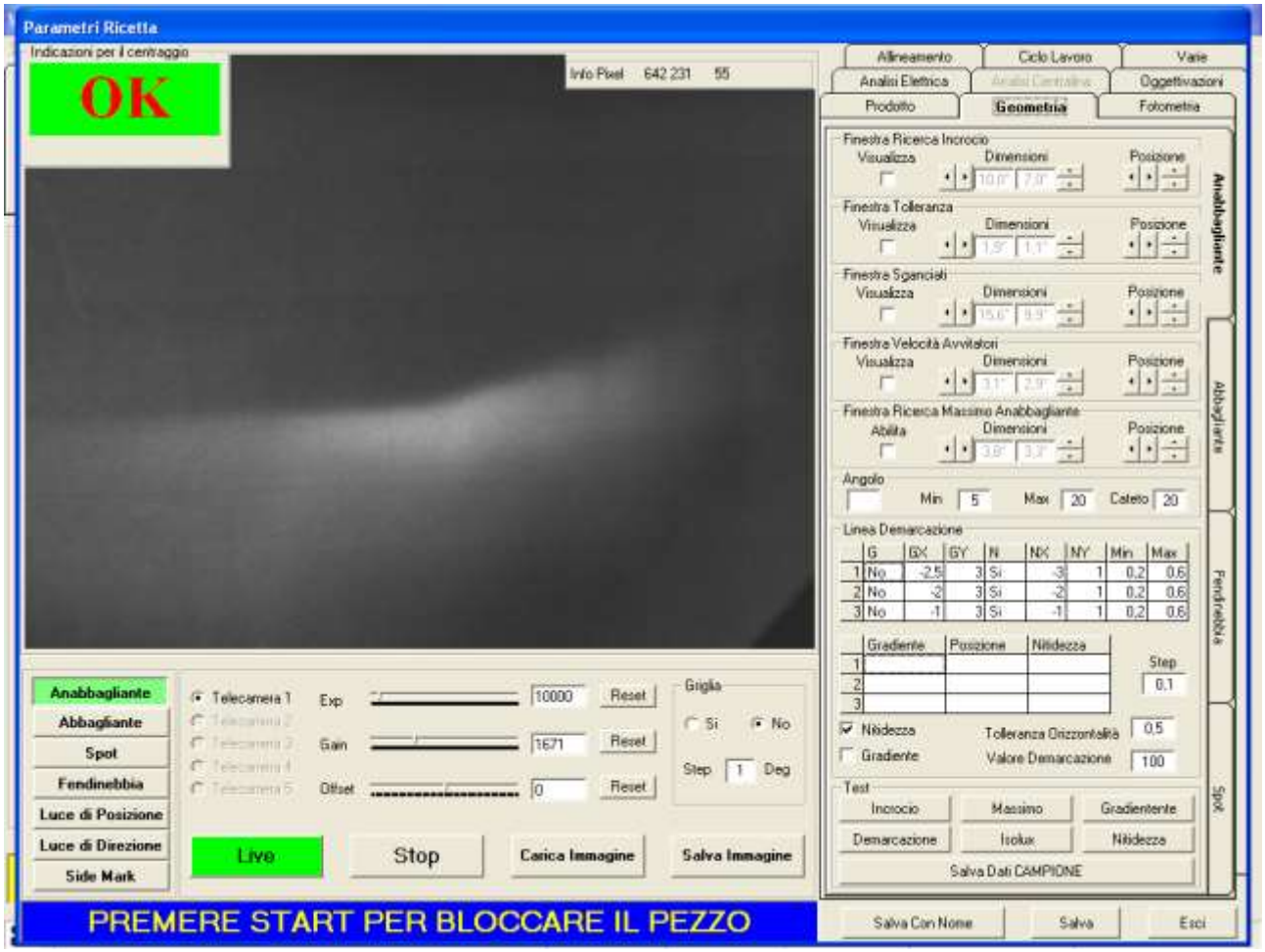
Project index: index relative to the product project.

Model: indicates to which model of the headlamp the recipe refers.
IMPORTANT: this parameter defines operation of the test unit in relation to the specific model. Erroneous specification of the model could prevent the system from functioning correctly.

The Code, Product description, Right or Left Headlamp and Circulation are displayed on the main screen.

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11.2.5 LOW BEAM GEOMETRY



After switching on the low beam, adjust the Exp (Exposition) and Gain values of the videocamera so as to have the best picture possible.
Pay particular attention to these two extremes:

- Videocamera saturation.
- Excessively faint picture.

Videocamera saturation occurs when the lens is open too far or the exposure is too high. This situation is characterized by parts of the picture appearing completely white (saturated).
When you move the cursor over those zones, the luminosity value displayed in the Pixel Info window is:

- 255 for 8 bit videocameras.
- 65535 for 16 bit videocameras.

The picture appears too faint when the lens is closed too much or the exposure is too low. This situation is characterized by the fact that the videocamera picture shows the headlamp beam incomplete, you see only part of it.
Adjust the exposure and gain so as to obtain the best compromise between these extremes.
At this point you can proceed to adjust the parameters of low beam geometry.

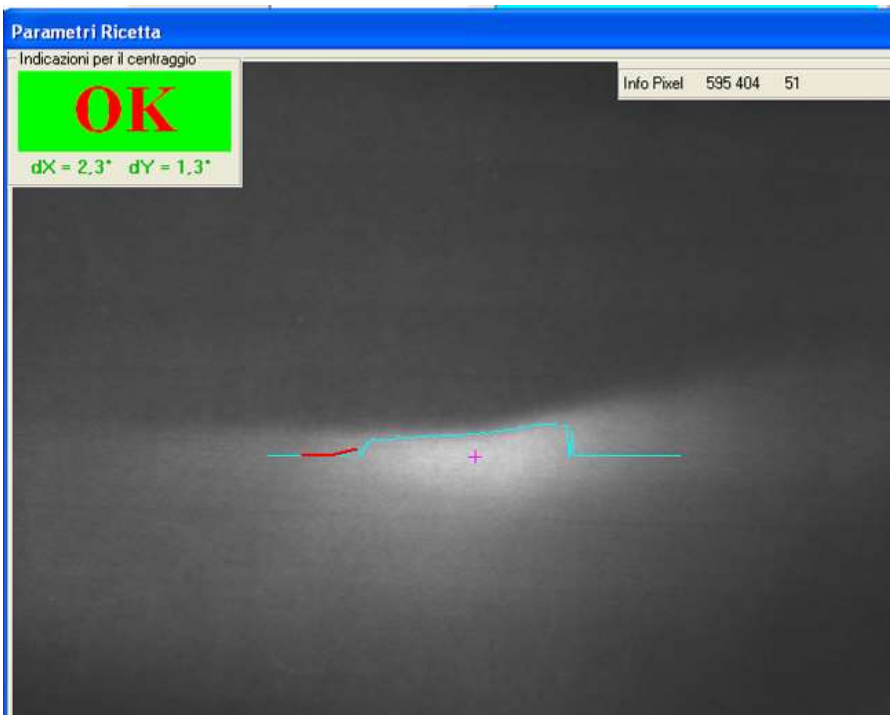
Press “Maximum” and “Intersection”; this will tell the system to search for the intersection of the demarcation line. The maximum low beam point and virtual intersection found are displayed on the screen.



The goal of the following settings is to enable the system to find the intersection (V point) of the low beam correctly.

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Press “Demarcation” to display the demarcation line found by the system.



In this case the virtual demarcation line does not correspond to the real line and the V point is not found correctly.

Correct the “Demarcation Value” field so that it overlaps the virtual demarcation line with the real beam demarcation.



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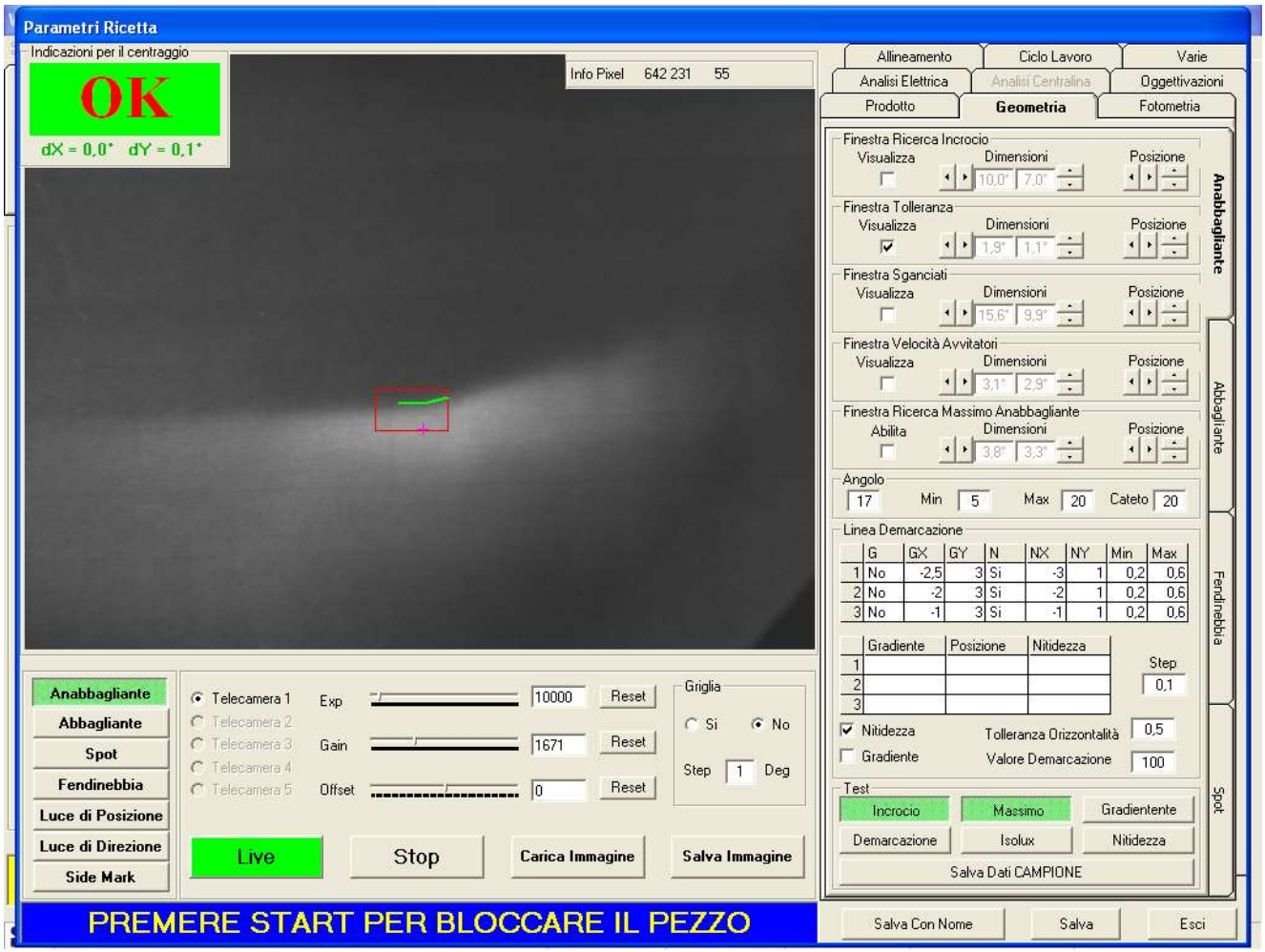


If the virtual demarcation line coincides with the real one, the system will find the intersection at the V point of the low beam.

When these conditions are correct, save the data.

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Enable display of the “Tolerance Window”.



This window indicates the tolerance for centering of the low beam. During the test procedure, the step “Find intersection” serves to adjust alignment of the low beam until the intersection of the beam is inside this window.
Adjust the position and size of the Tolerance Window and save the data to make the changes effective.
It is possible to make low beam alignment even more accurate than the sample with which the recipe was made. To do this, first press “Save SAMPLE data”. You will see the following window:



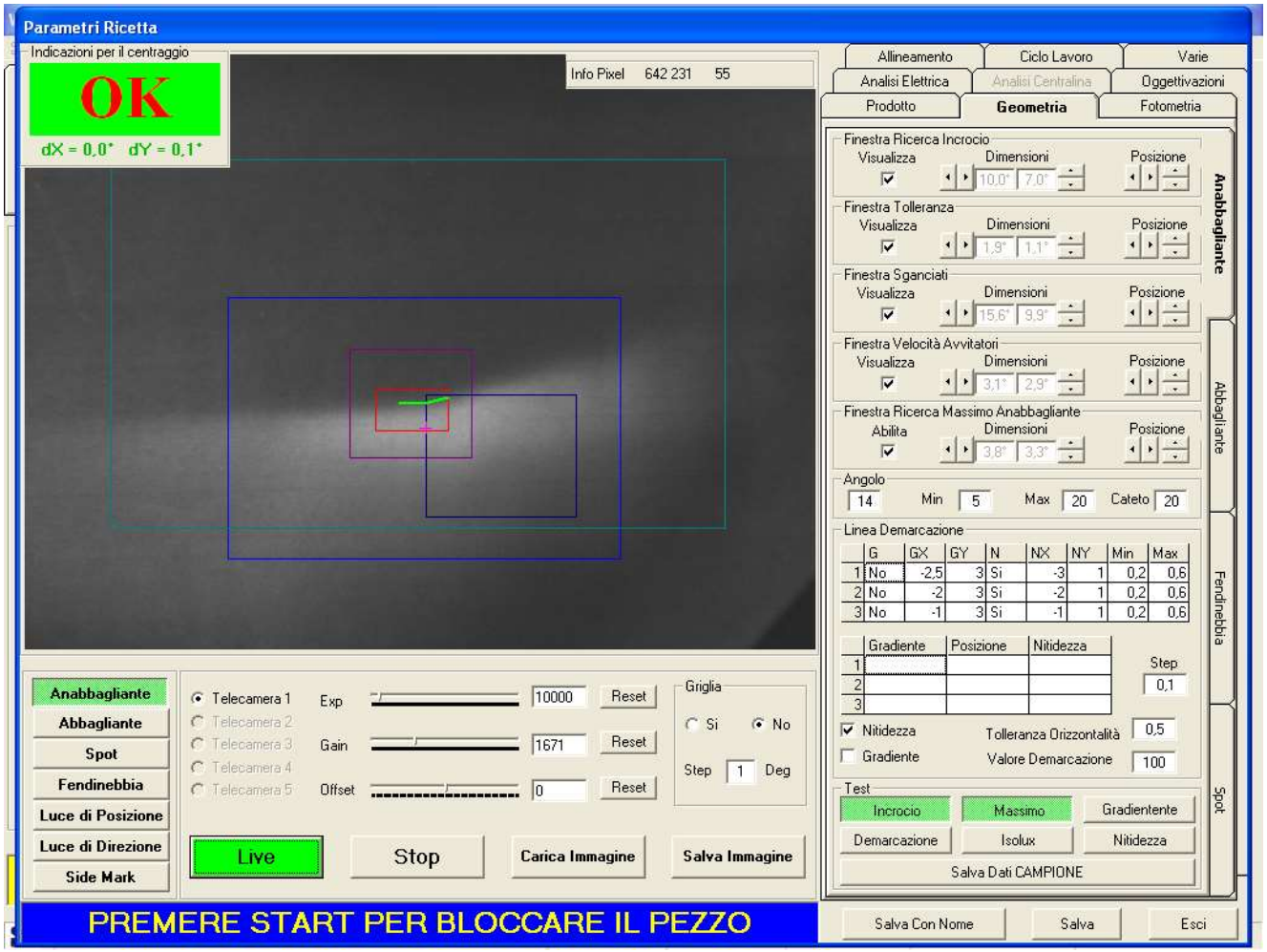
In this way, you save the exact position of the beam intersection displayed.
If you enable centering with the sample intersection (see alignment screen) the system will take the alignment of the headlamp you are testing not only within the tolerance window but toward the intersection point of the sample you saved in this way.

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IMPORTANT: Alignment with the sample intersection is slower than the search in the window and may be difficult if the headlamp beam is not sufficiently clear.

The window “Centering information” indicates the dX and dY deviations between the current V point and that of the sample.

We shall now describe the other functions shown on this "Low Beam Geometry" screen.



Intersection Search Window: Defines the zone of the search for the low beam intersection. This window is positioned around the low beam maximum. If the maximum of the low beam is in the correct position near the intersection, a smaller window can be defined; if not, if the maximum of the low beam is farther away from the intersection a larger window will have to be defined.

Tolerance Window: As we have already described, this indicates the window within which the alignment of the low beam can be considered correct.

Parabola Unfastened Window: If the intersection is outside this window, the system signals the possibility that the parabola of the headlamp is unfastened. Note: the system only checks the sides of this window inside the Intersection Search Window.

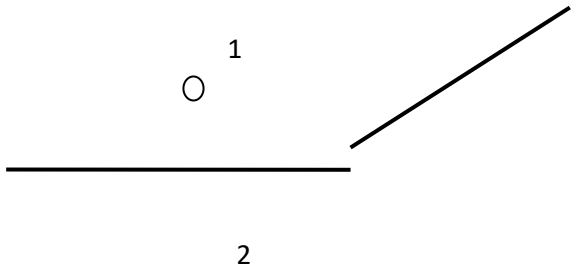
Screwdriver Speed Window: If required, automatic regulation of beam alignment is made with different speeds depending on where the intersection is with respect to this window: as long as the intersection is outside this window, adjustment is made as higher speed and as soon as the intersection enters this window the speed decreases for finer adjustment. See the beam alignment screen for speed specifications.

Low Beam Maximum Search Window: If enabled, the system checks the Low Beam Maximum position. This check is made after beam alignment: if the Low Beam Maximum is outside this window the system signals the position error.

Angle: During the Intersection Search step, the angle of the Demarcation Line is checked. This screen can be used to define the range of tolerance of this measurement and define the cathetus with which this angle is evaluated. This check is made after beam alignment.

Line of Demarcation: The system allows you to evaluate the sharpness and gradient of the line of demarcation.

Sharpness: Sharpness is a parameter that indicates the clarity of the low beam in terms of shading of the line of demarcation. Given the luminosity of the two points located one above and one below the line of demarcation, sharpness is calculated with the following formula:



$$N = (E1-E2) / (E1+E2)$$

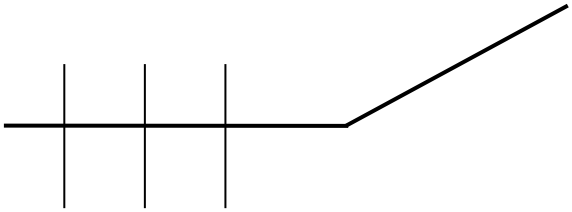
This value ranges between 0 and 1:

- The sharper the line of demarcation, the closer this value is to 1
- The more shaded the line of demarcation, the closer this value is to 0

It is possible to define 3 positions along the line of demarcation for making this calculation.

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Gradient: Calculating the Gradient will check the horizontality of the line of Demarcation.
Given two or three points along the horizontal section of the line of demarcation, the luminosity is determined along the segments displayed with a specific step. Along each segment the maximum sharpness is calculated for the scanning points, that is, the luminosity gradient.
The position of the gradient of all the segments is checked so as to determine the horizontality of the line of demarcation.



The table below can be used to define the parameters for sharpness and gradient.

Linea Demarcazione

	G	GX	GY	N	NX	NY	Min	Max
1	SI	-2,5	3	Si	-3	1	0,2	0,6
2	SI	-2	3	Si	-2	1	0,2	0,6
3	SI	-1	3	Si	-1	1	0,2	0,6

G: Enabling of points 1 – 2 – 3 to check the gradient.
GX: Position of X in degrees with respect to the intersection of the points for checking the gradient.
GY: Amplitude in degrees of the segment for calculation of the gradient.

N: Enabling of points 1 – 2 – 3 to check the sharpness.
NX: Position of X in degrees with respect to the intersection of the points for checking the sharpness.
NY: Distance in degrees of the points for calculation of the sharpness.
Min: Minimum sharpness admissible.
Max: Maximum sharpness admissible.

	Gradiente	Posizione	Nitidezza
1	0,07	-0,20°	0,27
2	0,06	-0,10°	0,30
3	0,06	-0,10°	0,33

Step 0,1

☑ Nitidezza — Tolleranza Orizzontalità 0,5

☐ Gradiente — Valore Demarcazione 100

Step: scan step of the segment for calculation of the gradient.

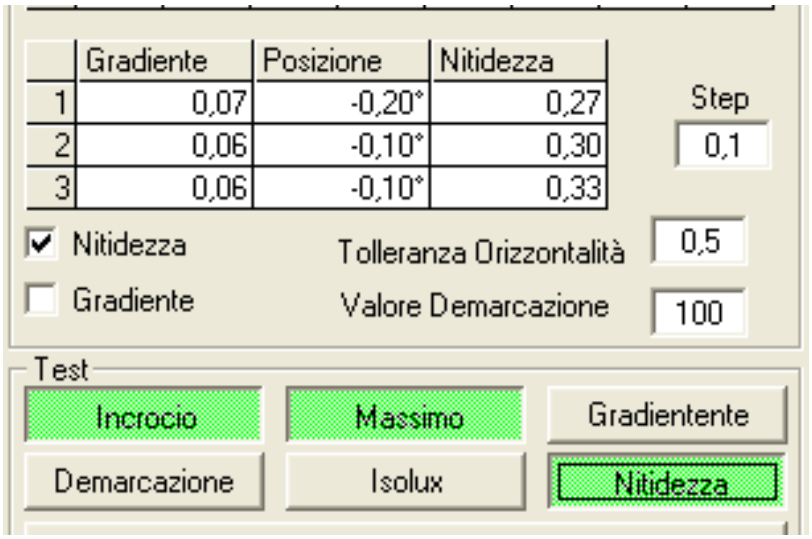
Tolerance on horizontality: deviation admissible for the position of the gradients (in degrees).

Sharpness: enabling of the sharpness check.

Gradient: enabling of the horizontality check.

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It is possible to enable both the sharpness and the gradient test.



During the tests in the table alongside the various gradient value points are displayed, indicating their position with respect to the intersection and the sharpness value.

The following pages illustrate the full screenshots during these tests.

SHARPNESS TEST

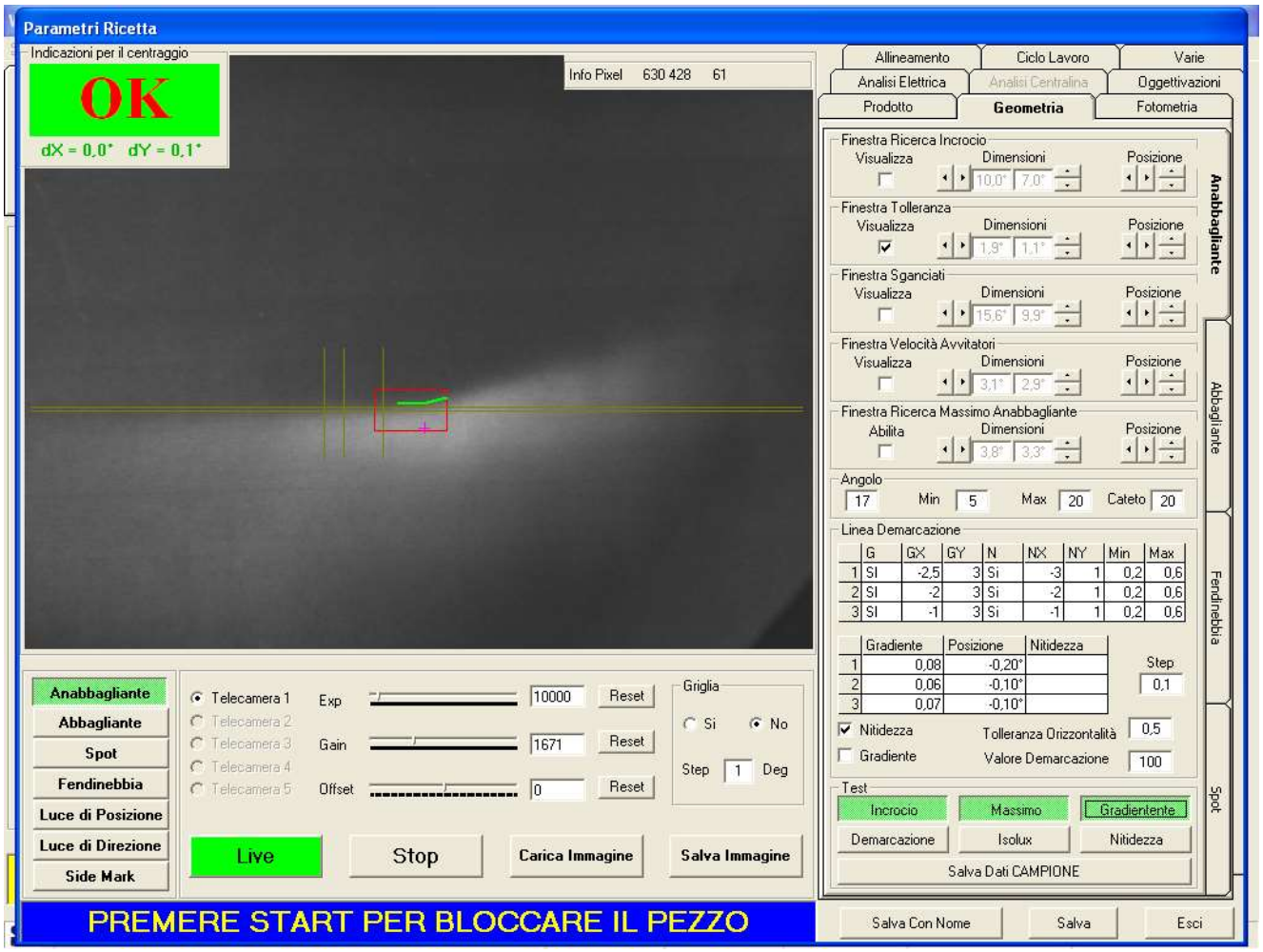
The window displaying the picture on the videocamera shows the points along the line of demarcation used to calculate sharpness. The extremities of the vertical segments are the points used for calculation.



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TEST OF HORIZONTALITY

The window displaying the picture on the videocamera shows the points along the line of demarcation used to calculate gradient. The vertical segments indicate the scan made to calculate the gradient. The vertical lines indicate the gradient positions found: if these lines are close together it means the line of demarcation is horizontal, otherwise, the farther apart they are the more the line is oblique.

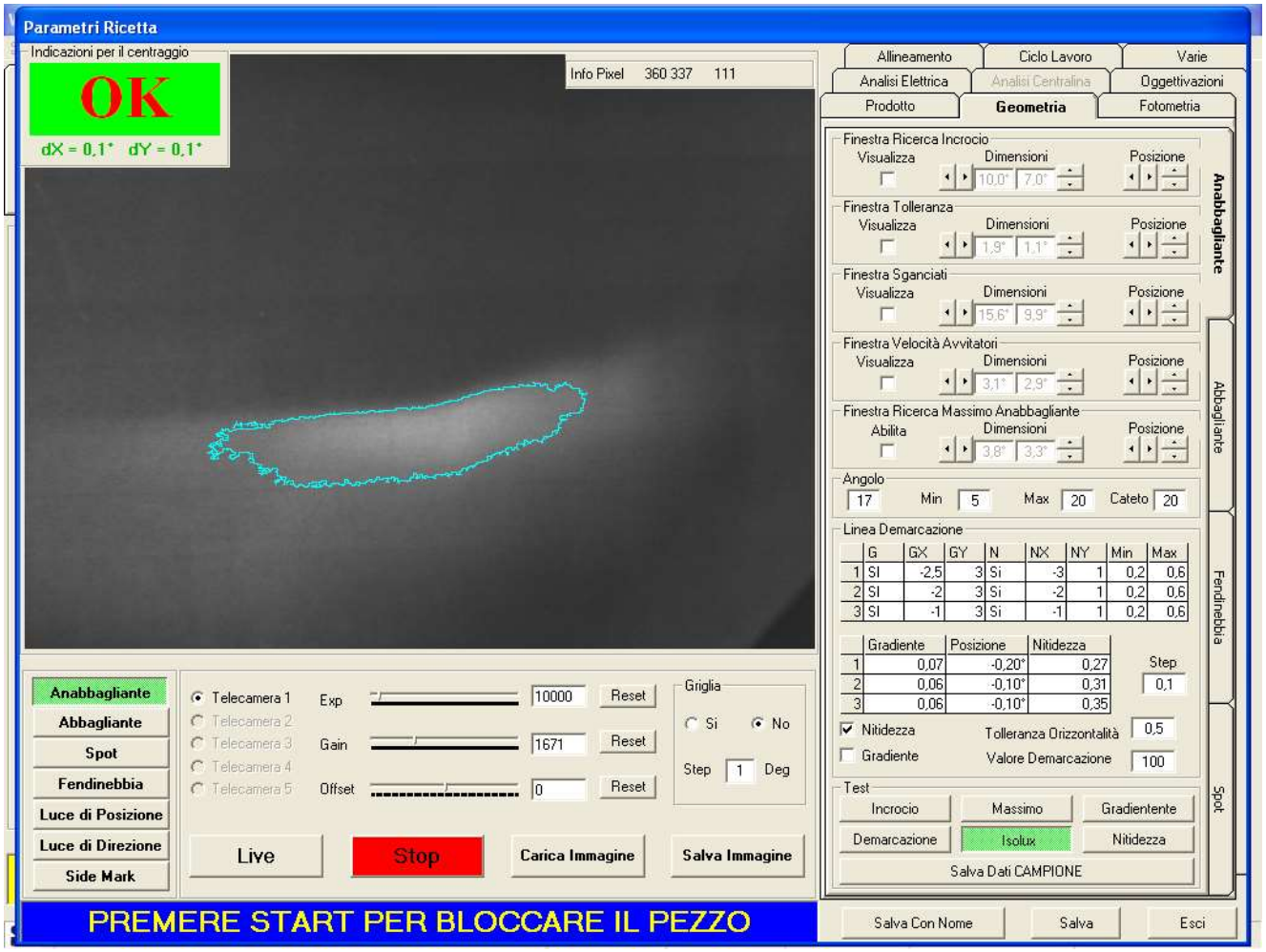


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Isolux: It is possible to make an evaluation of the Isolux of the low beam.
To test the Isolux, proceed as follows:

1. Place the videocamera in freeze-frame mode.
2. Press the Isolux key on the test window.
3. Position the mouse in the luminous zone and click the right mouse button.

This will display a border that gives the indication of the Isolux corresponding to the value of luminosity pointed with the mouse.
The Luminosity data is indicated in the Pixel Info Window.

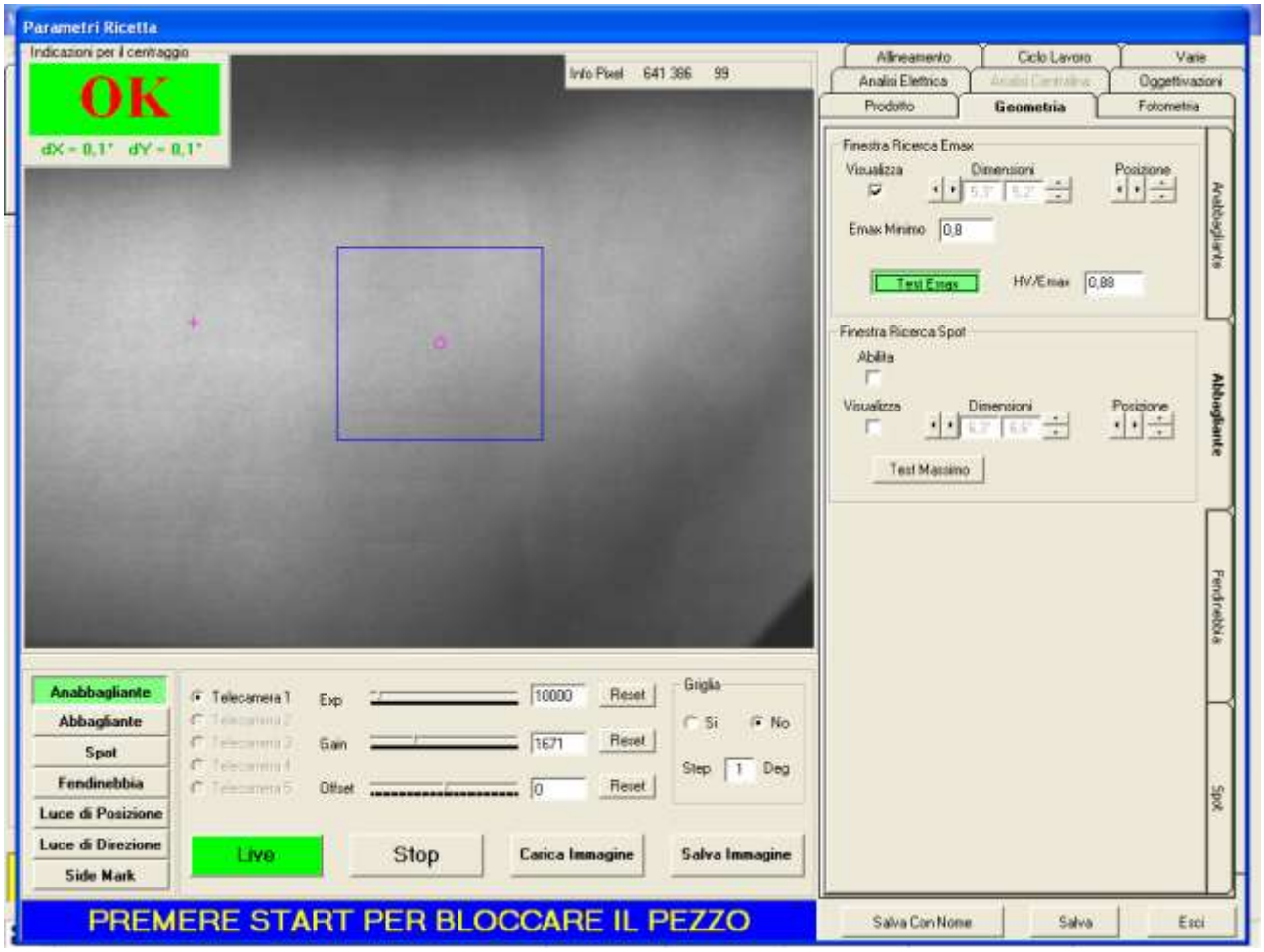


11.2.6 HIGH BEAM GEOMETRY

Enter the High Beam Geometry screen and switch on the high beam.

If a second videocamera is installed for analysis of the high beam, select Videocamera 2 from the list of videocameras installed.
Adjust the Exp (Exposition) and Gain values of the videocamera so as to have the best picture possible (follow the same instructions as for the low beam).

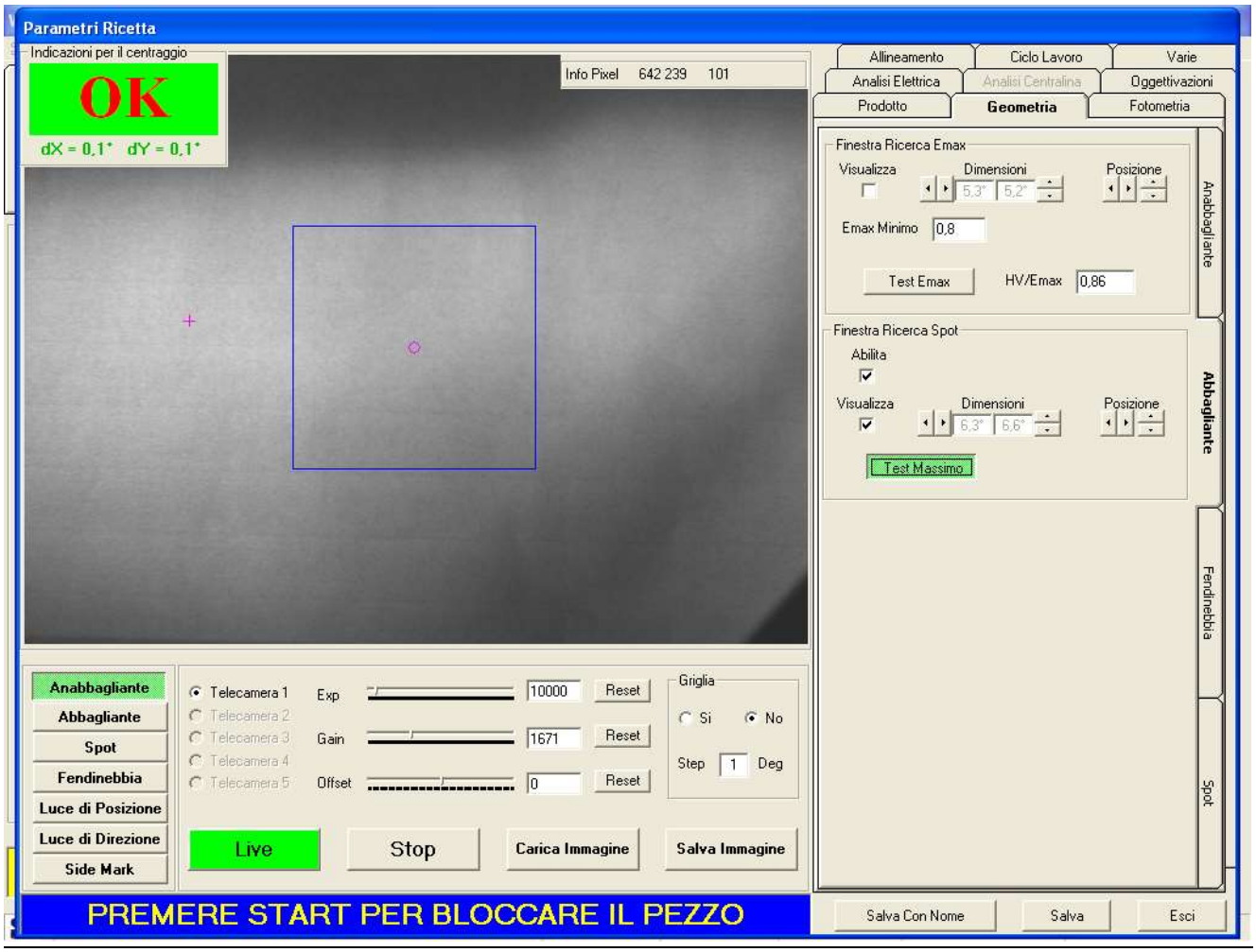
Emax Search Window: To check high beam alignment, the Emax is tested.
Using the Sample Piece, enable the window for viewing, resize it and position it around the Maximum High Beam. Lastly, save the data.
Press the "Test Emax" key to run the test. The HV/Emax box is displayed with a value that indicates the deviation of the Maximum High Beam with respect to the center of the window previously set. In the best case the value is close to 1, the more it moves away from the ideal condition the closer the value will tend toward 0.
Set the minimum value accepted during the Emax check in the test procedure.



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Spotlight Search Window: For headlamps with the spotlight function, you can check the position of this function.
After switching on the spotlight, enable the search window for display. After resizing and repositioning it properly, save the data. Press the “Test Maximum” key to run the test. If the spotlight search is enabled with a check in the “Enable” checkbox, during the test procedure the position of the spotlight with respect

to the window previously set will be checked: if the Maximum is inside the search window the test is considered positive.

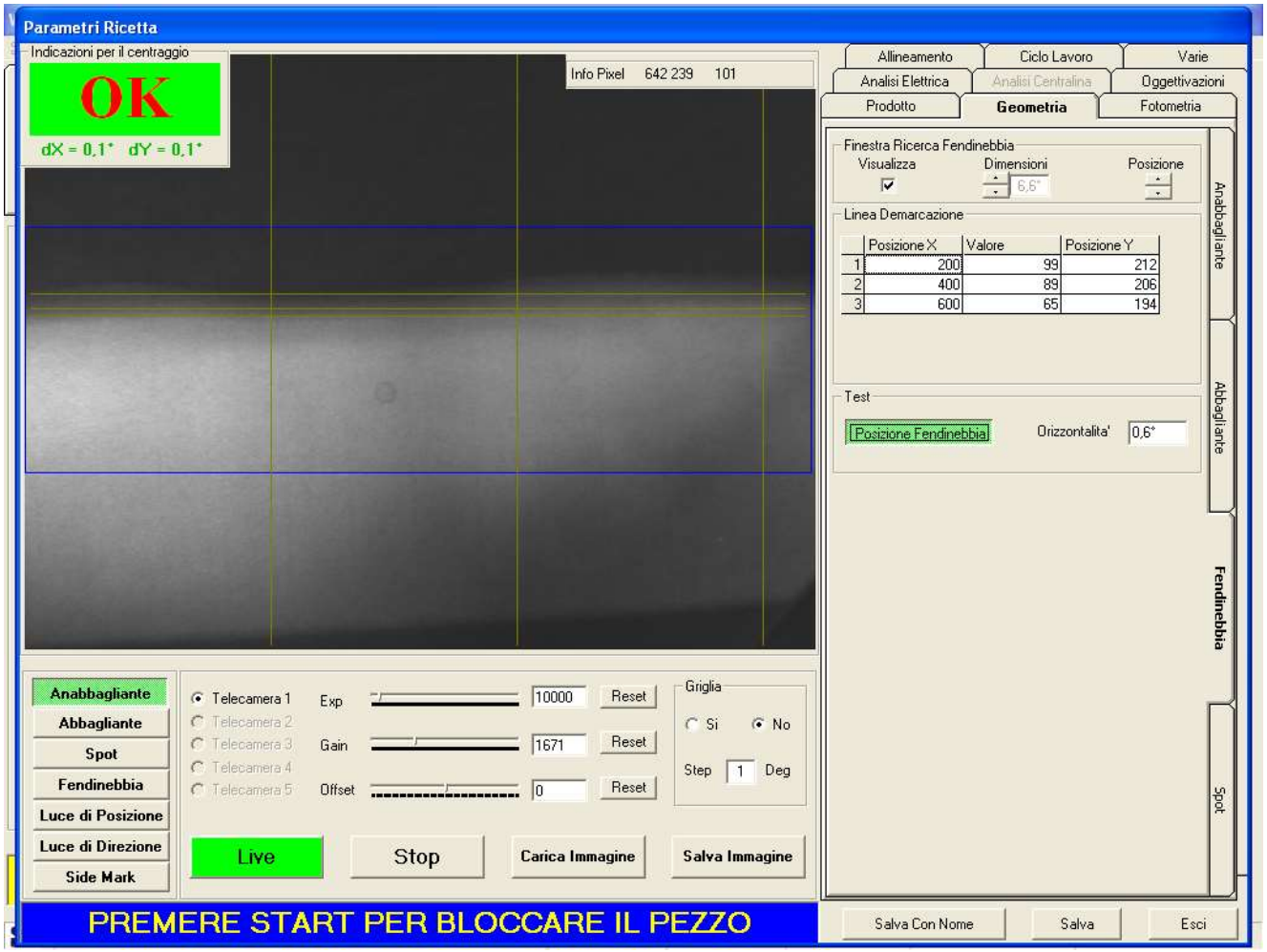


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11.2.7 FOGLIGHT GEOMETRY

Enter the Foglight Geometry screen and switch on the foglight.

Adjust the Exp (Exposition) and Gain values of the videocamera so as to have the best picture possible (follow the same instructions as for the low beam).



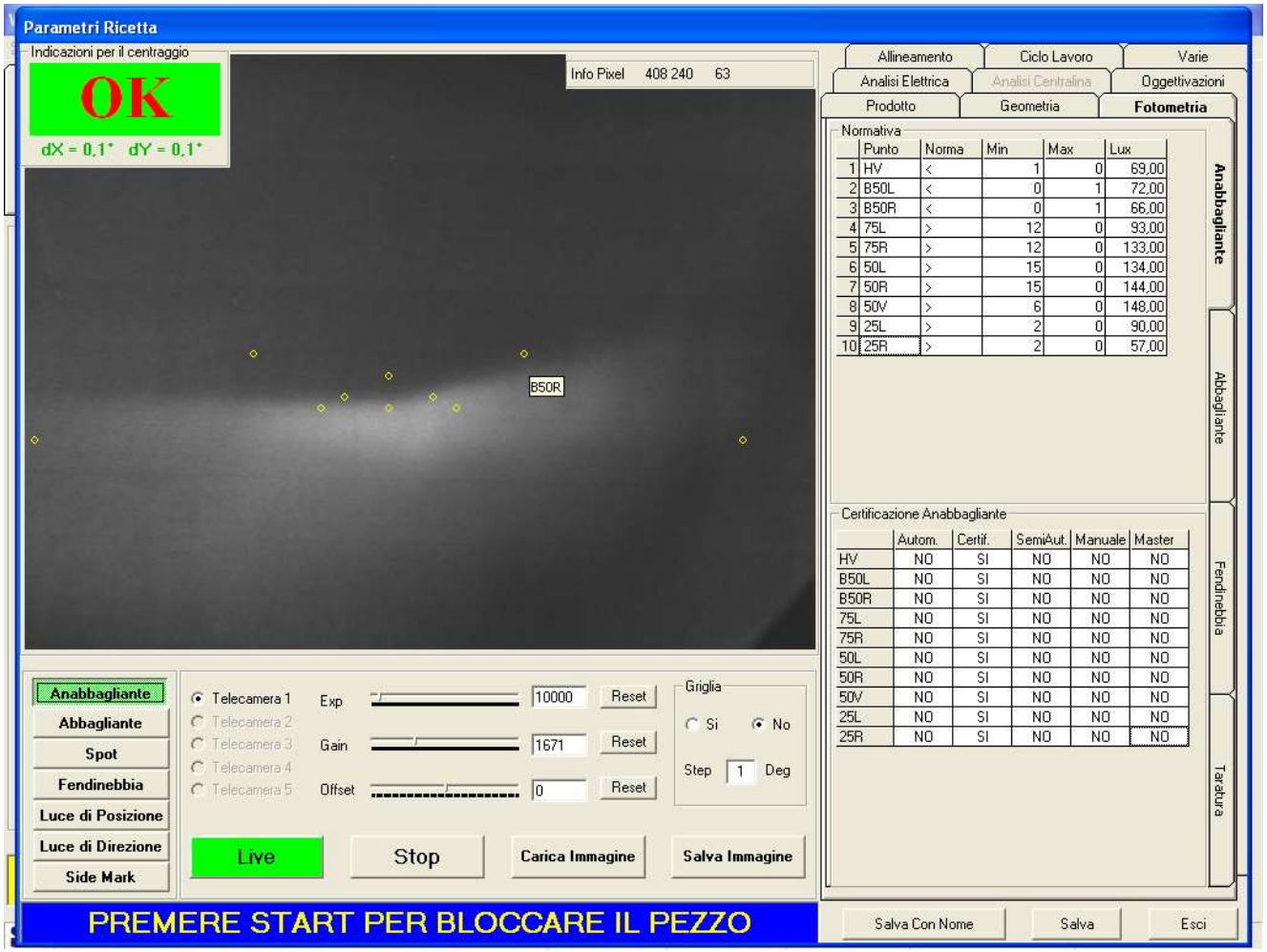
Foglight Search Window: You can define a zone in which the foglight demarcation should be found, to check the correct position of the beam.
Enable the window for viewing; compare the two horizontal lines in the videocamera picture that define the field of tolerance for the foglight beam. Resize and position the window adequately.
Press the “Foglight Position” key to run the test.
Use the table to set position X in which the test is made. The three horizontal yellow lines should be within the field of tolerance defined.

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11.2.8 LOW BEAM PHOTOMETRY

The system allows you to check the photometry of the low and high beams.

After entering the Low Beam Photometry screen and switched on the headlamp, the points provided for measuring photometry will be displayed on the picture of the beam. Move the mouse over these points to display the corresponding name.



There are two tables on the screen:

1. Standard.
2. Low Beam Certification.

The Standard Table lists the points provided for measurement and allows you to enter the limits envisaged by the standard.

The Certification Table allows you to define, for each procedure (Automatic, Certification, Semiautomatic, Manual, Master) which points to measure during the headlamp test.

We shall now describe the two tables in detail.

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STANDARD TABLE

Normativa

	Punto	Norma	Min	Max	Lux
1	HV	<	1	0	69,00
2	B50L	<	0	1	72,00
3	B50R	<	0	1	66,00
4	75L	>	12	0	93,00
5	75R	>	12	0	133,00
6	50L	>	15	0	134,00
7	50R	>	15	0	144,00
8	50V	>	6	0	148,00
9	25L	>	2	0	90,00
10	25R	>	2	0	57,00

Point: Indicates the name of the point (this item is not editable in this table).

Standard: Indicates the type of comparison with respect to the limits entered.

< The measurement must be below the Max limit entered (E.g.: B50L < 1).
< The measurement must be higher than the Min limit entered (E.g.: 75L > 12).
>< The measurement must be between the two Min and Max limits entered.

The comparison sign is edited by double-clicking on the corresponding box.

Min: Indicates the minimum limit of measurement (this item is edited by clicking on the box and entering the desired value).

Max: Indicates the maximum limit of measurement (this item is edited by clicking on the box and entering the desired value).

Lux: Indicates the value in Lux read (the item is not editable in this table).

IMPORTANT: use the comma (upper + U keys) to enter decimal values.

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LOW BEAM CERTIFICATION TABLE

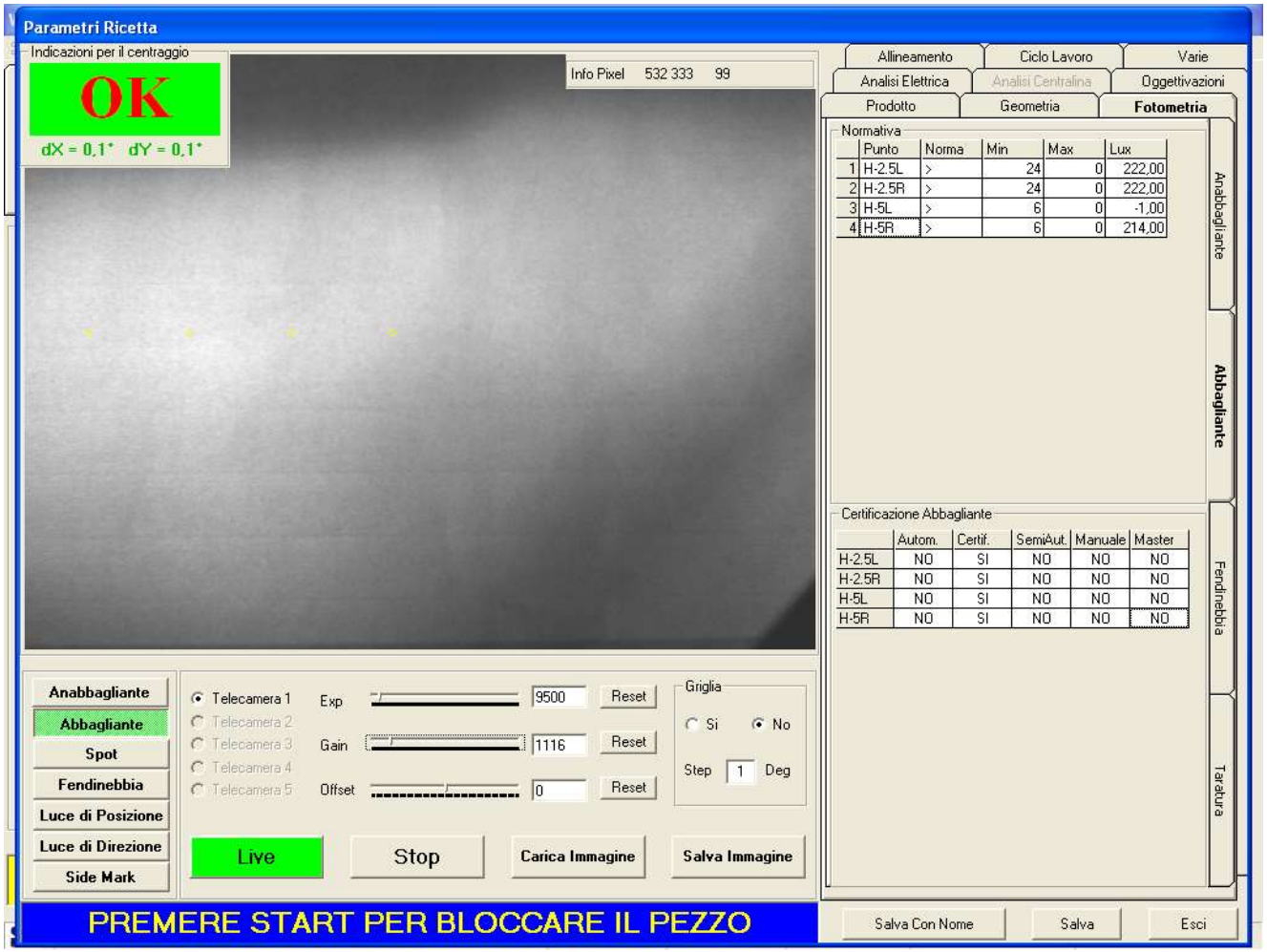
Certificazione Anabbagliante

	Autom.	Certif.	SemiAut.	Manuale	Master
HV	NO	SI	NO	NO	NO
B50L	NO	SI	NO	NO	NO
B50R	NO	SI	NO	NO	NO
75L	NO	SI	NO	NO	NO
75R	NO	SI	NO	NO	NO
50L	NO	SI	NO	NO	NO
50R	NO	SI	NO	NO	NO
50V	NO	SI	NO	NO	NO
25L	NO	SI	NO	NO	NO
25R	NO	SI	NO	NO	NO

The first column indicates the various points that can be selected for measurement, while the columns that follow indicate the type of test procedure (Automatic – Certification – Semiautomatic – Manual – Master)

For every procedure you can decide which points to measure during the test; double-click on the desired box to edit the item.

11.2.9 HIGH BEAM PHOTOMETRY



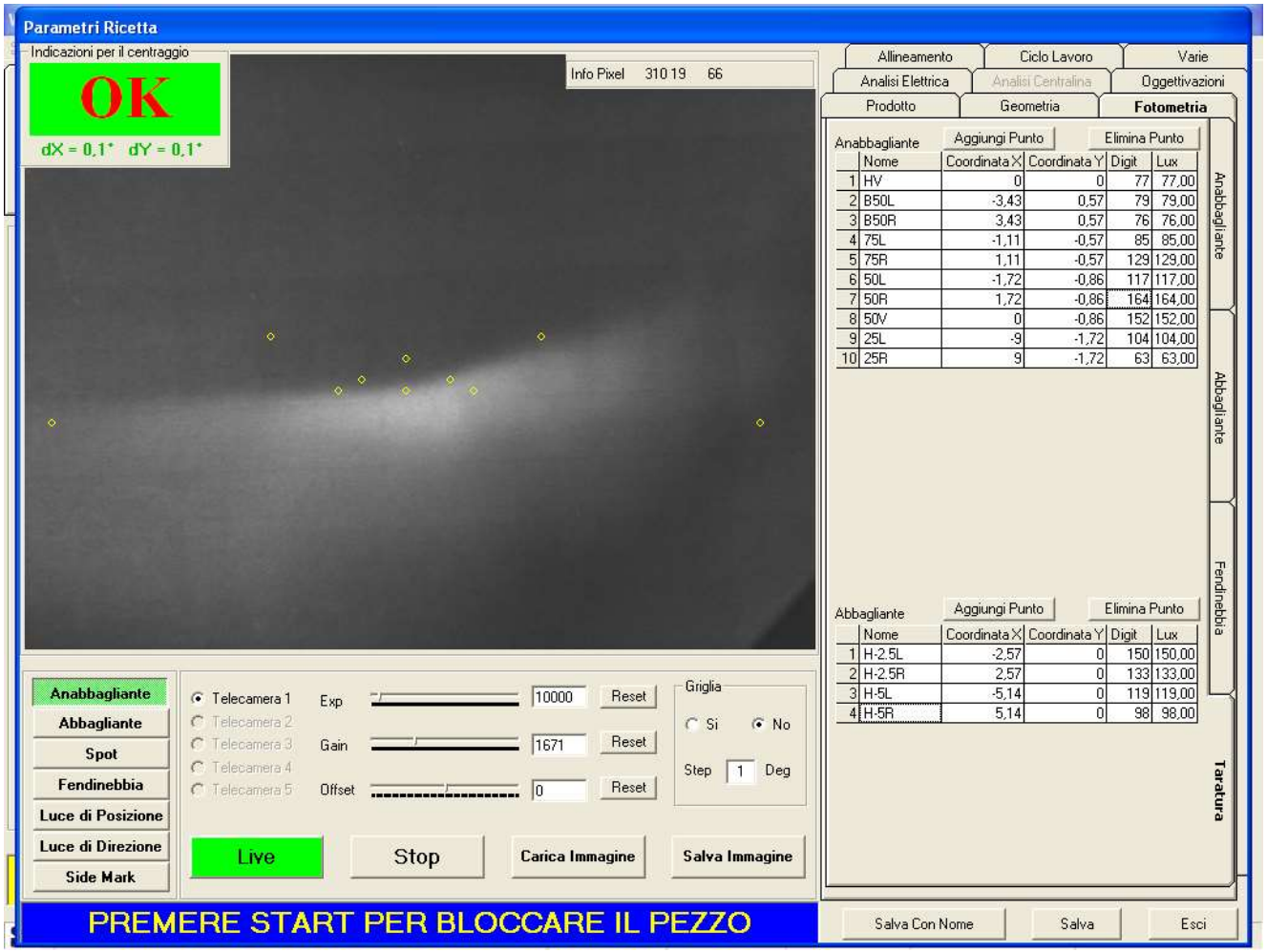
The High Beam Photometry screen contains the same information as described above, referring in this case, naturally, to the high beam.

Refer to the previous section for the description of the individual tables.

11.2.10 PHOTOMETRY CALIBRATION

The system is unable to supply absolute measurements for the photometric points, but is calibrated so as to associate the Lux reading by the videocamera with the points measured in the photometric laboratory.

Therefore, in order to calibrate the photometric points it is essential to have a sample measured in the laboratory.



Using the calibration table it is possible to define the name, coordinates and Lux reading of the point. Moreover, it is possible to add or remove points.

We shall now describe the photometric point calibration table. The procedure is the same for both high and low beam.

Photometric Point Calibration Table.

Anabbagliante | Aggiungi Punto | Elimina Punto

	Nome	Coordinata X	Coordinata Y	Digit	Lux
1	HV	0	0	77	77,00
2	B50L	-3,43	0,57	79	79,00
3	B50R	3,43	0,57	76	76,00
4	75L	-1,11	-0,57	85	85,00
5	75R	1,11	-0,57	129	129,00
6	50L	-1,72	-0,86	117	117,00
7	50R	1,72	-0,86	164	164,00
8	50V	0	-0,86	152	152,00
9	25L	-9	-1,72	104	104,00
10	25R	9	-1,72	63	63,00

Point Name: Indicates the name of the Point (the item is edited by clicking on the box and entering the name desired).

X Coordinate: Indicates the X Coordinate of the point with respect to the intersection of the beam (the item is edited by clicking on the box and entering the name desired).

Y Coordinate: Indicates the Y Coordinate of the point with respect to the intersection of the beam (the item is edited by clicking on the box and entering the name desired).

Digit: Indicates the reading in digits of the videocamera (this item is not editable).

Lux: Reading for calibration in Lux of the point. **Enter the videocamera reading in this box** (the item is edited by clicking on the box and entering the value required).

Add Point: Serves to enter a new point in the table; click the table in the position where you want to add the new point and press "Add Point" . A blank line is created where you can update the various items.

Eliminate Point: Serves to eliminate a point in the table; click the line to be eliminated and press "Eliminate Point". The selected point is eliminated from the table.



IMPORTANT: use the comma (upper + U keys) to enter decimal values.

The calibration table for the high beam has the same properties as we have just described.

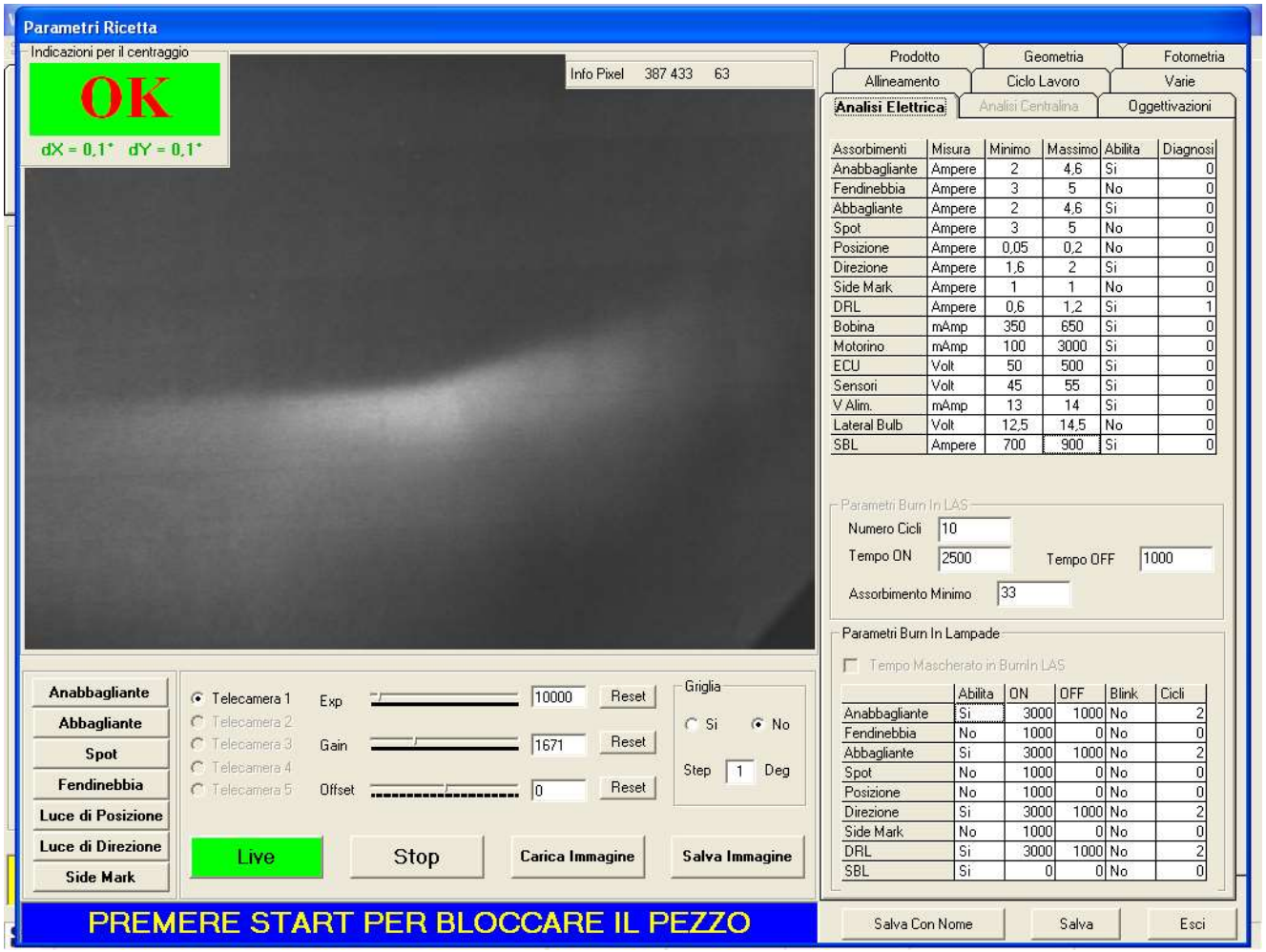
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11.2.11 ELECTRICAL ANALYSIS PARAMETERS

On this screen, you can enter all the parameters relative to the electrical test of the various functions. Typically, the test consists of the following:

- Lamp BurnIn procedure.
- Check of the absorptions or voltages of each function.

Note: For the BurnIn of discharge lamps refer to the product specifications.



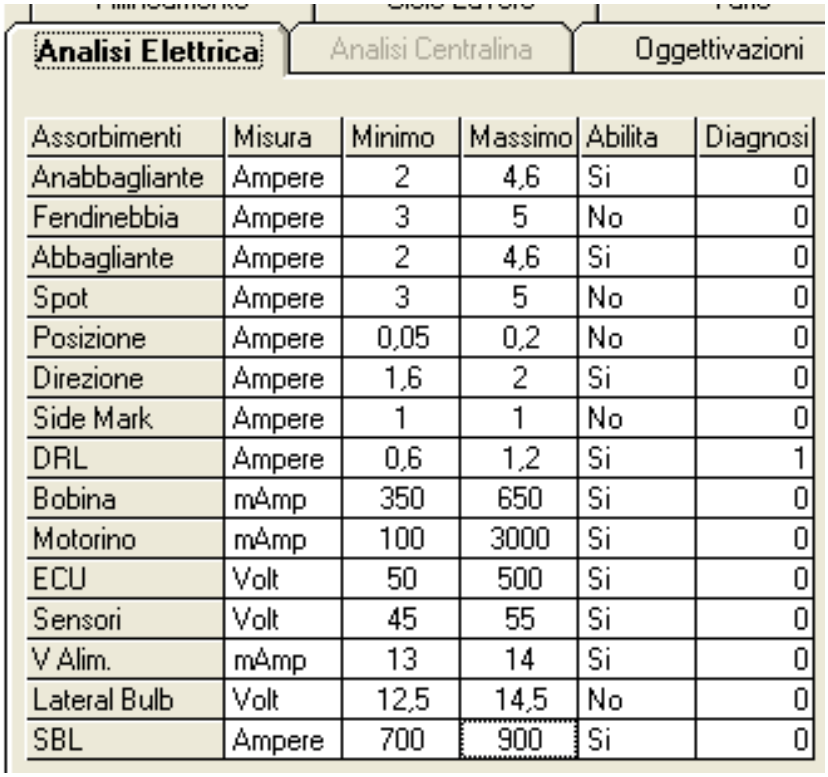
Lamp BurnIn service to detect any possible cases of "early deterioration" of the lamps.

The table of parameter settings will now be described in detail.

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TABLE OF ABSORPTIONS

In this table you can define the functions to test, the relative limits of tolerance and any digital diagnostic input if contemplated.



Analisi Elettrica | Analisi Centralina | Oggettivazioni

Assorbimenti	Misura	Minimo	Massimo	Abilita	Diagnosi
Anabbagliante	Ampere	2	4,6	Si	0
Fendinebbia	Ampere	3	5	No	0
Abbagliante	Ampere	2	4,6	Si	0
Spot	Ampere	3	5	No	0
Posizione	Ampere	0,05	0,2	No	0
Direzione	Ampere	1,6	2	Si	0
Side Mark	Ampere	1	1	No	0
DRL	Ampere	0,6	1,2	Si	1
Bobina	mAmp	350	650	Si	0
Motorino	mAmp	100	3000	Si	0
ECU	Volt	50	500	Si	0
Sensori	Volt	45	55	Si	0
V Alim.	mAmp	13	14	Si	0
Lateral Bulb	Volt	12,5	14,5	No	0
SBL	Ampere	700	900	Si	0

Absorptions: The first column identifies the function to be tested. The data in this column are not editable.

Measure: Indicates the unit of measurement of the test performed. The data in this column are not editable.

Minimum: Indicates the minimum limit of tolerance of the measure. The item is edited by clicking on the box and entering the value required.

Maximum: Indicates the maximum limit of tolerance of the measure. The item is edited by clicking on the box and entering the value required.

Enable: Indicates whether the function is controlled during the test. The item can be edited by double-clicking on the desired box.

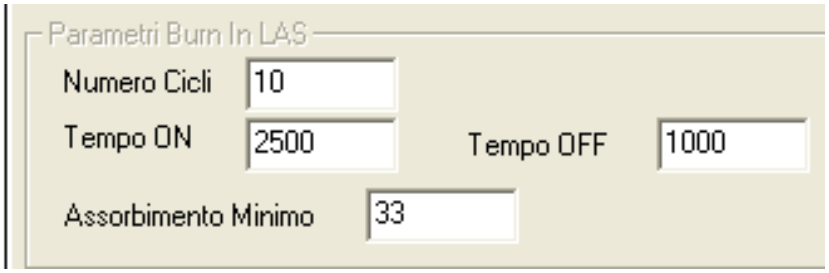
Diagnostics: Indicate the possible input of diagnostics relative to the function (see the technical specifications of the product). The item is edited by clicking on the box and entering the value required.

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IMPORTANT: use the comma (upper + U keys) to enter decimal values.

DL (DISCHARGE LAMP) BURNIN WINDOW

This window is enabled only if the discharge lamp function is enabled on the product screen.
This test serves to monitor performance of ignitions (priming) of the discharge lamps.
The electrical analysis is performed by a rapid acquisition card which evaluates the first instants of priming of the lamp.
Note: Refer to the product specifications for the BurnIn parameters.



Number of Procedures: Indicates the number of ON / OFF procedures required.

Time ON: Indicates the time ON of the lamp expressed in milliseconds.

Time OFF: Indicates the time OFF of the lamp expressed in milliseconds.

Minimum Absorption: indicates the minimum absorption in Amps admissible during the lamp priming stage.

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LAMP BURNIN TABLE



	Abilita	ON	OFF	Blink	Cicli
Anabbagliante	Si	3000	1000	No	2
Fendinebbia	No	1000	0	No	0
Abbagliante	Si	3000	1000	No	2
Spot	No	1000	0	No	0
Posizione	No	1000	0	No	0
Direzione	Si	3000	1000	No	2
Side Mark	No	1000	0	No	0
DRL	Si	3000	1000	No	2
SBL	Si	0	0	No	0

Enable: Indicates whether BurnIn is enabled for the function indicated.

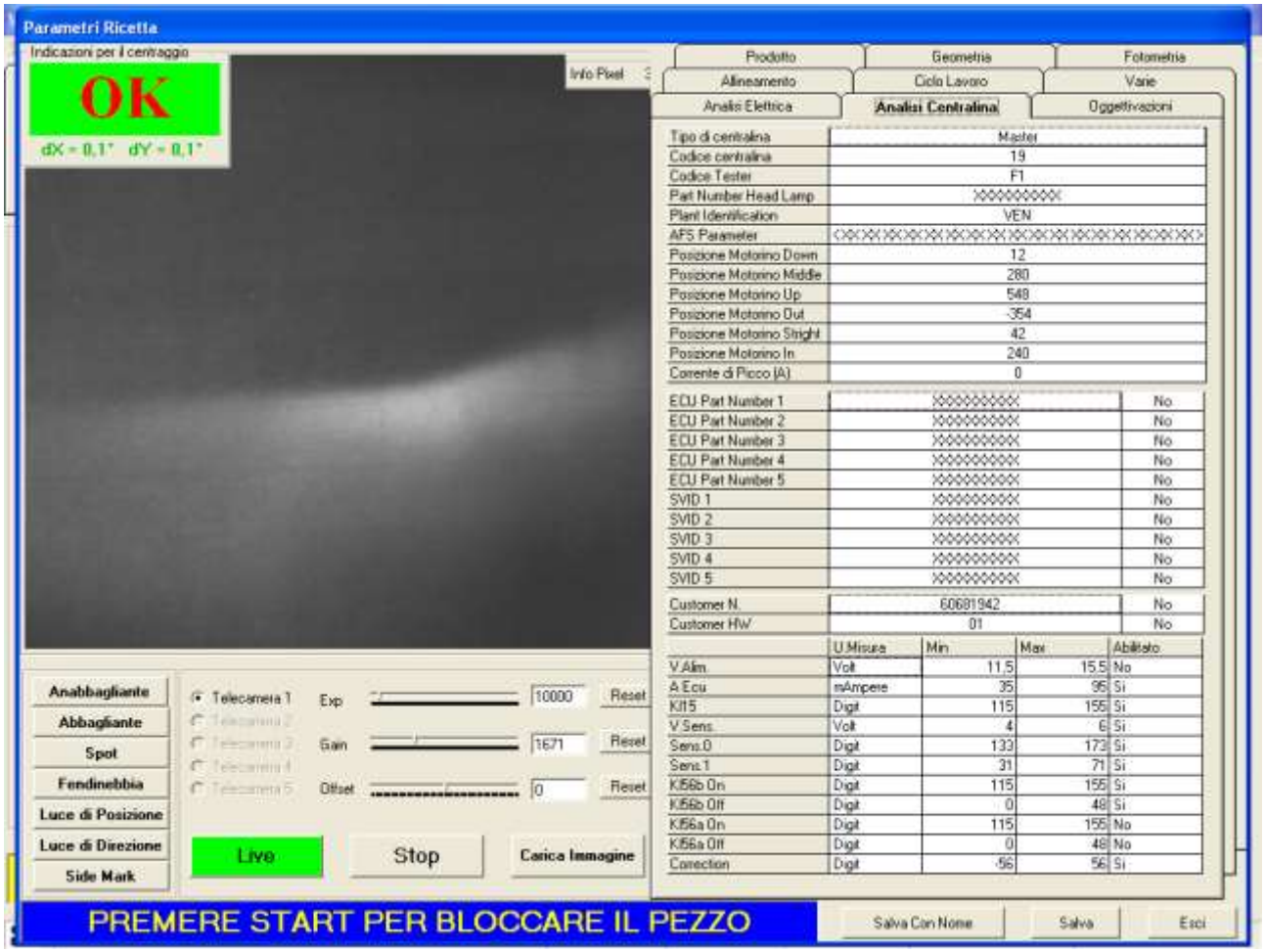
ON: Indicates the time ON of the lamp expressed in milliseconds.

OFF: Indicates the time OFF of the lamp expressed in milliseconds.

Procedures: Indicates the number of ON / OFF procedures required.

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11.2.12 CONTROL UNIT ELECTRICAL ANALYSIS

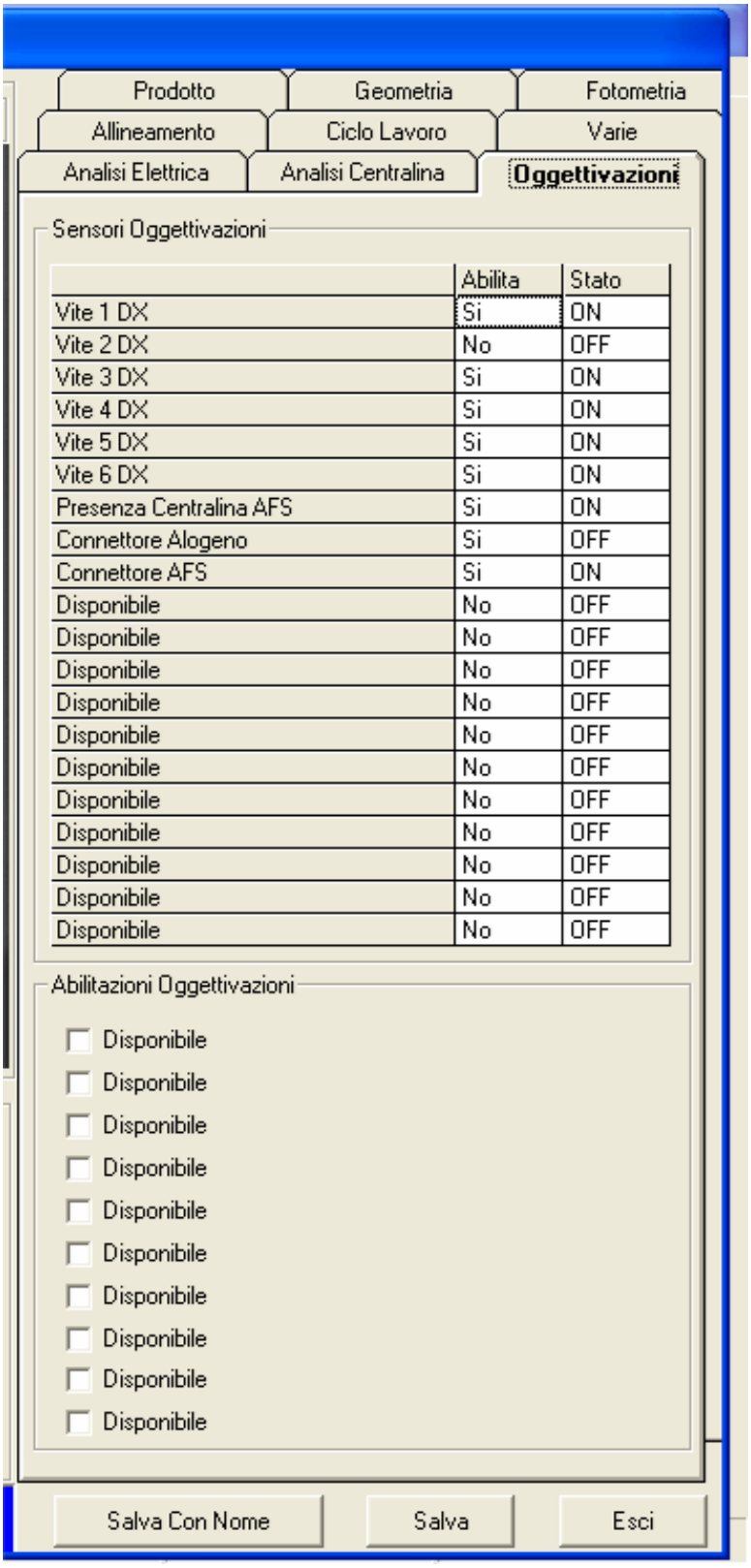


When testing headlamps with electronic control units it is possible to test the parameters of the control unit indicated in the product specifications.

Note: Refer to the product specifications for the parameters of the control units.

11.2.13 RECOGNITION PARAMETERS

On the Recognition Screen it is possible to enable or disable recognition of individual parts of the assembly and set the status of the input to be checked.



11.2.14 CORRECTOR PARAMETERS

On the Alignment Screen it is possible to enter the data relative to control of the corrector and automatic bear alignment adjustments.

We shall now describe the separate windows of the Corrector Screen.

The Analogue Corrector Parameters Window can be used to enter the data for the analogue headlamp corrector.
The analogue headlamp corrector is adjusted by means of the voltage settings made with the multiturn potentiometers installed on the control panel.



Dip: Indicates the dip angle required.

Maximum dip: Indicates the maximum dip tolerated for corrector movement.

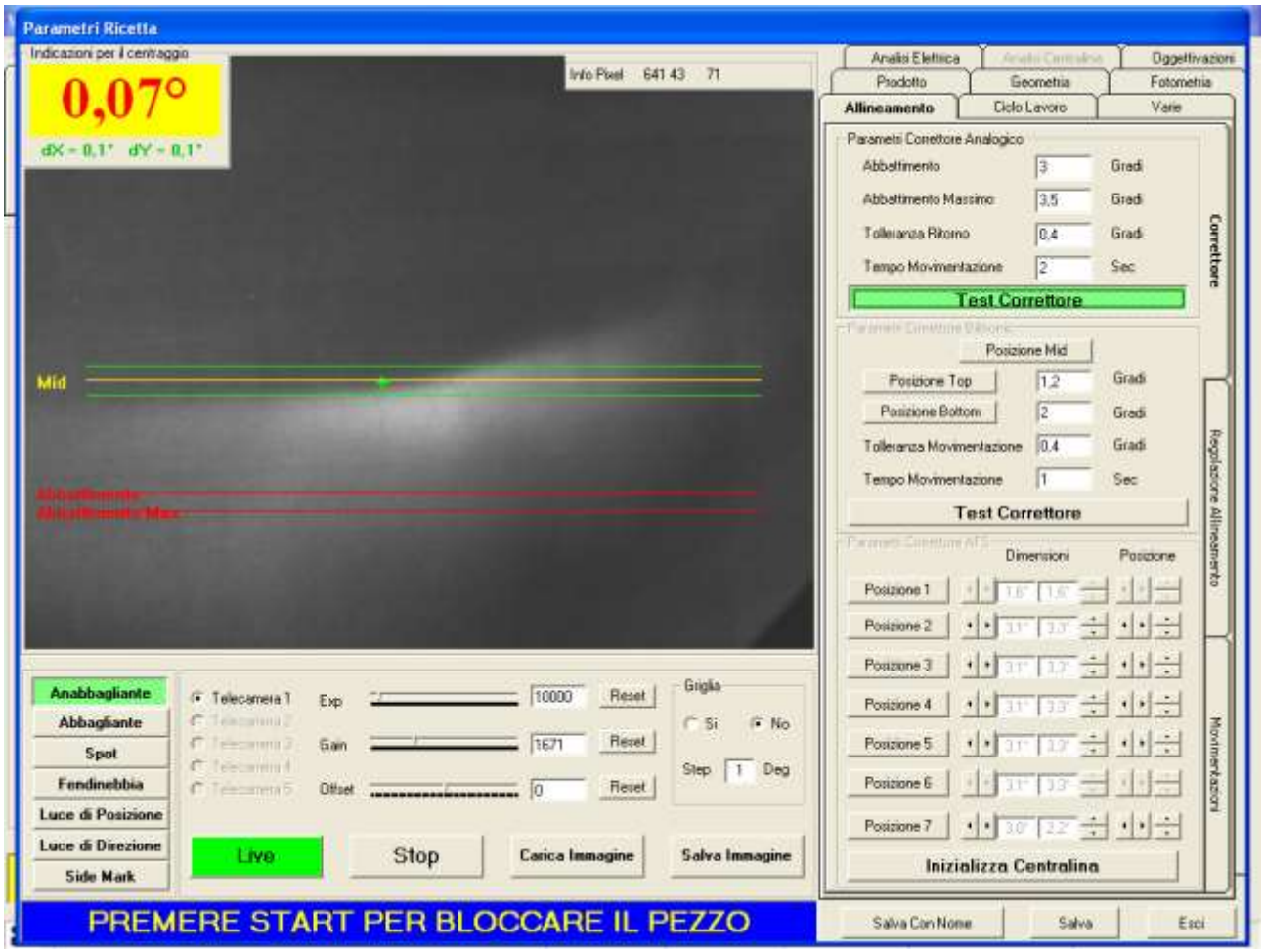
Tolerance on Return: Indicates the tolerance within which the intersection must remain at the end of corrector movement.

Movement time: Indicates the maximum time within which each of the two stages of movement must be completed: dip and return to zero.

Corrector Test: Press “Corrector Test” to check the settings made.

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In the example on the next page, the two red lines indicate the dip required (3 degrees) and the maximum admissible (3.5 degrees). During the first stage of the check, the corrector will dip the beam and the intersection will have to be within those limits.
The yellow line indicates the starting line of movement while the two yellow lines indicate the tolerance admissible for return to 0 (0.4 degrees).
During the second stage, the corrector will bring the beam up again and the intersection will have to be within those limits.
These two stages must both occur within the set time (2 seconds).



The Bilitronic Corrector Parameters Window can be used to enter the data for the corrector controlled by an electronic control unit. The difference with respect to an analogue corrector is that the movement is made both downward and upward.



Top Position: Indicates the required position for upward movement.

Bottom Position: Indicates the required position for downward movement.

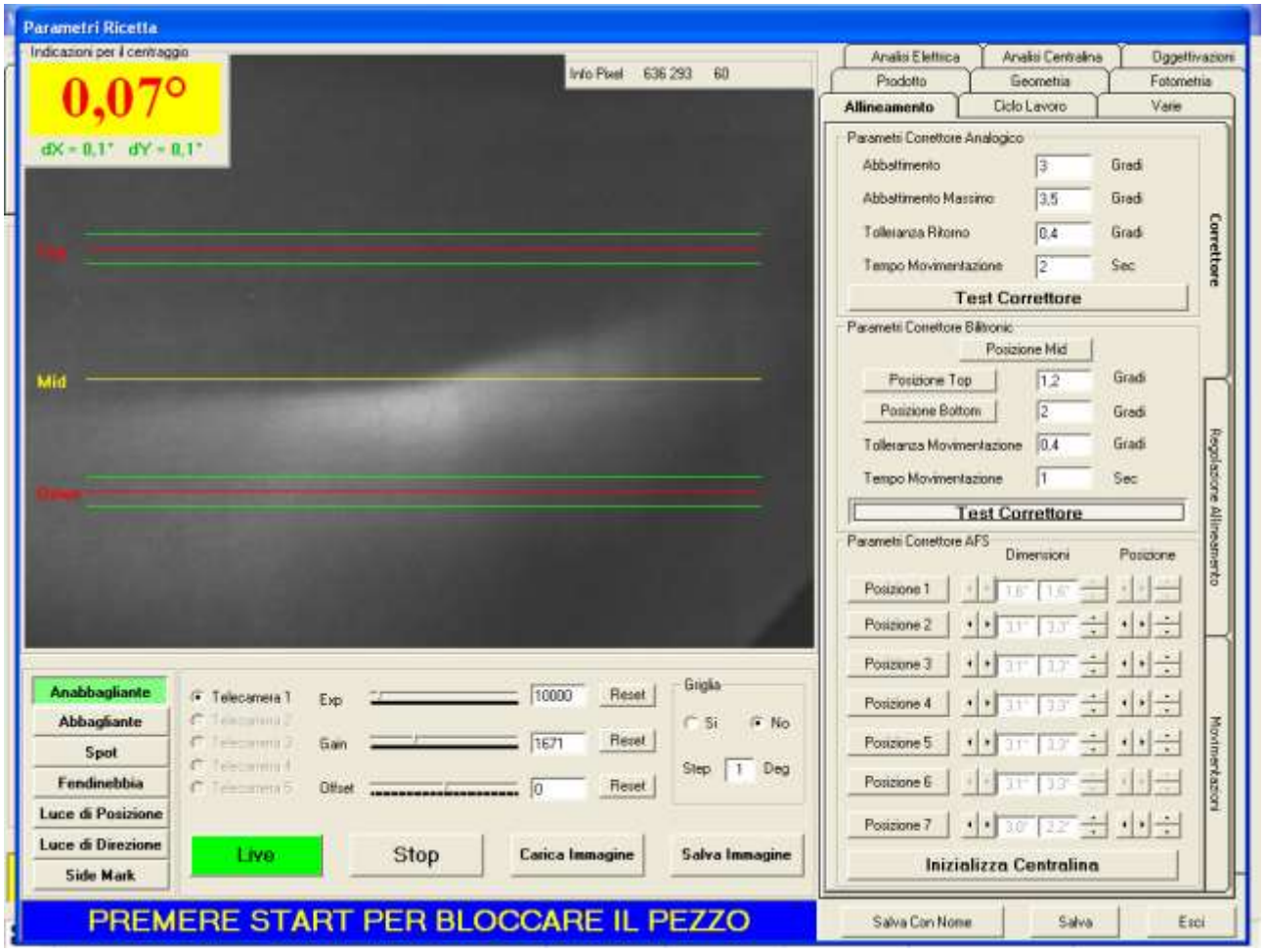
Tolerance on Movement: Indicates the tolerance admissible for the check of movement.

Movement time: Indicates the maximum time within which each of the two stages of movement must be completed: dip and return to zero.

Corrector Test: Press “Corrector Test” to check the settings made.

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In the example that follows, the two red lines indicate the intended objectives for the upward and downward movement, while the green lines indicate the admissible tolerance.
The yellow line indicates the starting line of movement.

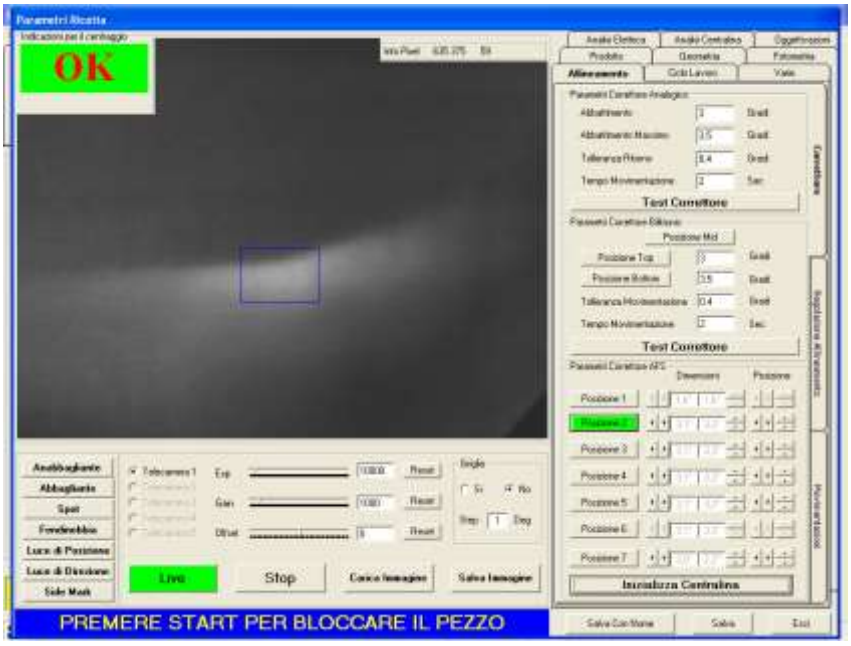


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The window *AFS Corrector Parameters* is used to make the settings in the window controlling AFS movement.



The procedure for movement of the headlamps with the AFS system requires, first of all, initializing the control unit. At this point, using the position keys, move the AFS beam to the desired position and select the appropriate videocamera to display the intersection. Next, move and resize the tolerance window around the intersection displayed. During the test, the system checks that at the end of each movement, the intersection is inside the required tolerance window.



1. Press “initialize control unit” and wait until it turns green.

2. Press “Position 2 to bring the beam to position 2 of the AFS movement.

3. Select the appropriate videocamera to display the beam intersection correctly.

4. Reside and move the tolerance window for the movement.

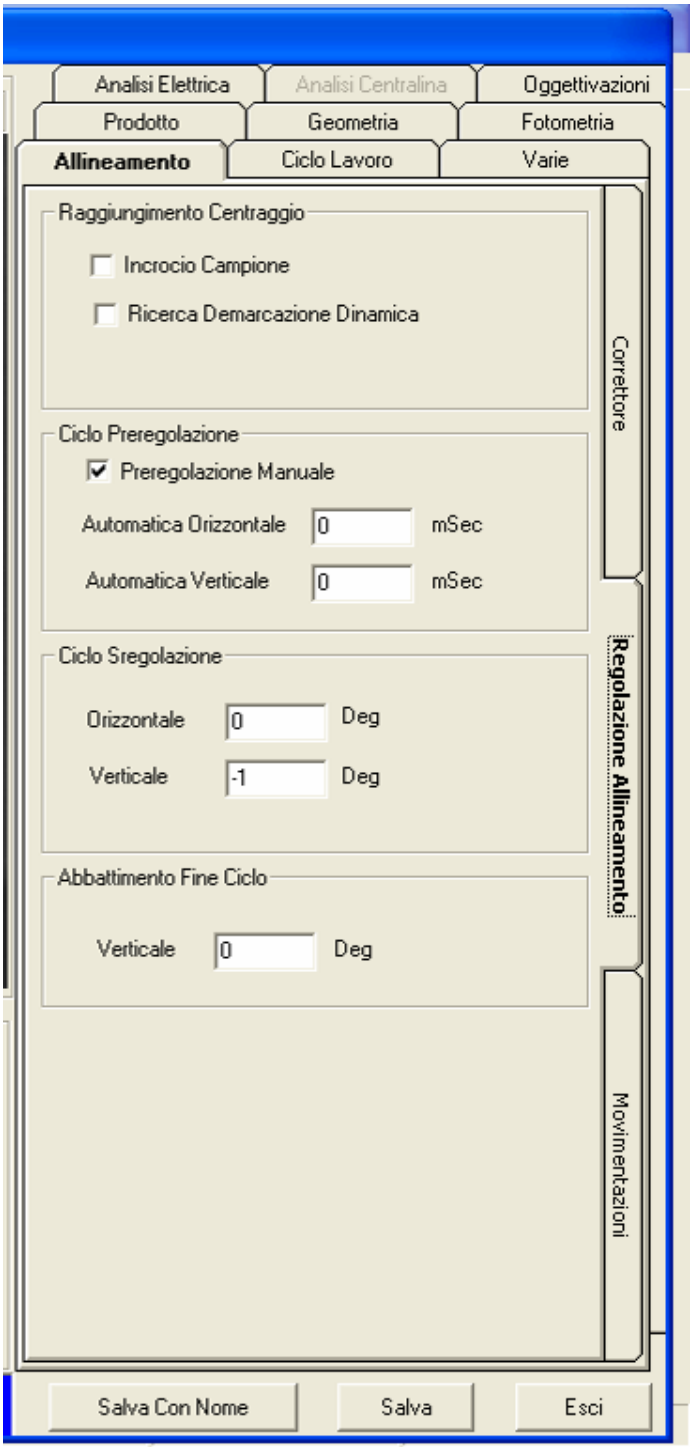
5. Repeat items 2 through 4 for all the positions required.

NOTE: Refer to the product specifications for details about AFS movement.

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11.2.15 ALIGNMENT ADJUSTMENT PARAMETERS

All the parameters relative to alignment of the low beam with automatic motors are set on the Alignment Adjustment screen .



Sample Intersection*:* Indicates whether beam alignment is made according to sample (see Low Beam Geometry Screen).

Dynamic Demarcation Search: Enables demarcation search throughout the entire test procedure. Otherwise, the demarcation saved during setting of Low Beam Geometry is used.

Manual Preadjustment: During the test procedure, it is possible to make preadjustment either manually, if there are no screwdrivers, or automatically.
If enabled, manual preadjustment permits the operator to perform a first adjustment of the beam, for example if the pieces arrive with the beam particularly deviated with respect to the tolerance.

Automatic Vertical Preadjustment: If the screwdrivers are present, it is possible to perform a timed preadjustment of the screwdrivers so as to bring a particularly deviated beam toward the window of tolerance. The screwdriver will move the beam vertically for the time set in milliseconds.

Automatic Horizontal Preadjustment: See Automatic Vertical Preadjustment.

Horizontal Maladjustment: Maladjustment serves to check correct operation of the headlamp centering bushings. Making it produce a given movement in degrees, the system will verify whether that horizontal movement is made correctly and thus whether the headlamp bushing is working properly.

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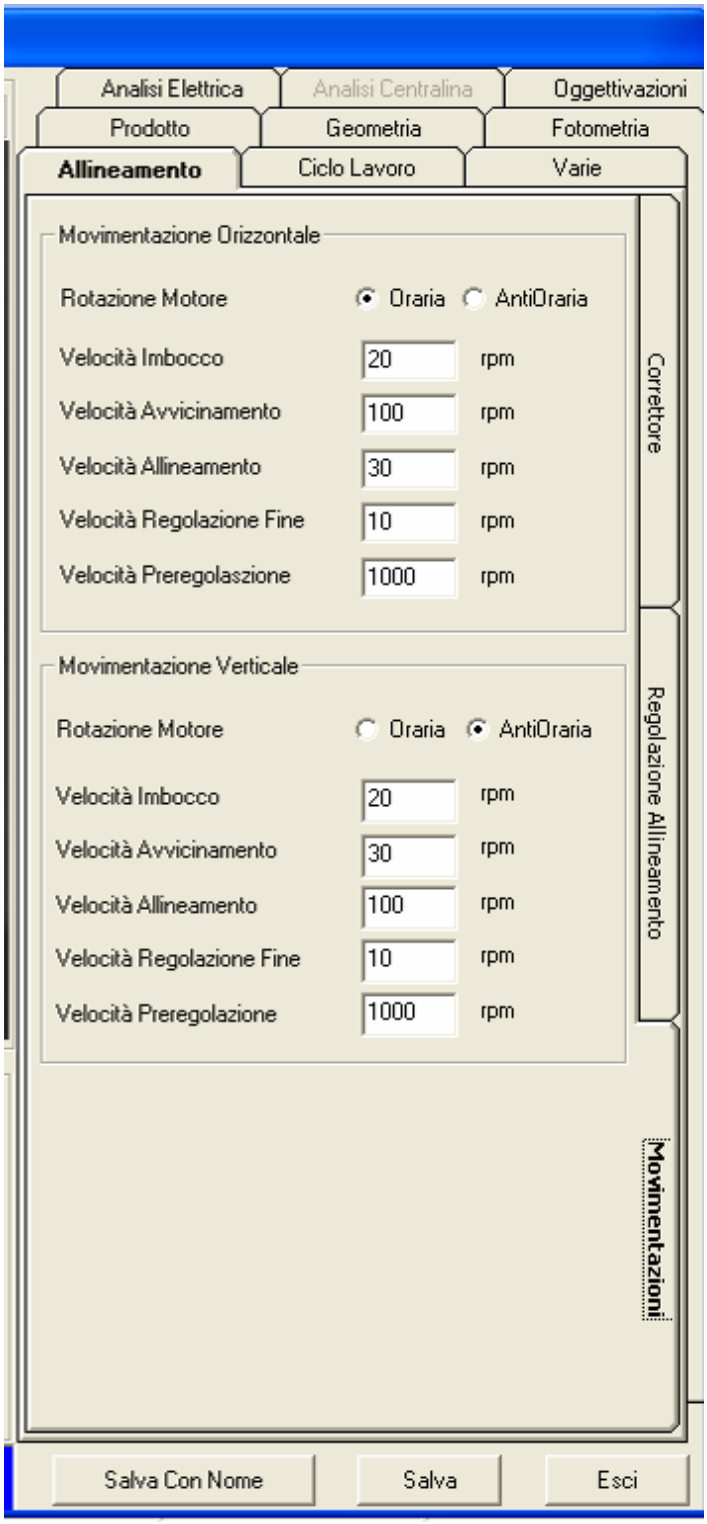
Vertical Maladjustment*:* See Horizontal Maladjustment.

Vertical Dip at the end of the procedure: If different from 0, this information indicates the dip made at the end of the procedure.

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11.2.16 ALIGNMENT MOVEMENT PARAMETERS

The characteristics of screwdriver movement are set on the Movement screen.



Motor Rotation: Indicates the rotation of the screwdrivers to bring the beam toward the window of tolerance.

Insert Speed: this is the speed used to insert the tools in the headlamp adjustment bushing at the start of the test procedure.

Approach Speed: during the Intersection Search step, this is the speed used to approach the window of tolerance if the intersection is outside the screwdriver window (see Low Beam Geometry Screen, *Screwdriver Speed Window*).

Alignment Speed: during the Intersection Search step, this is the speed used to approach the intersection to the window of tolerance if the intersection is outside the screwdriver window (see Low Beam Geometry Screen, *Screwdriver Speed Window*).

Fine Adjustment Speed: if Sample Intersection Search is enabled, this speed is used when the intersection enters the window of tolerance.

Preadjustment Speed: this is the speed used in the preadjustment step, if enabled.

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11.2.17 TEST PROCEDURE PARAMETERS

The sequence of test procedures is entered on the Test Procedure screen .



For the description of the individual steps, see the paragraph entitled Headlamp Test Procedure.

The above table displays the following information for every step:

Column 1: Performance index; 24 are listed. This column is not editable.

Column 2: Description of Step. This column is not editable.

Column 3: Timeout in seconds within which the check envisaged in the step has to be satisfactorily completed: if at the end of this Timeout the check is not positive, an error is signaled with regard to the step in progress.
To enter the value desired, double-click on the box and enter the value.

Column 4: Status of Step. There are 5 possible situations:

Disabled: The step is omitted.
Signal and block of test procedure: in case of negative outcome of the Step the test procedure is halted and an error is signaled.
Signal and continuation of test procedure: in case of negative outcome of the Step the error is signaled but the test procedure is not halted.
Repeat procedure in case of error: in case of negative outcome the Step is repeated.
Wait for Start from Operator: If the outcome of the Step is positive, the procedure stops and waits for the operator to press the start key on the unit.

For the Step "Print Label" the possibilities are:

Never: The label is never printed.
Only Good Piece: The label is printed only in case of positive test outcome.
Only Reject Piece: The label is printed only in case of negative test outcome.
Always: The label is always printed.

To edit the status of the step, double-click on the appropriate box to obtain the desired setting.



Use the keys "Move Line UP" and "Move Line DOWN" you can edit the sequence of steps.

It is also possible to copy procedures:

1. Press the "Copy Test Procedure" key
2. Select procedure to copy
3. Select destination procedure
4. Press "Confirm Copy" key

Note: on entering the Edit Recipe screen the current test procedure is displayed.

11.2.18 MISCELLANEOUS PARAMETERS

Check Master at Change of Shift.
These parameters force the operator to test the Master piece at every change of shift.



Enabled: You can enable or disable the test of the Master Piece at change of shift.

Associated Input: This function is not used.

Procedure Tolerance: if the control is enabled, the operator can test the Master Piece within the Procedure Tolerance.

Example: If the test is enabled, at the end of the shift or startup of the program the "Sample Piece" key begins to blink on the main screen with the wording “Test Master”. Within the number of procedures set in the Procedure Tolerance field, the operator must press the “Test Master” key and start the test of the Sample Piece. If this is not done, the following messages appear on the main screen:

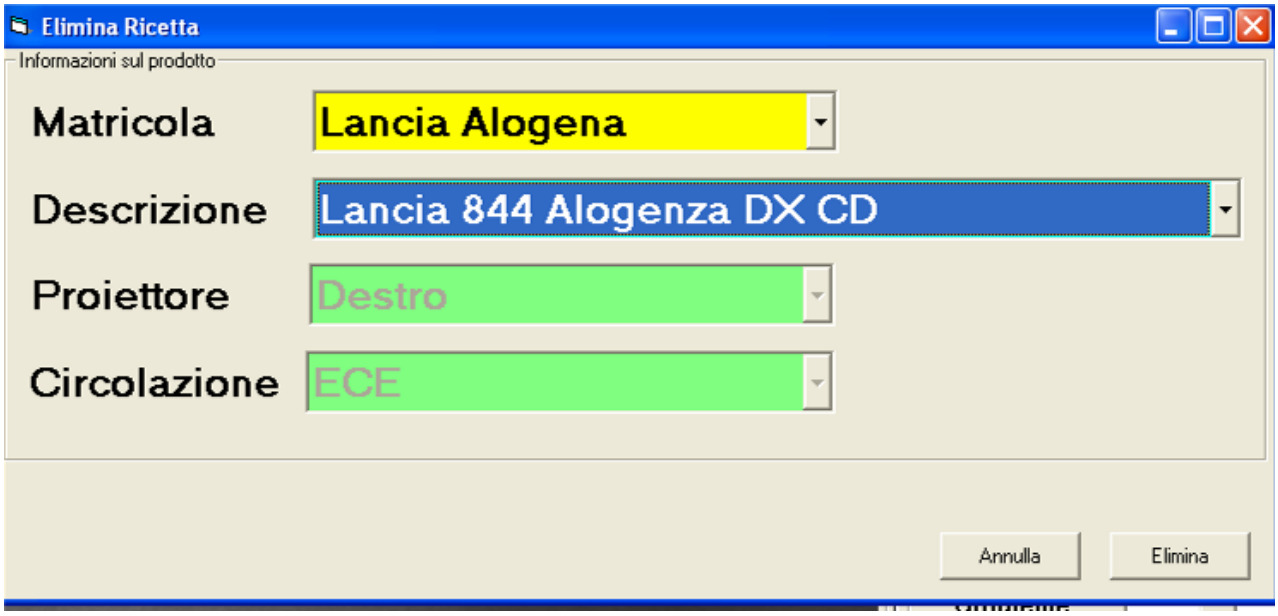


At this point it is no longer possible to start a procedure without first running the test of the sample piece.

11.3 DELETE RECIPE

Access level: Supervisor.

Use this command to delete recipes you are no longer using.
Select the recipe to delete from the window shown below and press “Delete”.



The request to confirm deletion appears.



If you try to delete the recipe in use, the following message appears.



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12 TEST PROCEDURES

Access level: Expert.



The system makes it possible to manage up to 5 different procedures. The procedures are all independent and thus can be used for specific needs.

- Automatic Procedure
- Certification Procedure
- Semiautomatic Procedure
- Manual Procedure
- Master Procedure

For example, you might use the Automatic Procedure for normal production and the Certification Procedure to check all the photometric points, while the Master Procedure can be programmed to check a piece with the rejects desired using the steps in the “Signal Error and Continue Procedure” mode.
The programming and use of individual procedures is at the discretion of the end user.

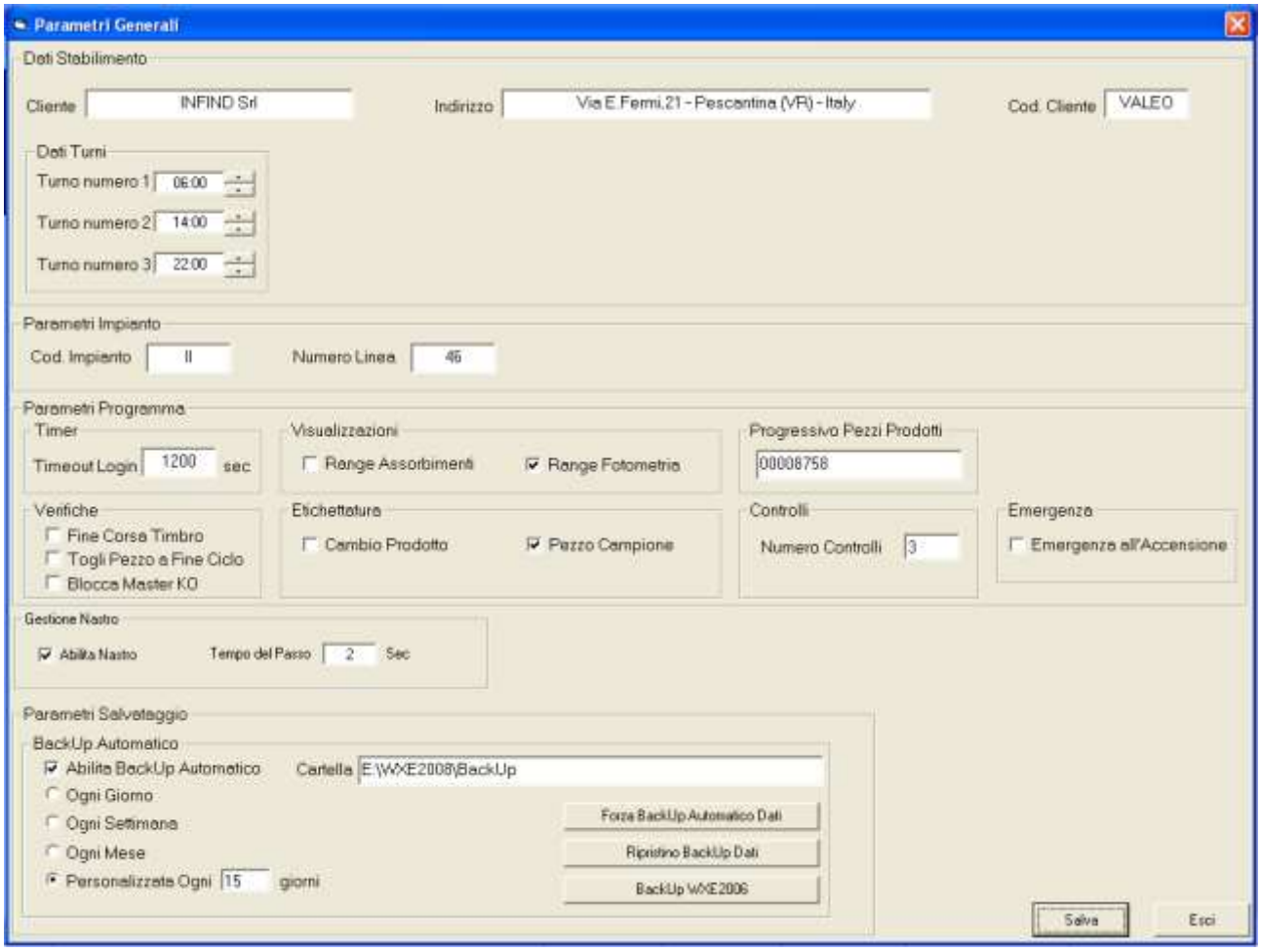
Note: to program test procedures see the section entitled Editing Recipes.

13 PARAMETERS



13.1 GENERAL PARAMETERS

Access level: Supervisor.



13.2 MACHINE PARAMETERS

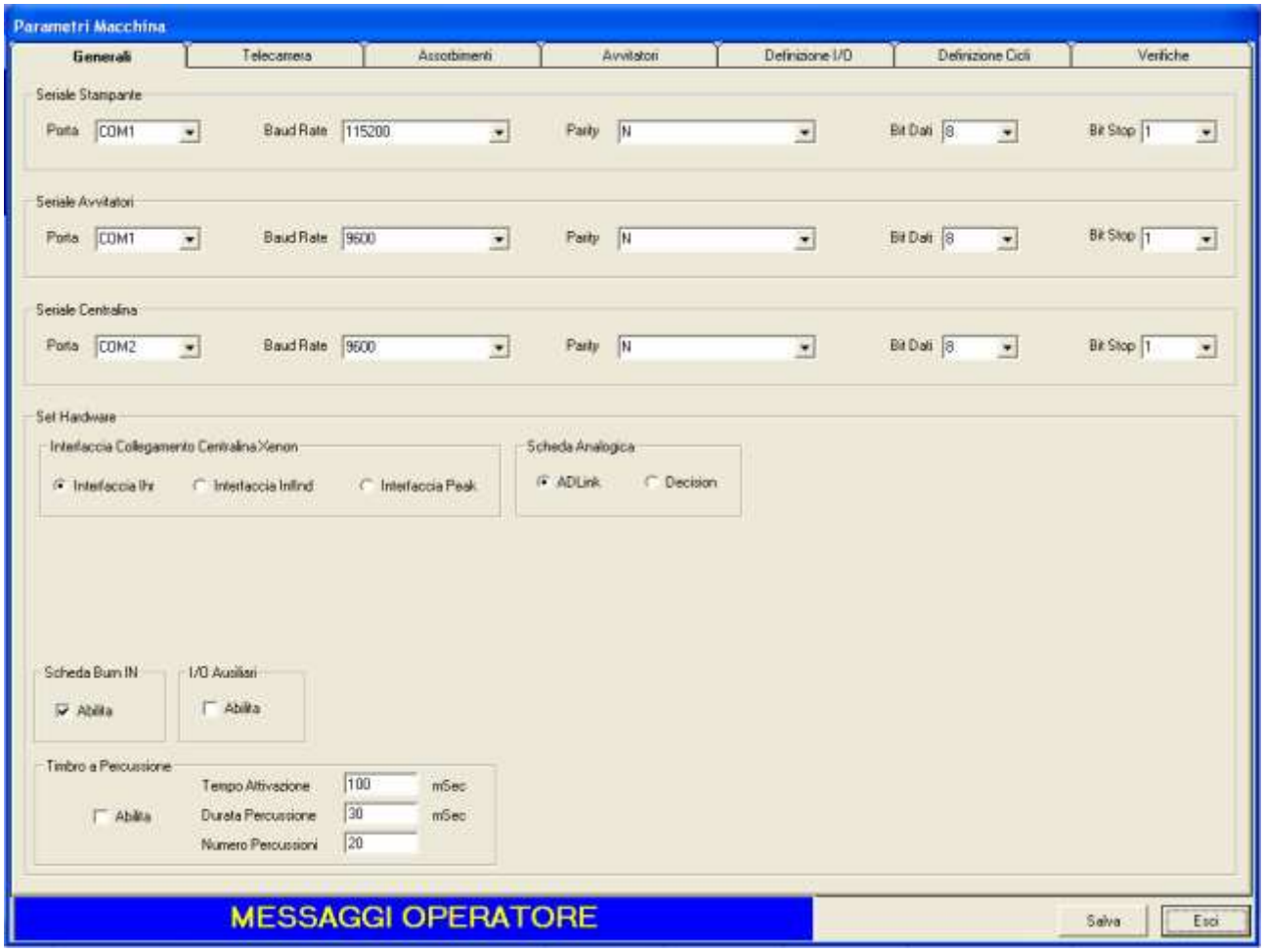
Access level: Installer.

The Machine Parameters define the configuration of the system and are linked to the specifications for the unit.
For this reason they can only be edited by expert installers.
Hereafter we display the screens of machine parameters without any explanation.

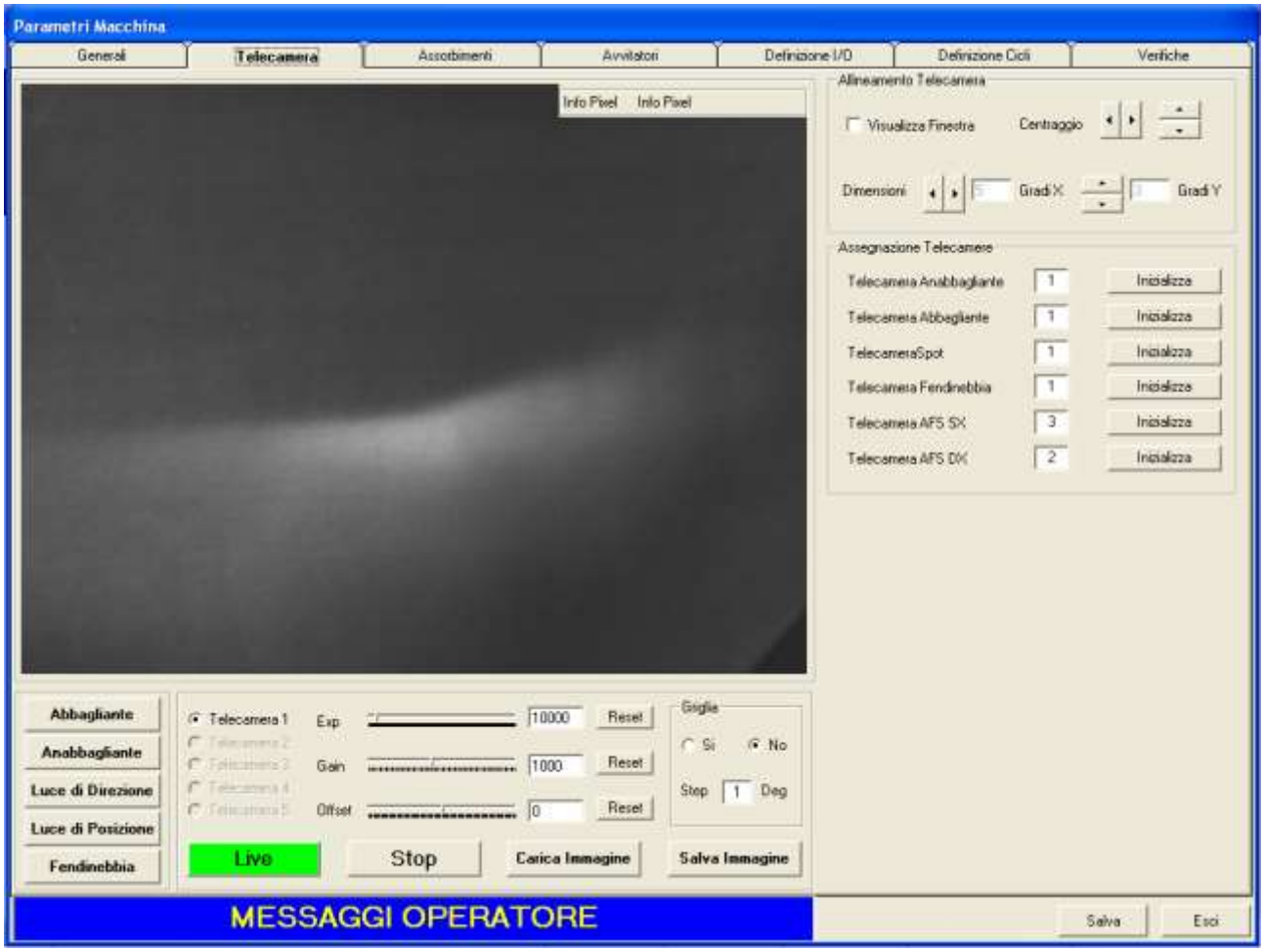
Contact the manufacturer in case of problems relative to their use.

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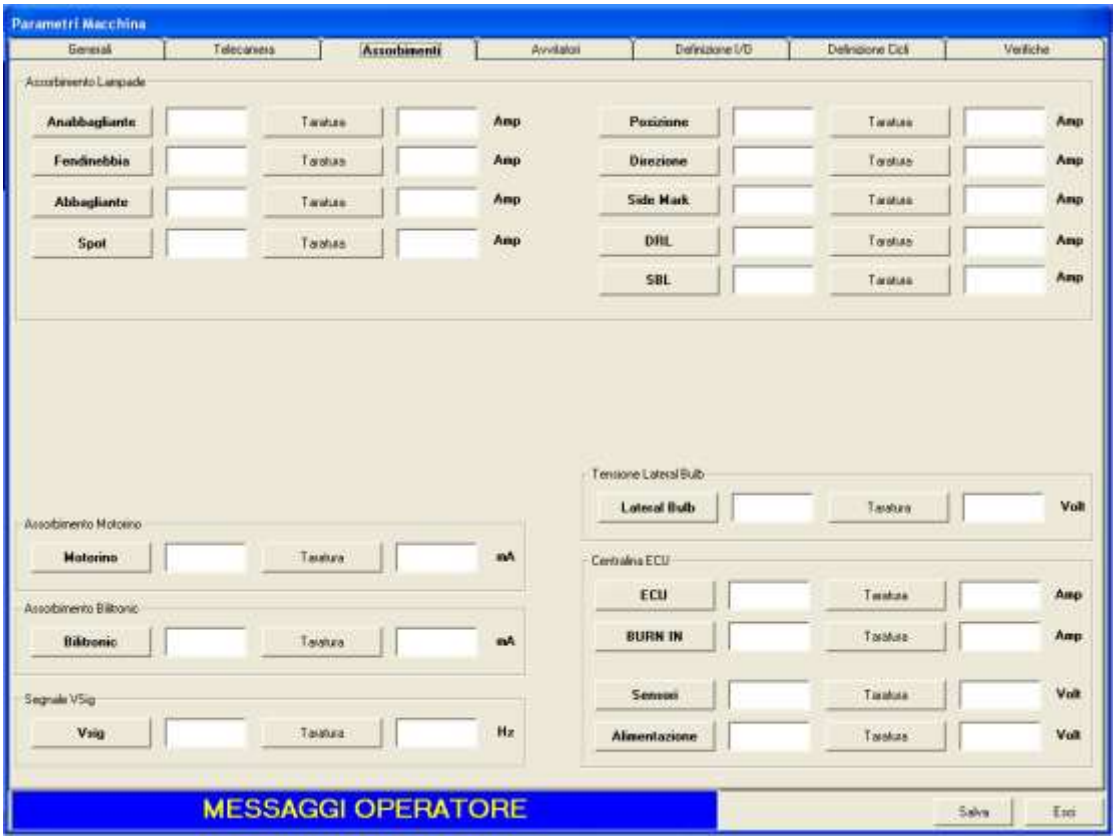
13.2.1 GENERAL MACHINE PARAMETERS

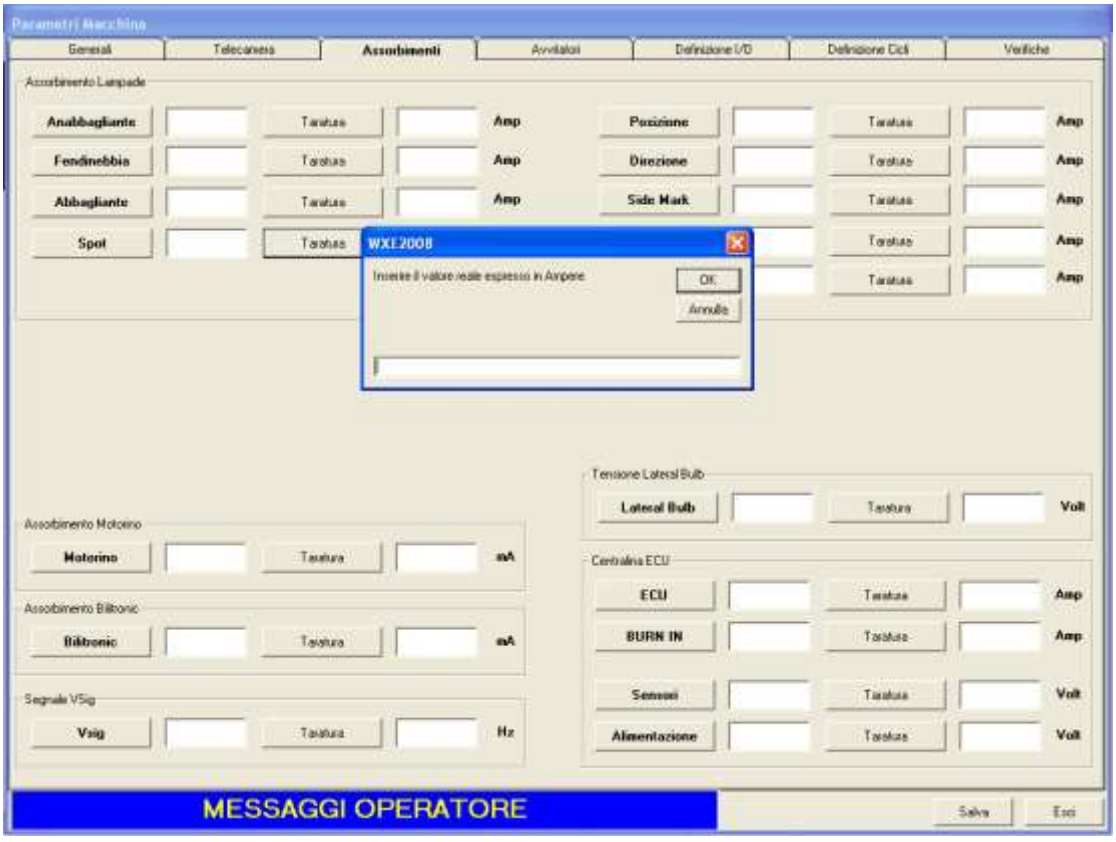


13.2.2 VIDEOCAMERA PARAMETERS

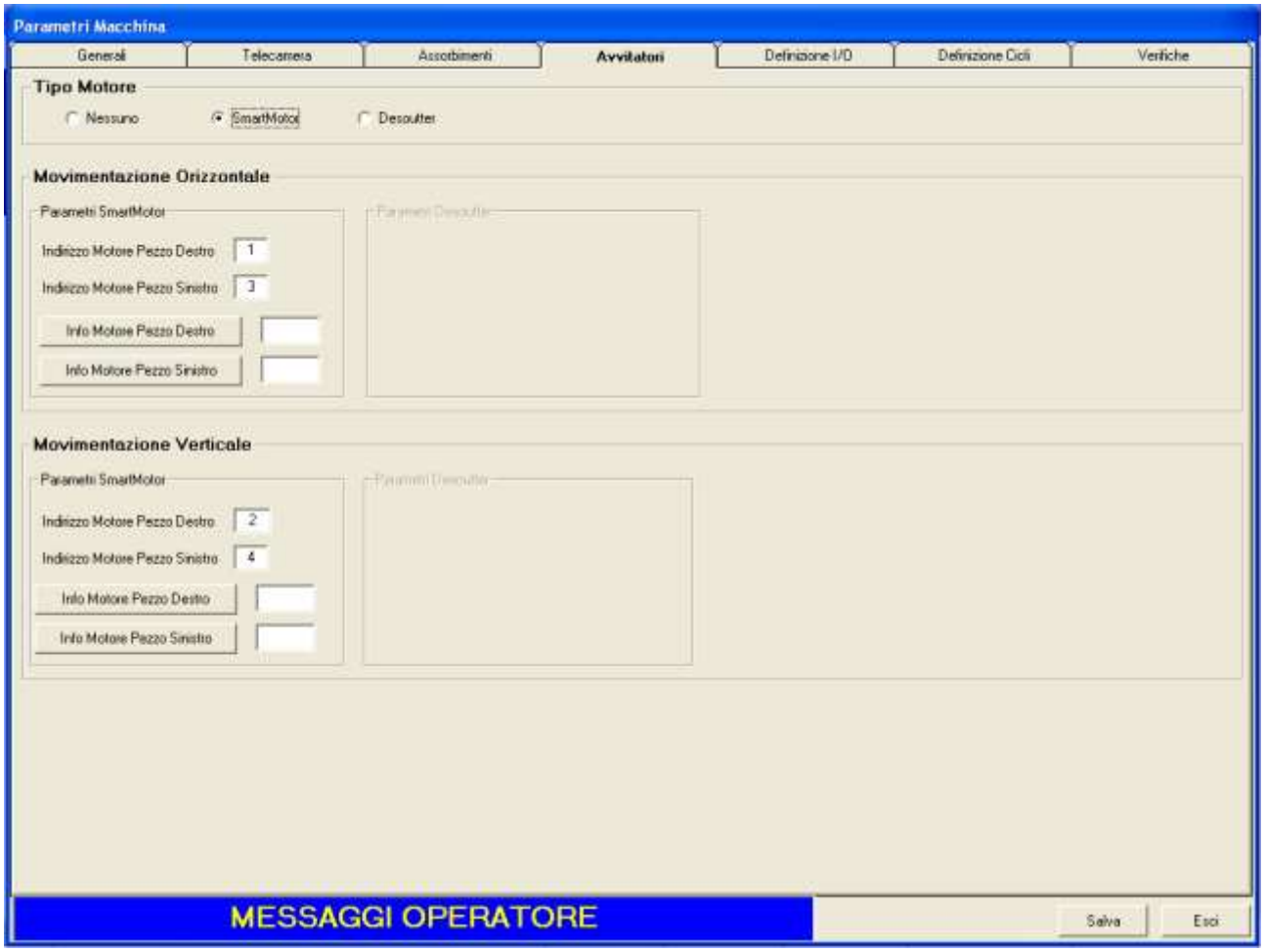


13.2.3 ABSORPTION PARAMETERS





13.2.4 SCREWDRIVER PARAMETERS



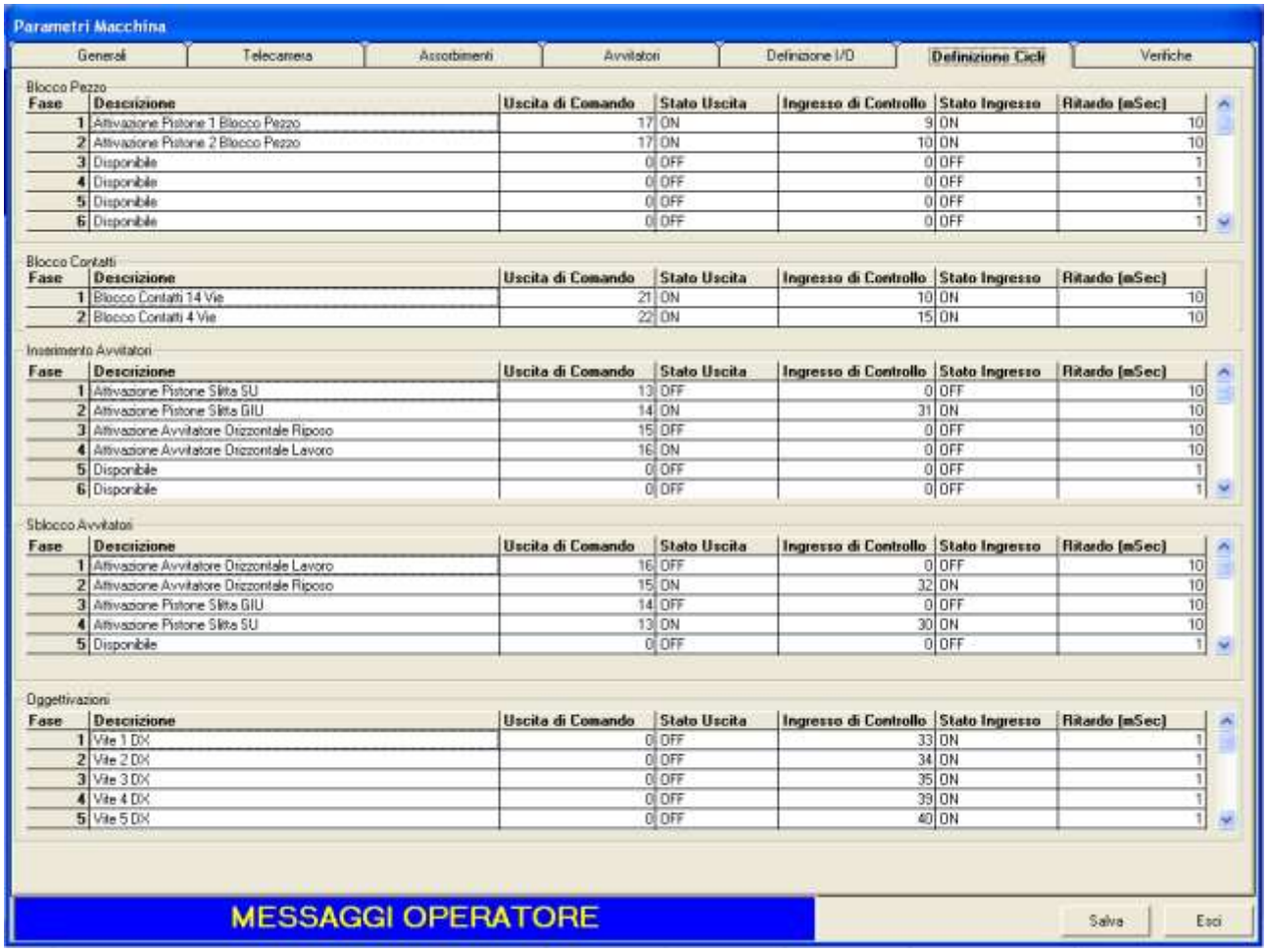
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13.2.5 I/O DEFINITION PARAMETERS



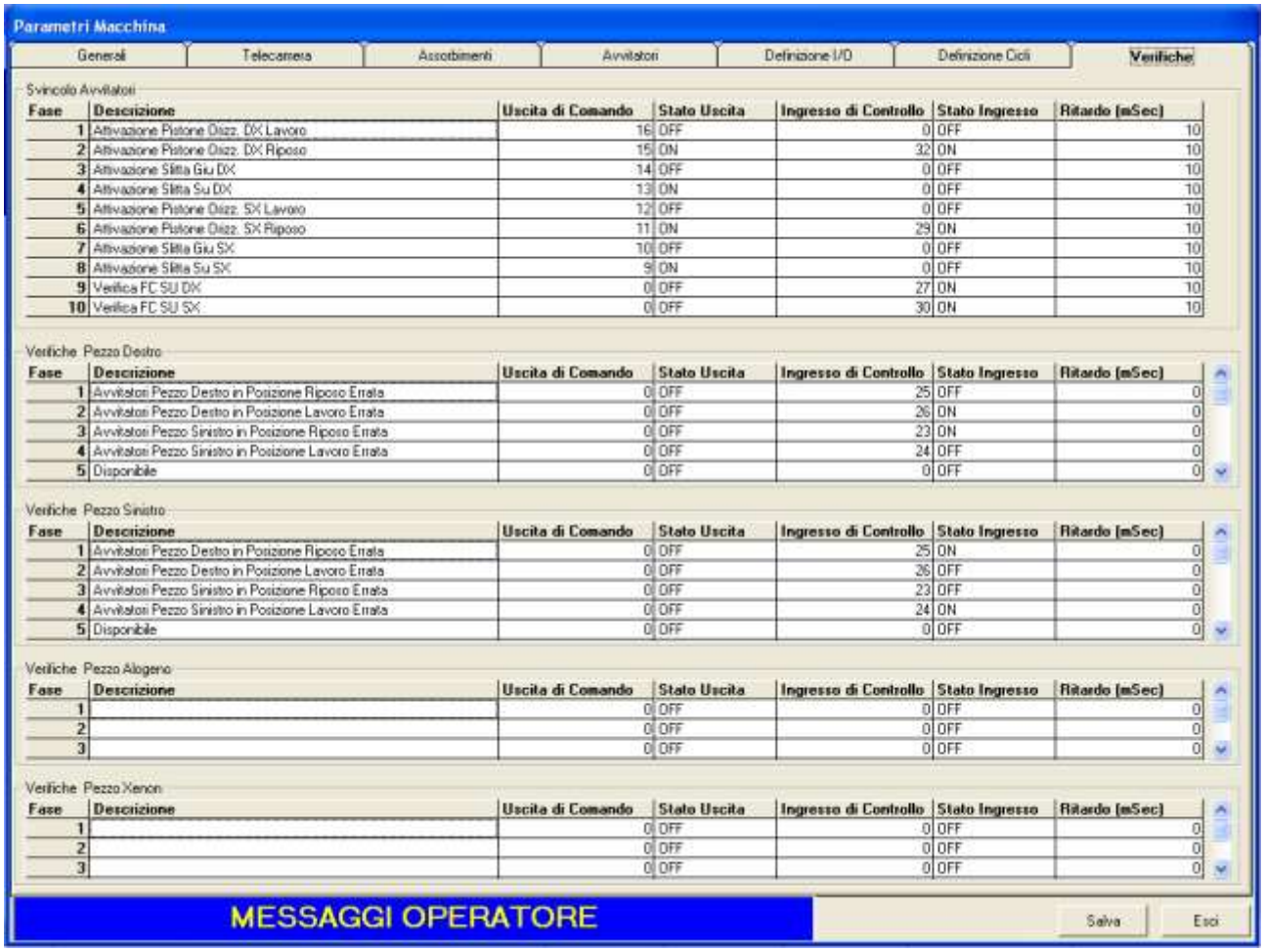
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13.2.6 PROCEDURE PARAMETERS



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13.2.7 TEST PARAMETERS



Svincolo Avvitatori

Fase	Descrizione	Uscita di Comando	Stato Uscita	Ingresso di Controllo	Stato Ingresso	Ritardo (mSec)
1	Attivazione Pistone Disz. DX Lavoro	16	OFF	0	OFF	10
2	Attivazione Pistone Disz. DX Riposo	15	ON	32	ON	10
3	Attivazione Slitta Giu DX	14	OFF	0	OFF	10
4	Attivazione Slitta Su DX	13	ON	0	OFF	10
5	Attivazione Pistone Disz. SX Lavoro	12	OFF	0	OFF	10
6	Attivazione Pistone Disz. SX Riposo	11	ON	29	ON	10
7	Attivazione Slitta Giu SX	10	OFF	0	OFF	10
8	Attivazione Slitta Su SX	9	ON	0	OFF	10
9	Verifica FC SU DX	0	OFF	27	ON	10
10	Verifica FC SU SX	0	OFF	30	ON	10

Verifiche Pezzo Destro

Fase	Descrizione	Uscita di Comando	Stato Uscita	Ingresso di Controllo	Stato Ingresso	Ritardo (mSec)
1	Avvitatori Pezzo Destro in Posizione Riposo Errata	0	OFF	25	OFF	0
2	Avvitatori Pezzo Destro in Posizione Lavoro Errata	0	OFF	26	ON	0
3	Avvitatori Pezzo Sinistro in Posizione Riposo Errata	0	OFF	23	ON	0
4	Avvitatori Pezzo Sinistro in Posizione Lavoro Errata	0	OFF	24	OFF	0
5	Disponibile	0	OFF	0	OFF	0

Verifiche Pezzo Sinistro

Fase	Descrizione	Uscita di Comando	Stato Uscita	Ingresso di Controllo	Stato Ingresso	Ritardo (mSec)
1	Avvitatori Pezzo Destro in Posizione Riposo Errata	0	OFF	25	ON	0
2	Avvitatori Pezzo Destro in Posizione Lavoro Errata	0	OFF	26	OFF	0
3	Avvitatori Pezzo Sinistro in Posizione Riposo Errata	0	OFF	23	OFF	0
4	Avvitatori Pezzo Sinistro in Posizione Lavoro Errata	0	OFF	24	ON	0
5	Disponibile	0	OFF	0	OFF	0

Verifiche Pezzo Alogeno

Fase	Descrizione	Uscita di Comando	Stato Uscita	Ingresso di Controllo	Stato Ingresso	Ritardo (mSec)
1		0	OFF	0	OFF	0
2		0	OFF	0	OFF	0
3		0	OFF	0	OFF	0

Verifiche Pezzo Xenon

Fase	Descrizione	Uscita di Comando	Stato Uscita	Ingresso di Controllo	Stato Ingresso	Ritardo (mSec)
1		0	OFF	0	OFF	0
2		0	OFF	0	OFF	0
3		0	OFF	0	OFF	0

14 DIAGNOSTICS

Access level: Expert.



Using the items on the Diagnostics menu it is possible to access the screens that serve to check the operation of the different functions of the system.

Hereafter we shall describe the various diagnostic screens.

Note: Depending on the functions and/or hardware enabled, some screens may be disabled.

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14.1 INPUTS/ OUTPUTS



This screen can be used to test both the digital and analogic Input and Output functions.

Digital Inputs: The enabled input, with ON status, is highlighted by the corresponding green key, while OFF status is indicated by the gray key.

Digital Outputs: The outputs for switching on the lamps can be enabled simply by clicking on the corresponding key, while movements have to be confirmed with the Start Procedure key on the unit. Movement of the screwdrivers occurs only if the operator is outside the safety barrier.

Analogue Inputs: The data relative to absorptions are displayed on the right-hand side.
For every function, the value in digits read by the screen is displayed with the corresponding value in the correct unit of measurement.

Analogue Outputs: For systems with control of the signal of the control unit, VSig, it is possible to command the analogue output that drives the Voltage/Frequency converter.

To enable the outputs, place the switch on the keyboard in the Step or Manual position, select the output desired and press the Step key.
In Manual Position only the output selected is enabled/disabled.
In Step position, if the output enables/disables a movement of a cylinder with dual command, the second output is enabled so that you can make the movement.
IMPORTANT: See the wiring diagram for details of the I/O for the specific unit.

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14.2 TEST OF SCREWDRIVERS



Check screwdriver function using the speed selection keys.
It is possible to select the direction of rotation of the screwdrivers in the rotation window.

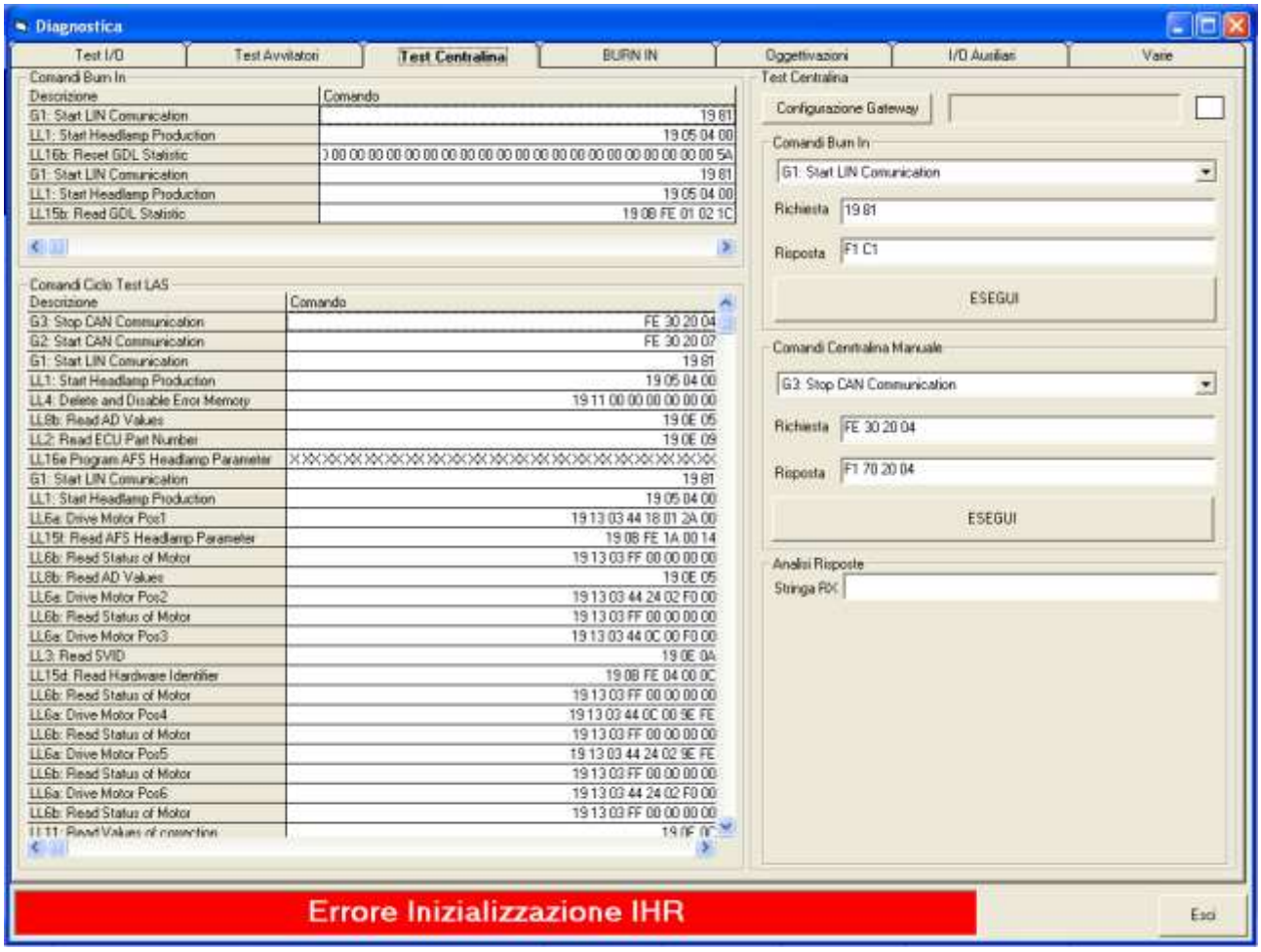
If necessary, press “Screwdriver Stress Procedure” to perform continuous clockwise and counter clockwise rotations.

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14.3 TEST OF CONTROL UNIT

The test screen for discharge lamp control units is divided into two zones:

- On the left, the commands to be sent to the control unit are displayed with the expected response.
- On the right, you can send data to the control unit and view the response.

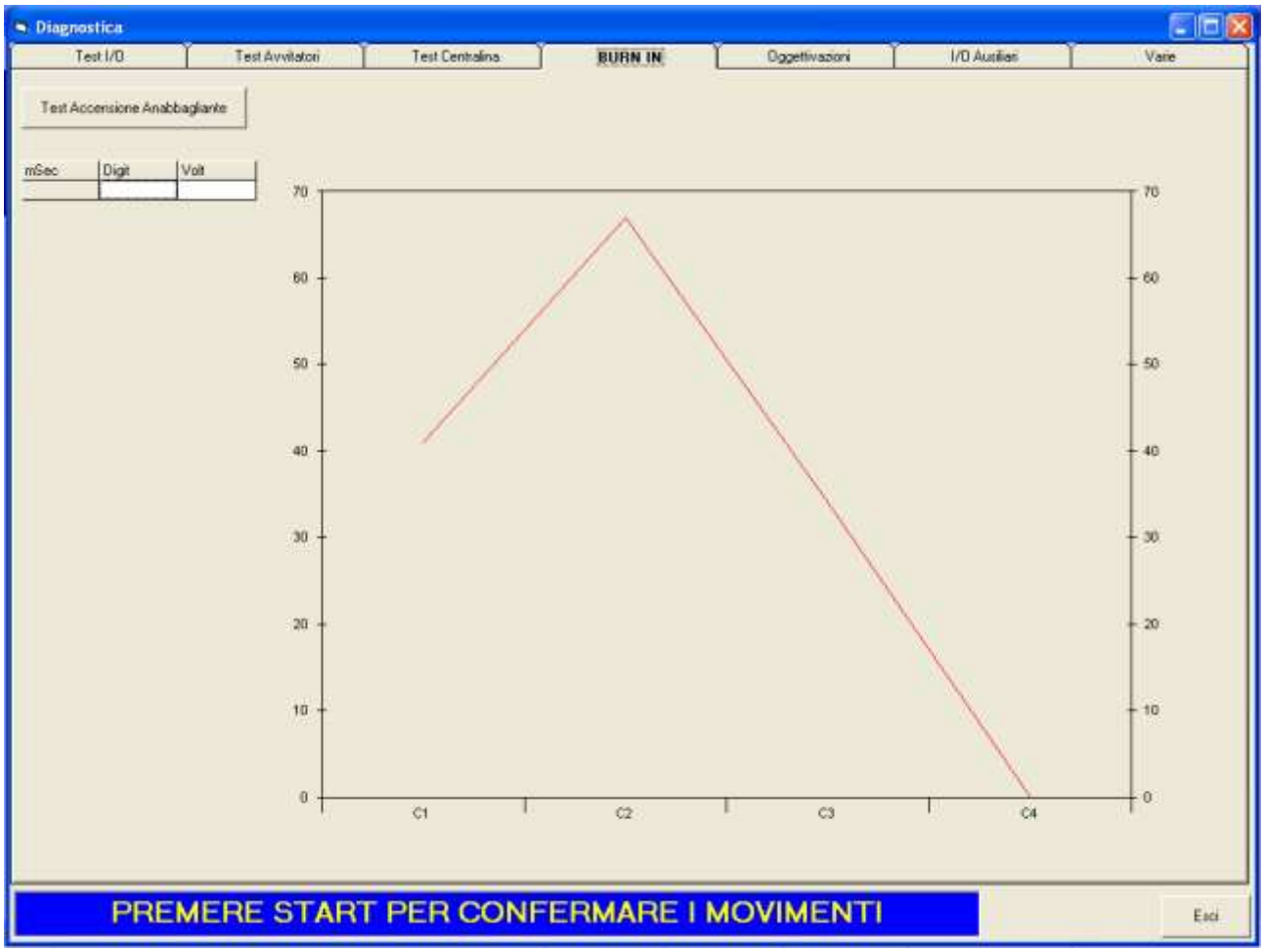


Press “Configuration of Gateway” to initialize the control unit.

Select the desired command and press “Run”.

Note: Refer to the control unit specifications for details of the commands and test sequence.

14.4 BURNIN TEST

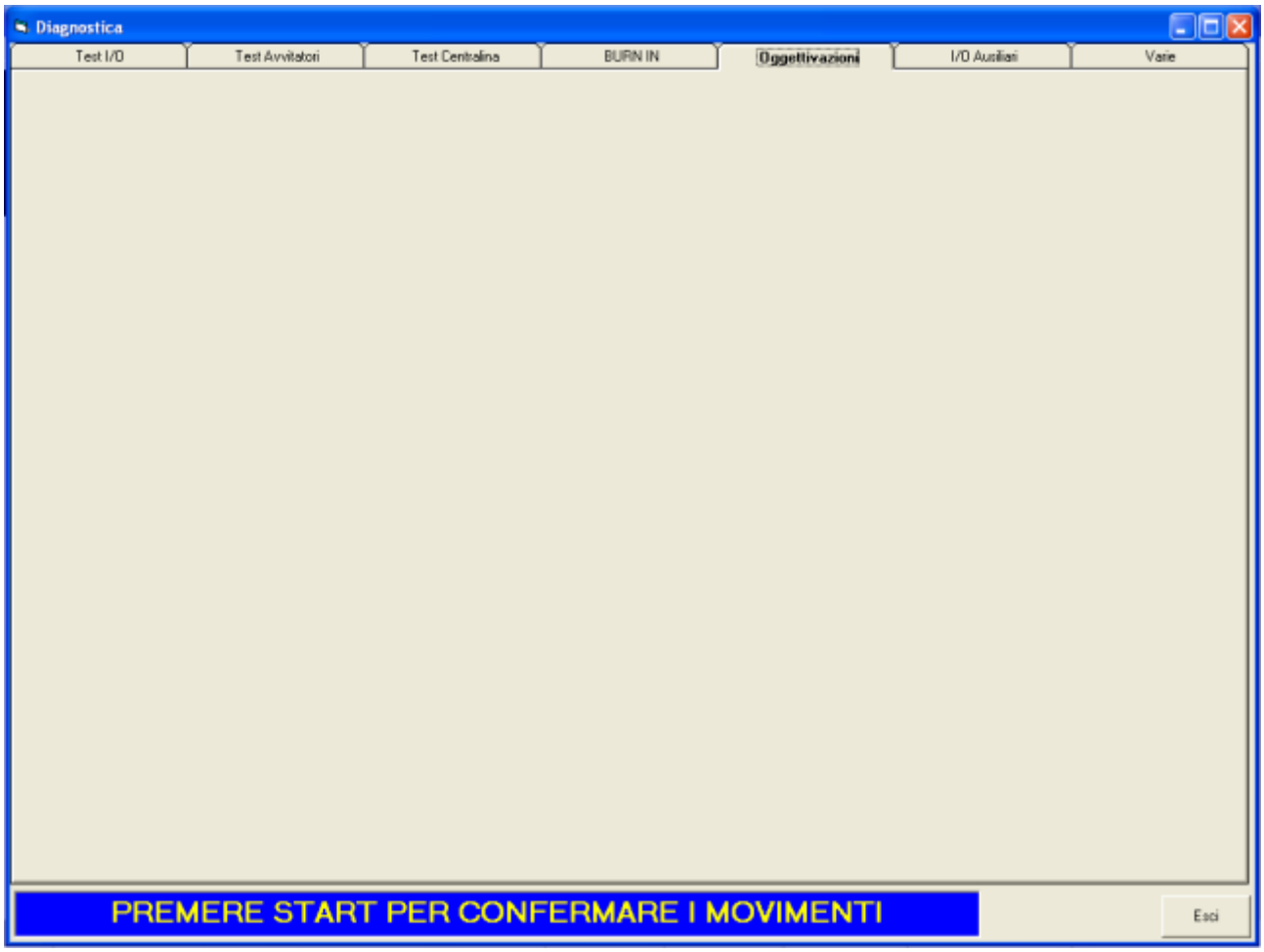


This screen can be used to check the function of the analogue screen for rapid acquisition, used for the BurnIn test of discharge lamps.

After locking the piece and inserting the electric connector from the I/O screen, press “Test Low Beam Ignition”. The low beam will be switched on and acquisition of absorptions of the discharge lamps will begin. At the end of the test, the data will be displayed on the table on the left-hand side with the graph of absorption performance.

The BurnIn test data for discharge lamps are set on the Edit Recipes screen.

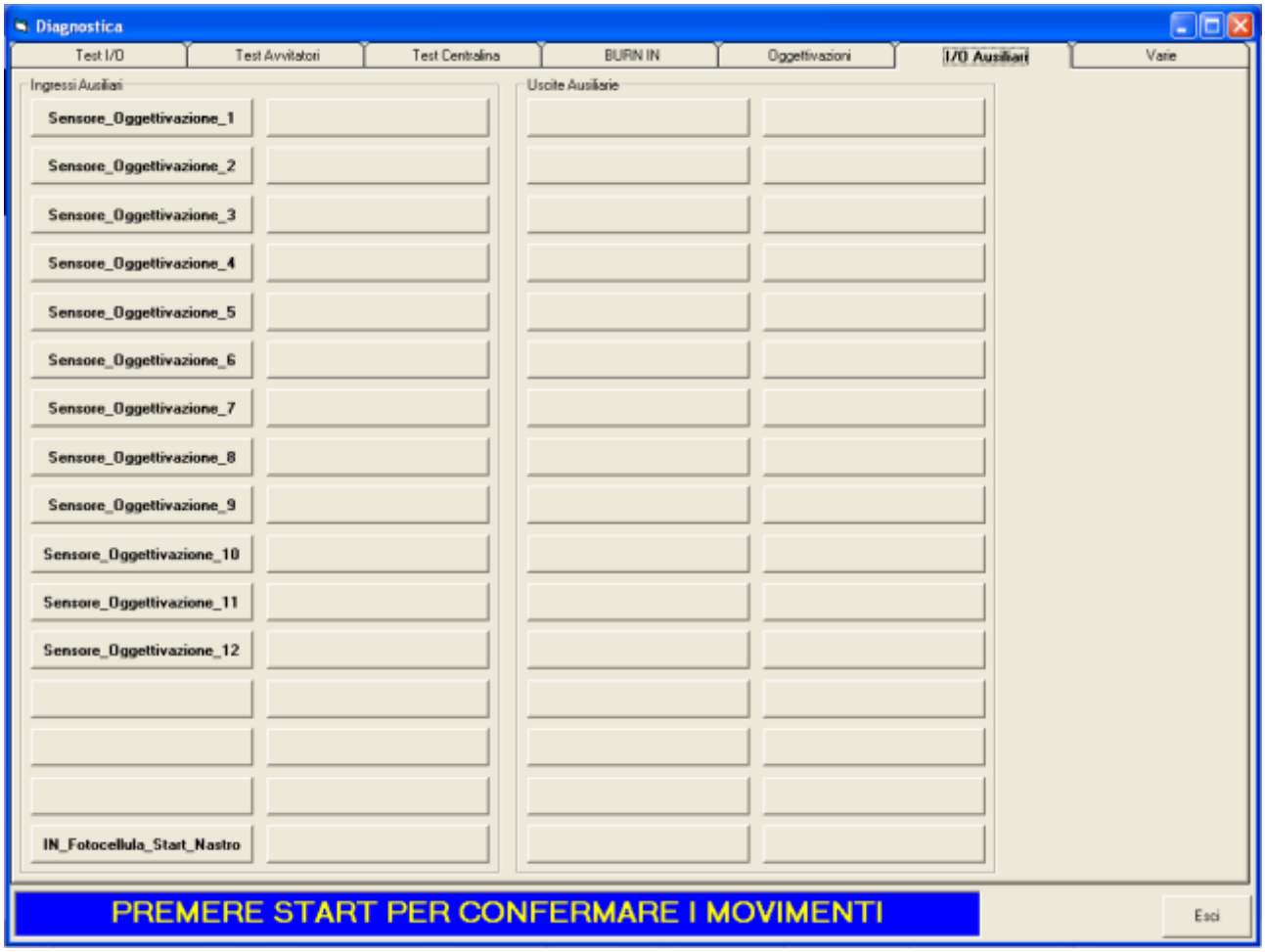
14.5 RECOGNITION



Not used.

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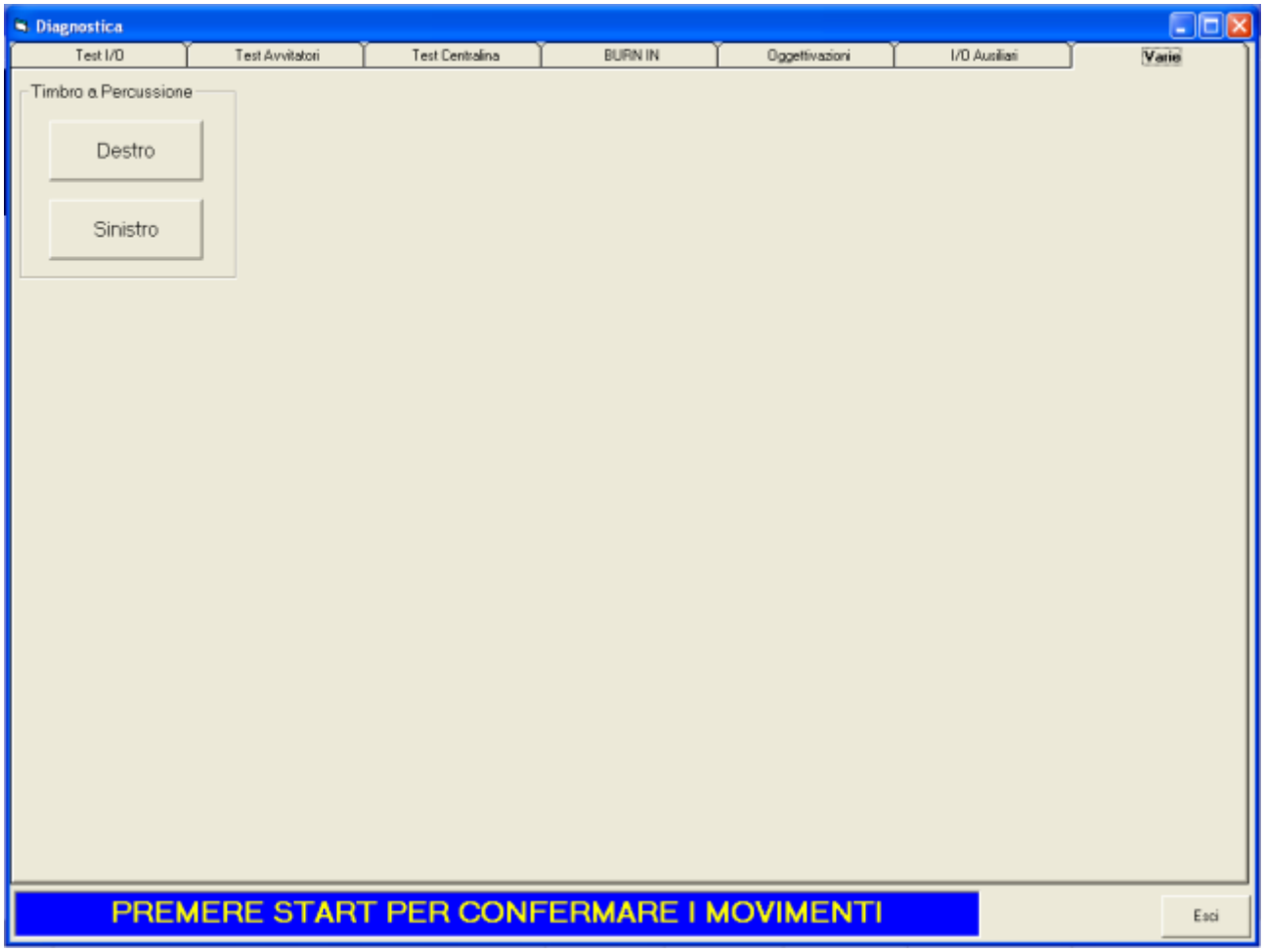
14.6 AUXILIARY I/O



On this screen you can check any auxiliary inputs / outputs if the relative screen is present and enabled.

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14.7 MISCELLANY



Use this screen to test the percussion stamp if enabled.

15 SECURITY


WXE 2008 Versione: 1.1.0
Sistema Ricette Cicli Lavoro Parametri Diagnostica Sicurezza About
Login
Gestione Supervisori
Funzionamento Automatico
Telecamera
Produzione Percentuali Statistica
Codice 00460809999
Descrizione Lancia 844 Alo DX CD
Proiettore Destro
Circolazione ECE
MANUALE
PEZZI BUONI 000012
PEZZI SCARTI 000003
Misure Certificazioni LAS
Assorbimenti
Misura
Anabbagliante
Abbagliante
Posizione
Direzione
DRL
Bobina Abba
Correttore (mA)
SBL
Geometria
Angolo
Nitidezza
Gradiente
Emax
Pezzo Campione
Carica Ricetta
Start
Debug
Posizionare il Pezzo
INFIND Srl - Via E.Fermi,21 - Pescantina (VR) - Italy
29/08/2008
15.59
Turno: 2
Operatore

This program has 4 levels of use, each of which permits access to specific functions of the system.
Hereafter we list the functions activated by accessing the function mode specified:

- *Operator Level*
  - Switching off the system
  - Test of the Master Piece
  - Loading recipe relative to the product to be tested
  - Resetting partial production counters (Good Pieces/ Rejects)
- *Expert Level*
  - Loading Test Procedures
  - Access to Manual Function screen
  - Access to videocamera test screen
  - Access to Diagnostics screen
  - Display of production percentages
- *Supervisor Level*
  - Access to Edit Recipes screen
  - Access to General Parameters screen
- *Installer Level*
  - Access to screen relative to setting machine parameters

15.1 LOGIN

Access level: Operator
At the Login request the following window appears:



Select User Name and enter the Password
If the Password is not correct the following message appears:



15.2 USER MANAGEMENT

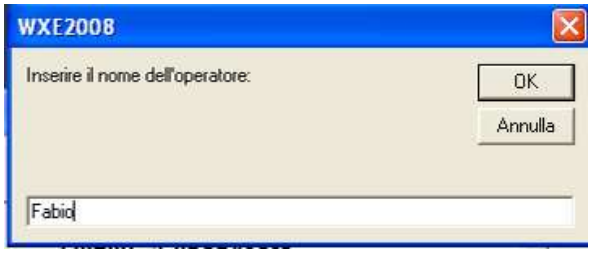
Access level: Supervisor

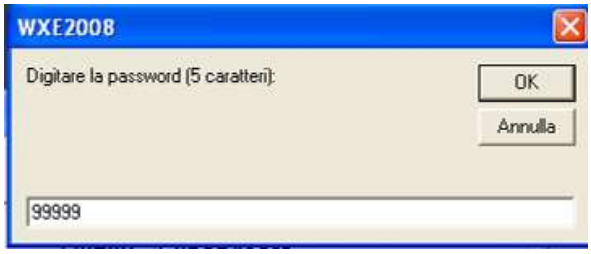
It is possible to create users who have access to the various functions, assigning the appropriate security level to each.

Select the security level for the new user and press “Create New Supervisor”.



You are asked to enter the User Name and then to enter the new password:

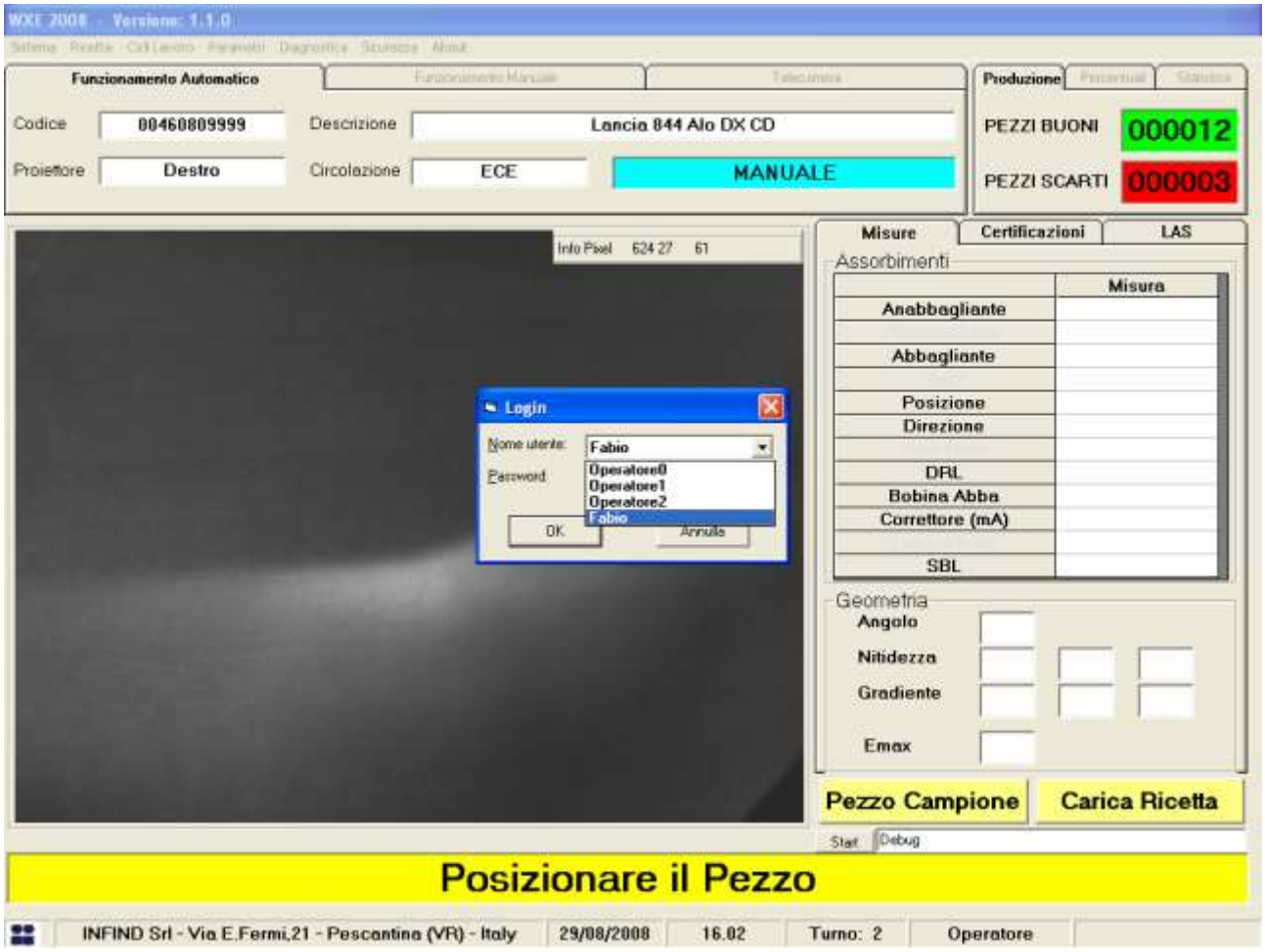




At this point the new User has been created.

When you request access to protected screens the new User will appear in the list. After entering the password assigned, the user will have access to the screens visible to that user's security level.

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In the main program directory a file Log.txt records all accesses to the Login.

Installer: Operator 0 Date: 28/08/2008 10.19.20
Supervisor: Fabio Date: 28/08/2008 10.22.04

To eliminate a User, select the Level and Name, enter the password and enter “Delete Supervisor”.



16 ABOUT

Security Level: Operator.



Displays the general information.

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17 TAKING THE UNIT OUT OF SERVICE

In case of prolonged disuse, it is necessary to disconnect the power supply from the main power panel and all the other supplies (pneumatic and/or hydraulic) necessary to the machine.

17.1 LOTO PROCEDURE

LOTO is a safety procedure used in automotive lighting to ensure that machinery is adequately insulated from its energy sources and cannot be started or excited unexpectedly, before the completion of maintenance or service work.

Below are listed Energy source present on “AIMING S220” 2K23048_01bench

- Electricity supply 220Volt
- Supply 6 Bar pneumatic compressed air

The energy insulation devices used are manually operated with the possibility of inserting a padlock in the locking position of the energy source.

The color of this device is RED.

The following documentation indicates the position of these devices on the bench.

COMPRESSED AIR SUPPLY 6 BAR

Compressed air treatment group formed by General Pneumatic Valve type **SMC cod. VHS30-F03A** with opening/closing system function and possibility of locking with lock in the closing position, and a progressive starting valve type **SMC cod. AV3000-F03-5YZ-Q** with discharge system function. The pneumatic valve discharges the entire system and since the cylinders are all double-acting in the system there is no residual energy. It is advisable to check on the line gauge that the pressure is zero

In the diagram Pneumatic the points of interruption are indicated and highlighted . After the interruption in the pneumatic systems no residual energy remains





Position pneumatic general valve



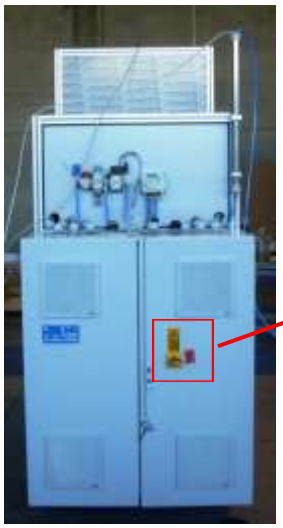
General pneumatic valve lockable during maintenance operations

Interruption point is marked with pictogram P

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ELECTRICAL GENERAL SWITCH

Electrical power 220 Volt
Electrical main switch







General electricity switch that can be padlocked during maintenance operations

Interruption point is marked with pictogram **E**

General electric switch**: SIEMENS cod. 3LD20170TK13** with function to stop electrical system and possibility of locking option with lock in OFF position.
This operation has to be made during clearing and maintenance of bench.
General switch stops incoming electricity therefore it can be assumed that all system is off. There are no capacitors in the electrical system so there is no residual energy. It is advisable to check with a multimeter the absence of voltage by measuring it on the clamps n. 100.0, n. 101.0 indicated in the electrical diagram.



ATTENTION: On clamp of input of main switch, there is voltage of line on clamps indicated with N and U. Clamps are IP56 protected, but it is advisable to disconnect the voltage on plant's line. In the electrical diagram the points of interruption and isolation are indicated and highlighted. After the interruption in the electrical systems there is no residual energy.

Protection of incoming clamps

18 MAINTENANCE

18.1 MAINTENANCE OPERATIONS

Maintenance should be carried out by qualified personnel trained in the types of jobs to be done. When performing any maintenance, we recommend isolating the area around the machine to prevent unauthorized personnel from accessing the zone. The following types of maintenance tasks should be done by authorized operators only.

	**ATTENTION:** All maintenance or repair procedures must be done after applying the LOTO procedure (chap. 17.1)

18.2 SPECIAL PRECAUTIONS

In performing works of maintenance or repairs the following recommendations should be implemented:

- Before starting work display a sign “MACHINE UNDER MAINTENANCE” in a clearly visible position.
- Do not use solvents or flammable materials.
- Take care not to spill lubricants or coolants.
- To access the higher parts of the machine use suitable means for the operations to be done.
- Do not climb on the parts of the machine as they are not designed to withstand the weight of a person.
- After completing any works replace and fasten any guards or casings removed.



The manufacturer is not liable for failure to comply with these recommendations or for any other improper use or activity not mentioned in these instructions.

18.3 ROUTINE MAINTENANCE



IMPORTANT! Before performing any maintenance the operator must disconnect electrical power using the main circuit breaker, closing it with the padlock supplied and keeping the key in his pocket.

Always use the special personal protection equipment (PPE):

- Gloves.
- Safety footwear.
- Suitable clothing.

The operations described hereafter should be performed with the timing indicated.
These operations, though simple, must be done by qualified personnel.
Routine maintenance includes inspections, controls and actions to prevent stoppages and breakdowns of the machine.

OPERATION	TIMING
Check fastening of mechanical parts in motion	Monthly
Status of machine lubrication	Monthly
Clean fan filters for ventilation of control panel	Bimonthly
Empty condensation from air conditioner filter	Quarterly
Clean fan filters for ventilation of computer	Bimonthly
Clean lens with water and a NON abrasive cloth	Monthly
Check electrical connector (Pin Clamp - Connector Cleaning)	Monthly
Replace keyboard batteries	As needed



Failure to comply with these requirements exonerates the manufacturer of any responsibility under the warranty.

18.4 SPECIAL MAINTENANCE

For works of special maintenance connected with repairs following the breakdown of functional parts, do not refer for instructions and type to the preceding sections.
Contact the manufacturer for any alterations and/or adaptations to agree on possible alterations to the machine.

19 SPARE PARTS AND ACCESSORIES

19.1 ASSISTANCE SERVICE

For any type of information relative to the use, maintenance and installation of the machine, the manufacturer is always at your disposal.
Clients should frame their queries in clear terms, referring to this manual and to the instructions it contains.

19.2 SPARE PARTS

ALWAYS USE ORIGINAL SPARE PARTS.
It is never advisable to use non-original spare parts and if this is done, any warranty still in force will be immediately invalidated and the manufacturer will have no responsibility with regard to use of the machine.



The manufacturer is not liable for breakage, malfunctions or damages to persons or property deriving from the use of non-original spare parts.

20 DISPOSAL



Since different legislation is in force in individual countries, for the disposal of the machinery the requirements imposed by the laws and bodies responsible for waste and packaging in the country in which the machine is in use must be observed.





Upon disposal, it is necessary to destroy the identification plate (if present), as well as the documentation provided upon purchase.